Biochemistry






Section H
Protein synthesis

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H1 The genetic code

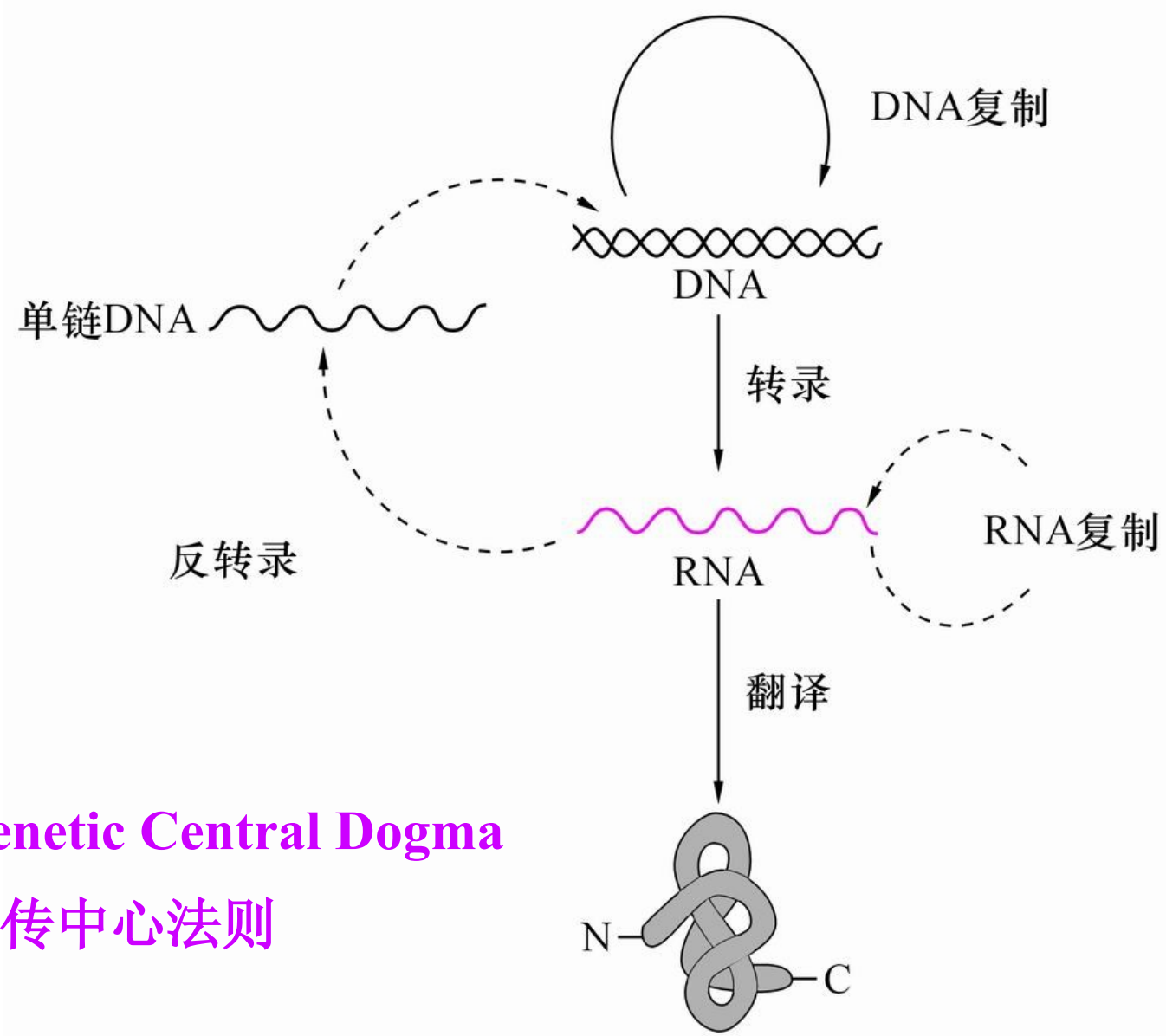
H2 Translation in prokaryotes

H3 Translation in eukaryotes

H4 Protein Targeting

H5 Protein Glycosylation





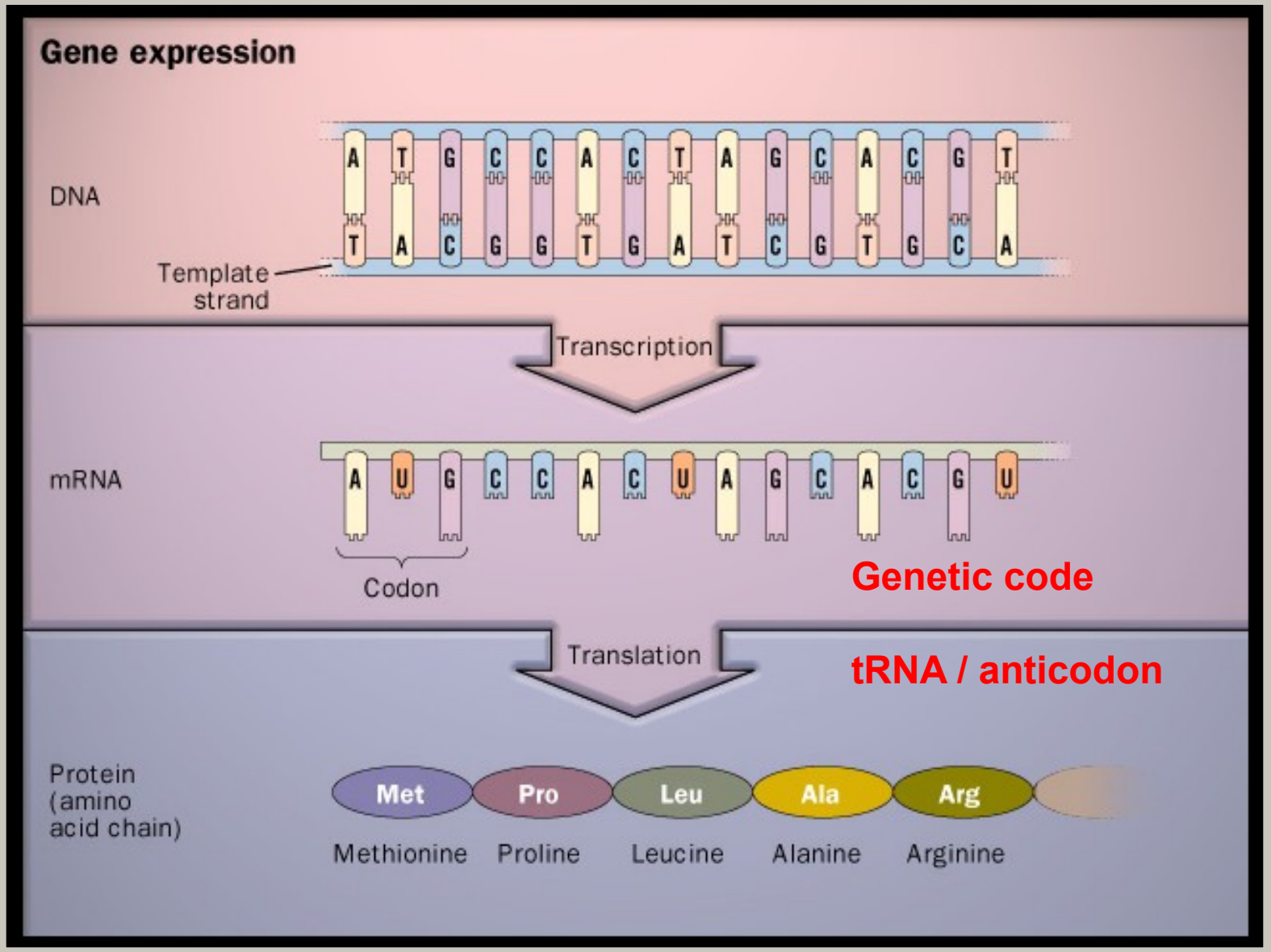
Genetic Central Dogma

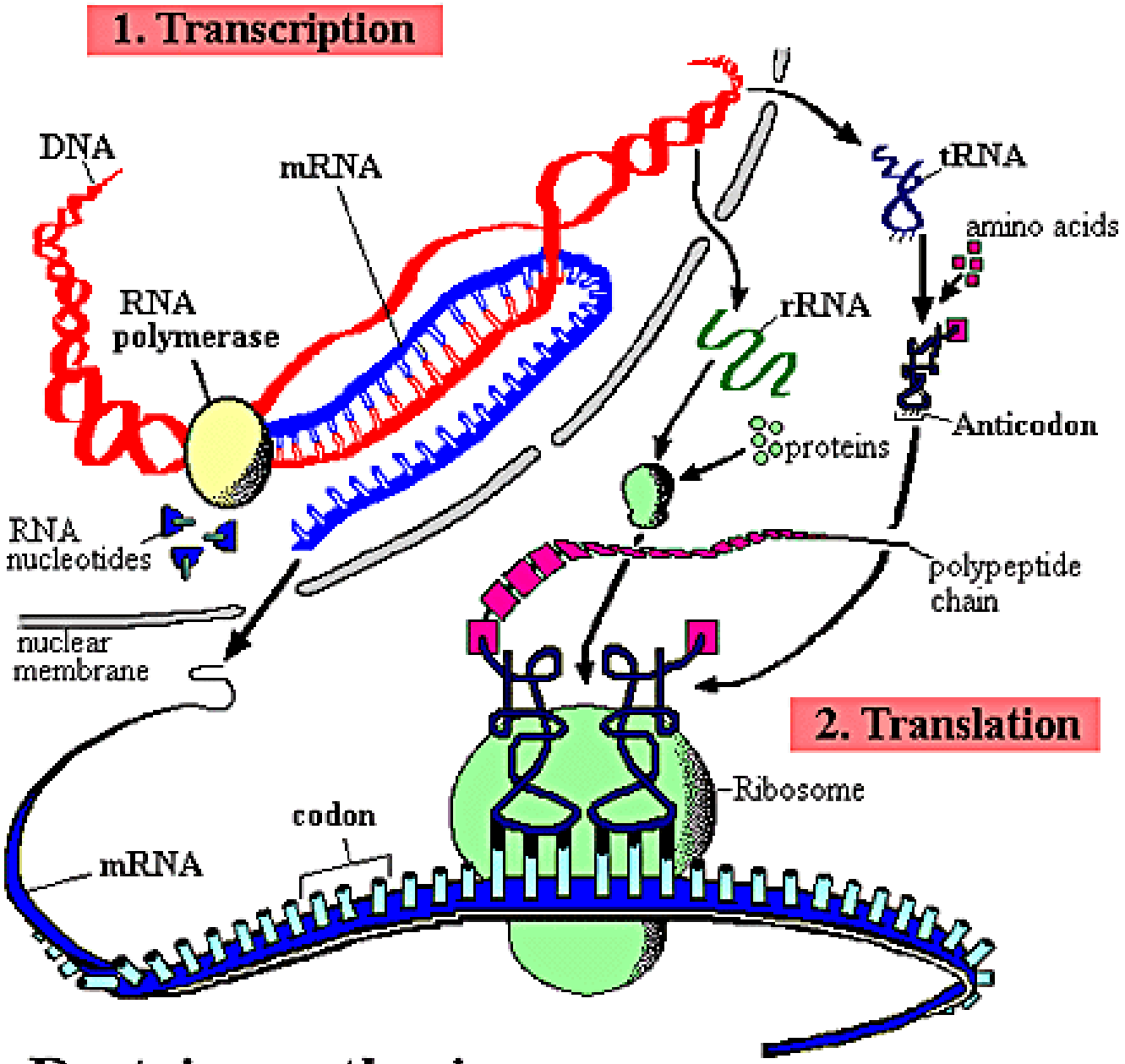
遗传中心法则

Brief history of discovery of genetic central dogma

1865 年，孟德尔完成长达 8 年的豌豆实验。
1902 年，美国生物学家萨顿和鲍维里通过观察细胞的减数分裂时又发现染色体是成对的，并推测基因位于染色体上。
1928 年 ，摩尔根发现了基因的连锁与交换。
1944 年， Oswald T. Avery 首次证明 DNA 是基因的载体（遗传信息的载体）
1950,1953 年， Barbara Mclintock 发现跳跃基因 transposon.
1953 年， Watson and Crick DNA double helix
Semiconservative replication–for high fidelity of DNA replication
1955 年， Francis Crick raised a purely theoretical adaptor hypothesis to explain how sequences of four nucleotides in a nucleic acid template determine sequences of 20 amino acids in proteins.
1958 年， Francis Crick genetic central dogma (The concept of messenger RNA (**mRNA**) was boldly formulated by Jacob and Monod in 1961)




Gene expression
DNA
Template strand
A T G C C A C T A G C A C G T
T A C G G T G A T C G T G C A
Transcription
mRNA
A U G C C A C U A G C A C G U
Codon
Genetic code
Translation
tRNA / anticodon
Protein (amino acid chain)
Met
Pro
Leu
Ala
Arg
Methionine
Proline
Leucine
Alanine
Arginine



Protein synthesis

1. Translation (protein synthesis) necessitates the coordinated interplay of about 300 macromolecules in the cells

- The most complex of all biosynthetic pathways.
  - 60 to 90 macromolecules for making up the protein-synthesizing machine ribosomes
  - Over 20 enzymes for activating the amino acids.
  - Over 10 auxiliary proteins for the initiation, elongation and termination of the polypeptide chains.
  - Account for up to 90% of the chemical energy used by a cell for all biosynthetic reactions.



The molecules used for translation account for more than 35% of the cell's dry weight. However, proteins are synthesized with very high efficiency: a complete polypeptide chain of 100 residues is synthesized in about 5 seconds in an *E.coli* cells at 37°C.



2. The molecular mechanism of protein synthesis was mainly revealed during the 2nd half of the 20th century

Ribonucleoprotein particles (were later called **ribosomes**) were revealed to be the site of protein synthesis in rat liver cells, using radioactively labeled amino acids and immediate subcellular fractionations (*early 1950s, by Zamecnik*).

Amino acids were found to be activated by attaching to a special form of heat-stable RNA molecules (later called tRNAs) before being incorporated into polypeptides (*1950s, by Hoagland and Zamecnik*).

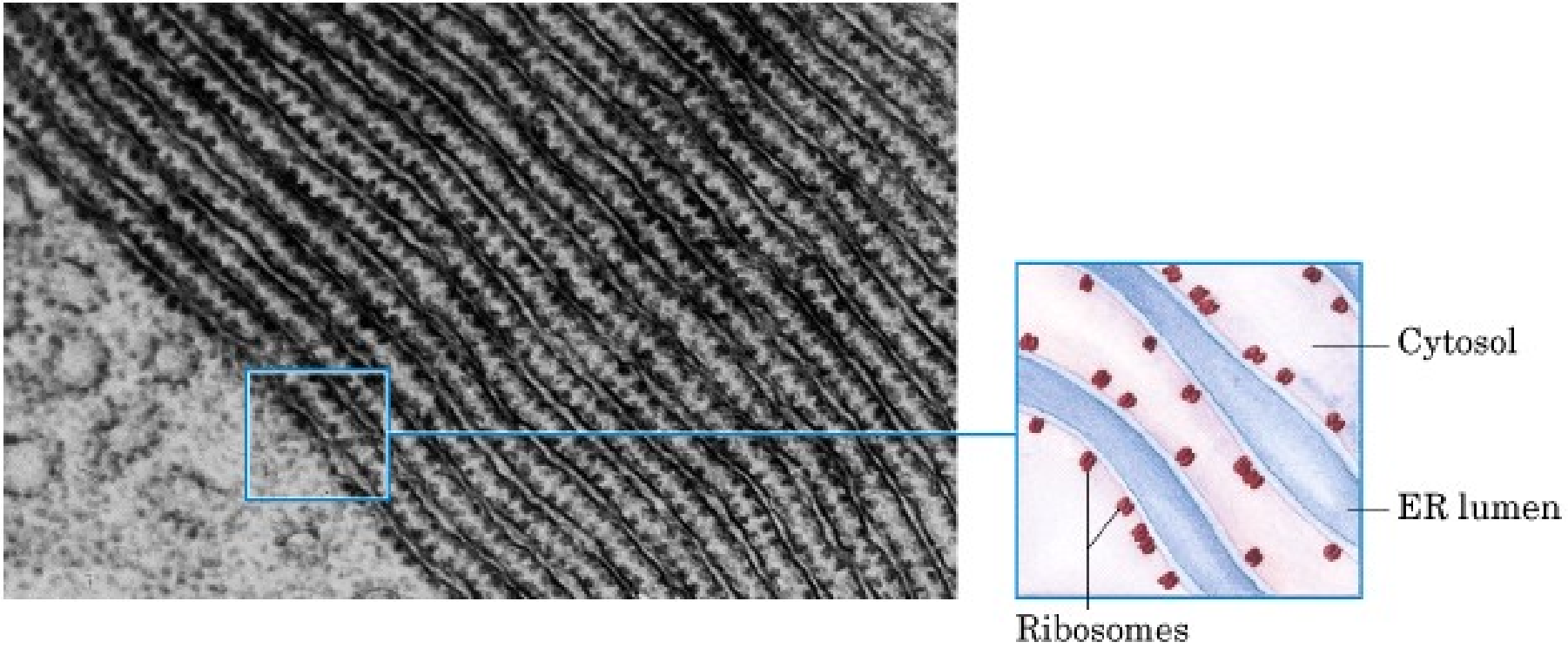


Each tRNA molecule was found to function as an **adapter** (originally hypothesized by Francis Crick), carrying a specific amino acid with one site and recognizing a specific site on a template with another site.

The concept of messenger RNA (**mRNA**) was boldly formulated by Jacob and Monod in 1961: a short-lived RNA should serve as the **information carrier** between gene and protein (to explain the quick induction of proteins in *E.coli*).

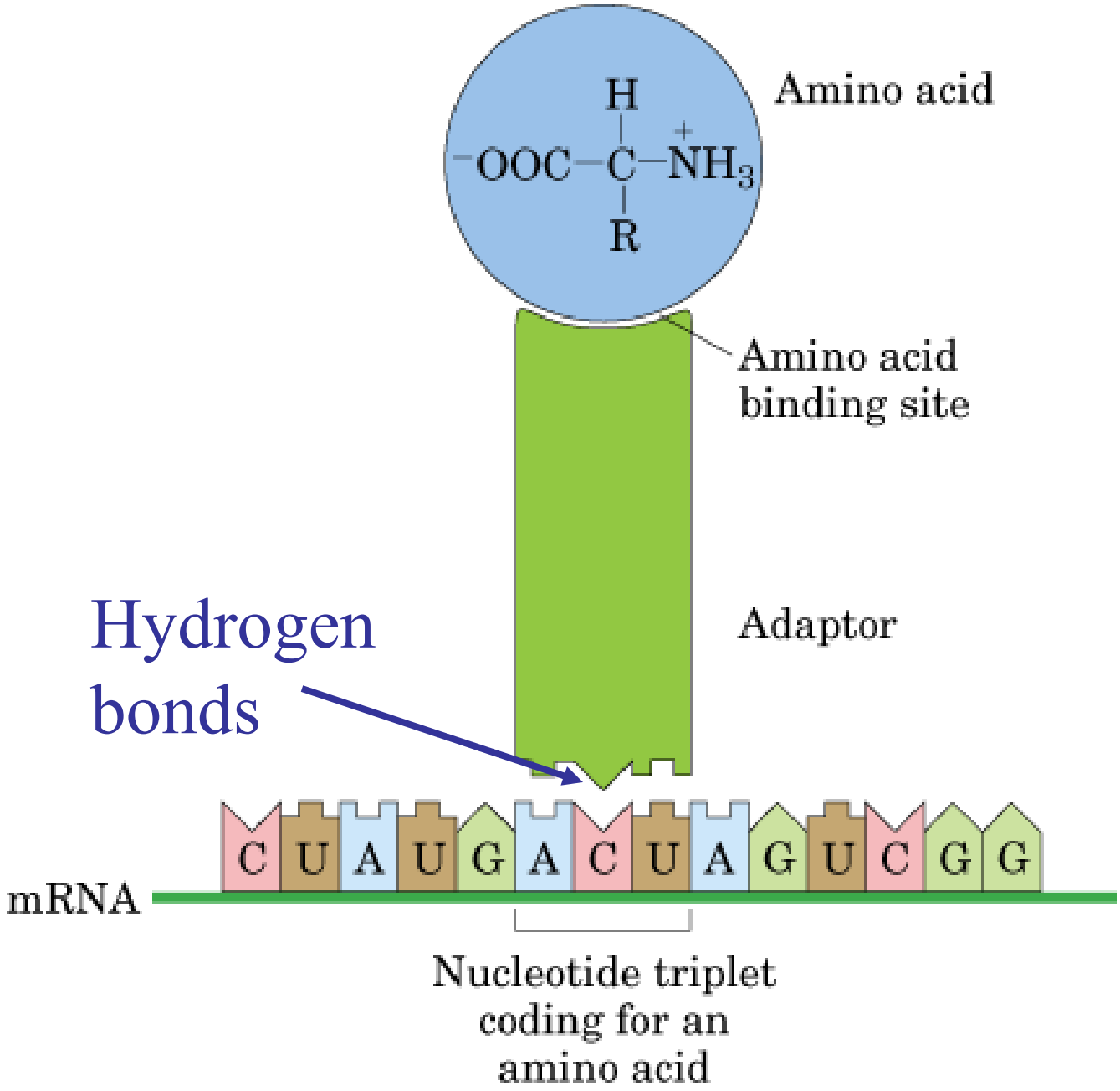
This bold hypothesis was quickly confirmed by studies of *E.coli* cells infected by T_2 phages .





Ribosomes were revealed to be the site of protein synthesis in early 1950s (pulse labeling with radioactive amino acids and subcellular fractionations).

Crick's adapter hypothesis



H1 The genetic code

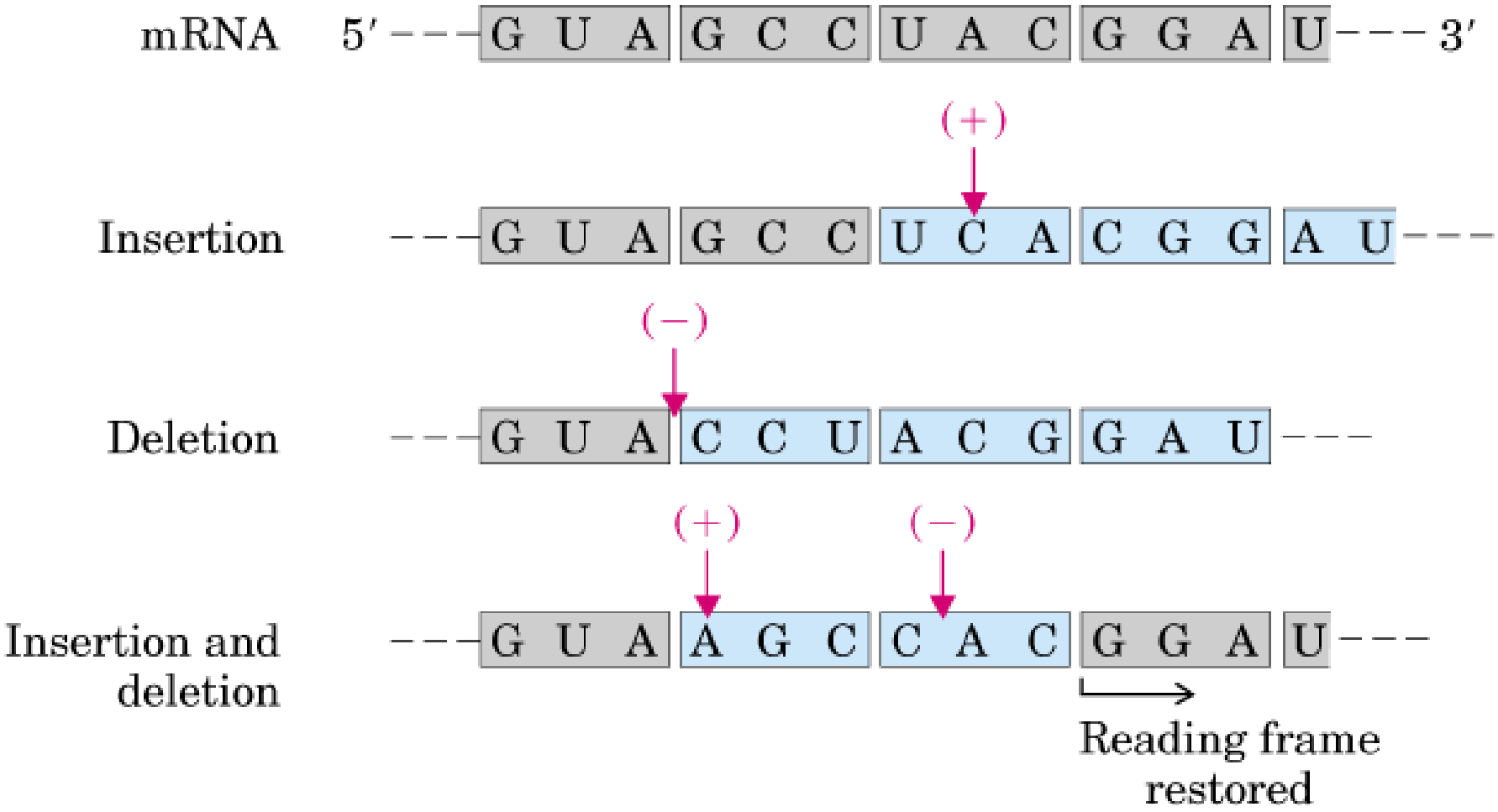
1. Amino acids in a polypeptide chain were found to be coded by groups of three nucleotides in a mRNA

Simple calculation indicated that three or more bases are probably needed to specify one amino acid.

Genetic studies of insertion, deletion, and substitution mutants showed **codons** for amino acids are **triplet** of nucleotides; codons do not overlap and there is no punctuation between codons for **successive** amino acid residues.

The amino acid sequence of a polypeptide is defined by a linear sequence of contiguous codons: the first codon establishs a **reading frame**.





Genetic studies showed that genetic codons are successive triplets of nucleotides

Nonoverlapping code

A U A C G A G U C √

1 2 3

Overlapping code

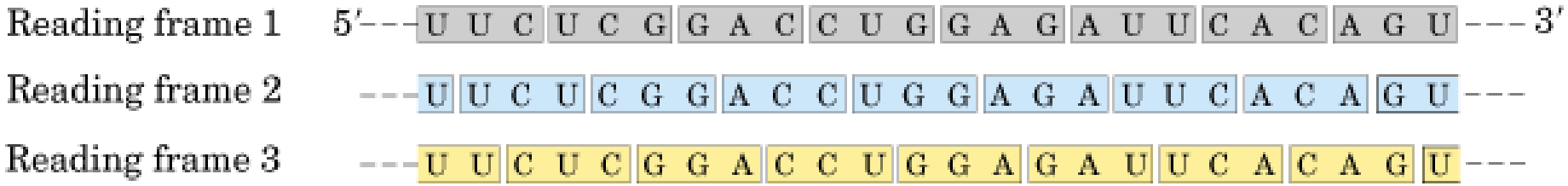
A U A C G A G U C X

1

2

3

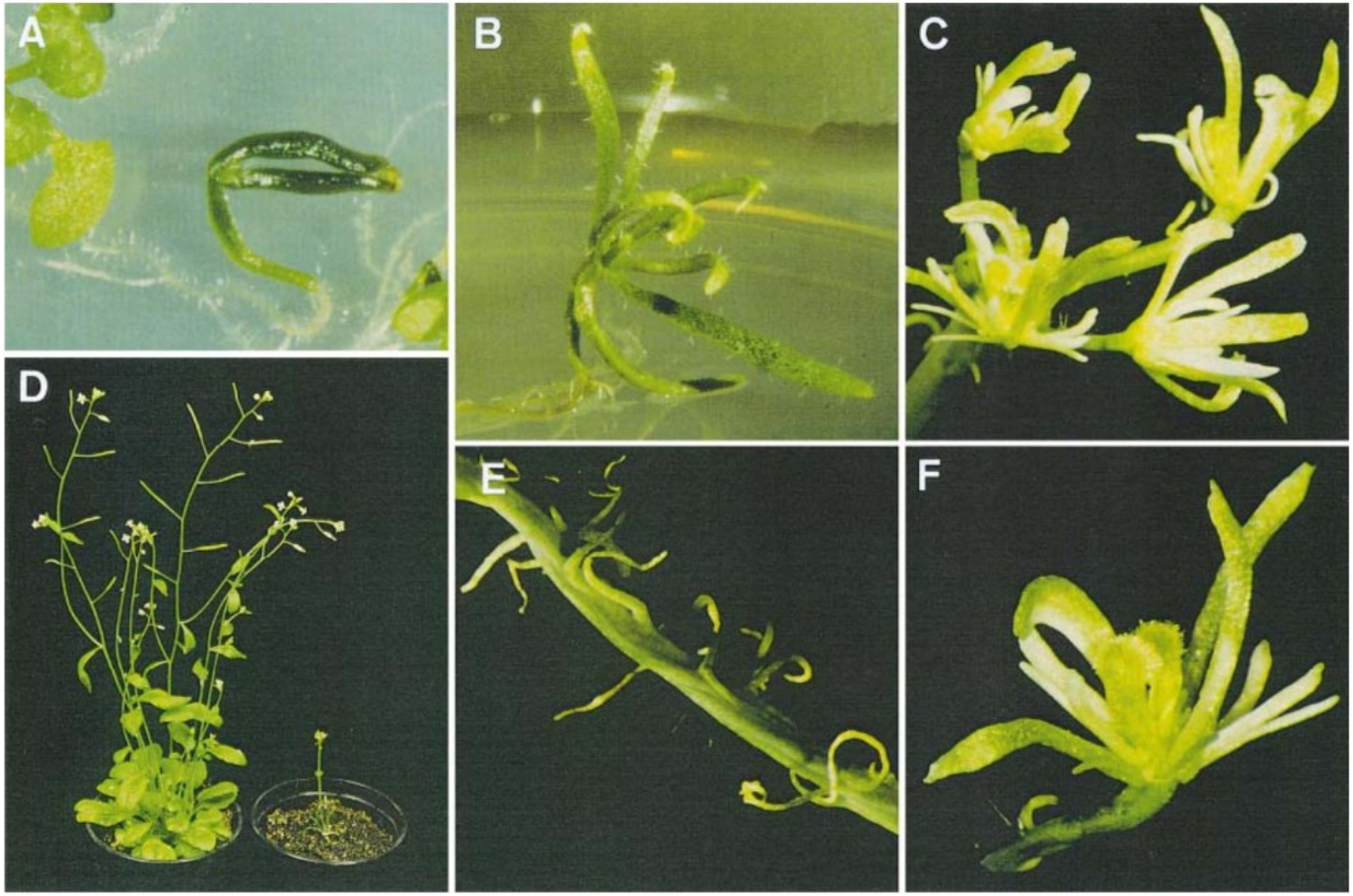




Each mRNA molecule would have three potential reading frames (*but only one usually codes for a polypeptide chain*) .



Arabidopsis Argonaute 1 (AGO 1)



WT mutant

ago1 mutant phenotype

ARGONAUTE_Translation

```
Translation of ARGONAUTE(1-3153)
Universal code

                  10        20        30        40        50        60
1         ATGGTGAGAAAGAGAAGAACGGATGCTCCATCTGAAGGAGGTGAAGGCTCTGGGTCTCGT
1          M  V  R  K  R  R  T  D  A  P  S  E  G  G  E  G  S  G  S  R
1           W  *  E  R  E  E  R  M  L  H  L  K  E  V  K  A  L  G  L  V
1            G  E  K  E  K  N  G  C  S  I  *  R  R  *  R  L  W  V  S

                  70        80        90       100       110       120
61        GAAGCTGGTCCAGTCTCAGGTGGTGGACGTGGTTCACAGCGAGGTGGTTTCCAGCAGGGA
21         E  A  G  P  V  S  G  G  G  R  G  S  Q  R  G  G  F  Q  Q  G
21          K  L  V  Q  S  Q  V  V  D  V  V  H  S  E  V  V  S  S  R  E
20        *  S  W  S  S  L  R  W  W  T  W  F  T  A  R  W  F  P  A  G

                 130       140       150       160       170       180
121       GGAGGACAACACCAAGGTGGAAGGGGTTATACTCCTCAACCTCAACAGGGAGGTCGTGGT
41         G  G  Q  H  Q  G  G  R  G  Y  T  P  Q  P  Q  Q  G  G  R  G
41          E  D  N  T  K  V  E  G  V  I  L  L  N  L  N  R  E  V  V  V
40        R  R  T  T  P  R  W  K  G  L  Y  S  S  T  S  T  G  R  S  W
```

Biochemistry

2. All 64 triplet codes were deciphered by 1966

61 of the codons code for the 20 amino acids and three (**UAA, UAG, UGA**) for chain termination (called **termination codons**, stop codons, or nonsense codons).

AUG is a dual codon coding for initiation and Met.

18 of the amino acids are coded by more than one codon: the genetic codes are **degenerate**（兼并性）.

(Having more than one codon specify the same amino acid.)

Condons that specify the same amino acid (**synonyms**) **often differ only in the third base**, the **wobble**（变偶 or 摆动）position, where base-pairing with the anticodon may be less stringent than for the first two positions of the condon.

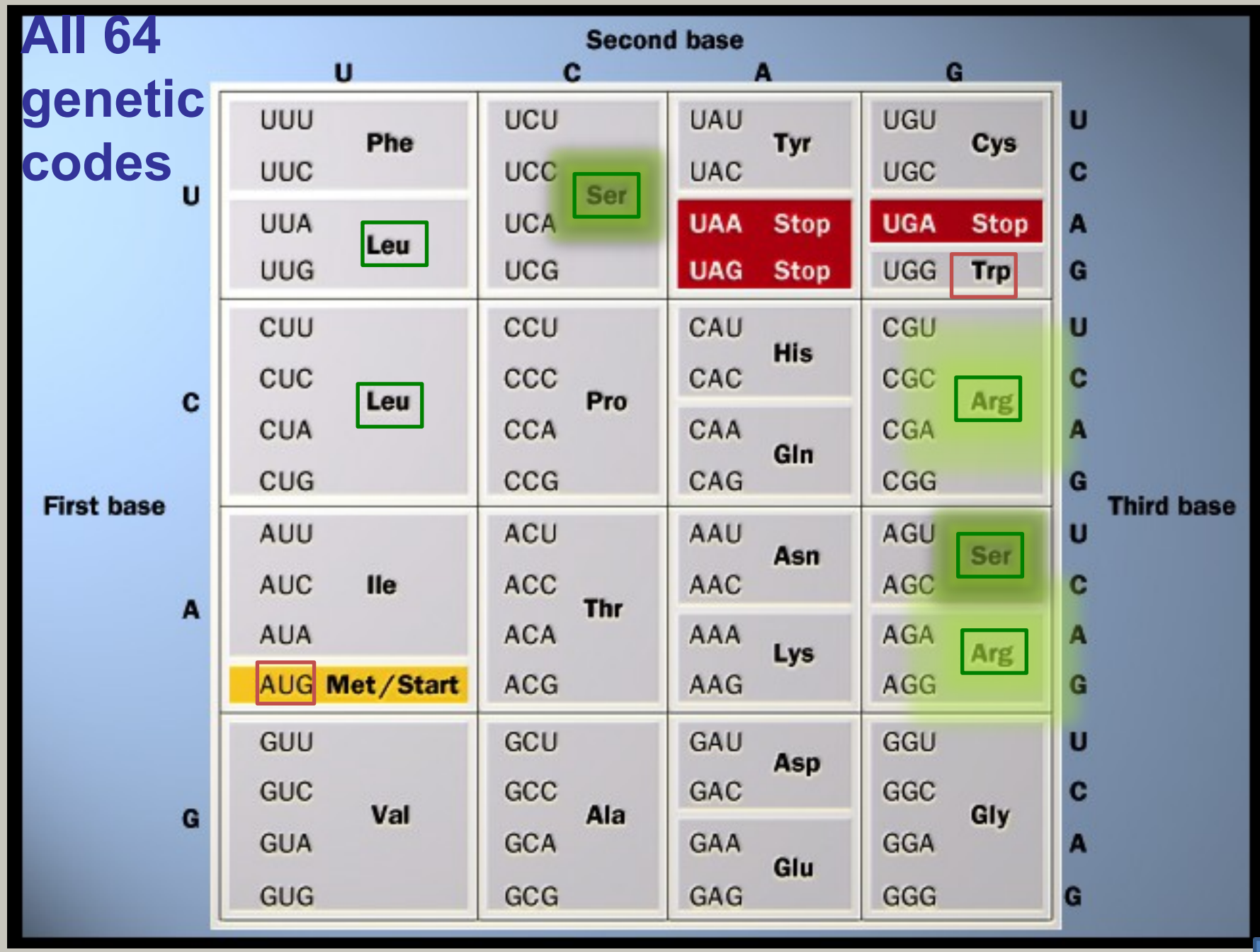


The codes seem to have evolved in such a way to minimize the deleterious effects of mutations, especially at the third bases: **XYU and XYC** always encode the same amino acid **XYA and XYG** usually code for the **same amino acid**.

A reading frame codes for **more than 50** amino acids without a stop codon is called an **open reading frame(ORF)**, which has the potential of encoding a protein.



All 64 genetic codes



The Nobel Prize in Physiology or Medicine 1968

"for their interpretation of the genetic code and its function in protein synthesis"

Presentation Speech

Established the chemical structure of tRNA

Robert W. Holley

USA

Cornell University
Ithaca, NY, USA

1922 - 1993

Biography



Devised methods to synthesize RNAs with defined sequences

Har Gobind Khorana

USA

University of Wisconsin
Madison, WI, USA

1922 -





Marshall W. Nirenberg

USA

National Institutes of Health
Bethesda, MD, USA

1927 -

Established the *in vitro* system for revealing the genetic codes



3. The genetic code has been proved to be nearly(not absolutely) universal

Direct comparisons of the amino acid sequences of proteins with the corresponding base sequence of their genes or mRNAs, as well as recombinant DNA technologies, proved that the genetic codes deciphered from *in vitro* studies were correct and almost universally applicable.

A small number of "**unusual codes**" have been revealed in many mitochondria genomes and nuclear genome of a few organisms.



Table 1 Known variant codon assignments in mitochondria

	Codons*				
	UGA	AUA	AGA AGG	CUN	CGG
Normal code assignment	Stop	Ile	Arg	Leu	Arg
Animals					
Vertebrates	Trp	Met	Stop	+	+
Drosophila	Trp	Met	Ser	+	+
Yeasts					
Saccharomyces cerevisiae	Trp	Met	+	Thr	+
Torulopsis glabrata	Trp	Met	+	Thr	?
Schizosaccharomyces pombe	Trp	+	+	+	+
Filamentous fungi	Trp	+	+	+	+
Trypanosomes	Trp	+	+	+	+
Higher plants	+	+	+	+	Trp
Chlamydomonas reinhardtii	?	+	+	+	?

The features of genetic codon：

① 方向性 (5’-3’ direction)

② 读码的连续性，非重叠 (successive triplets/ no overlap)

③ 兼并性 (degenerate)

④ 变偶性 (wobble)

⑤ 起始密码和终止密码 (start and stop codons)

⑥ 通用性与例外 (universal /unusual codes)



H2 Translation in prokaryotes



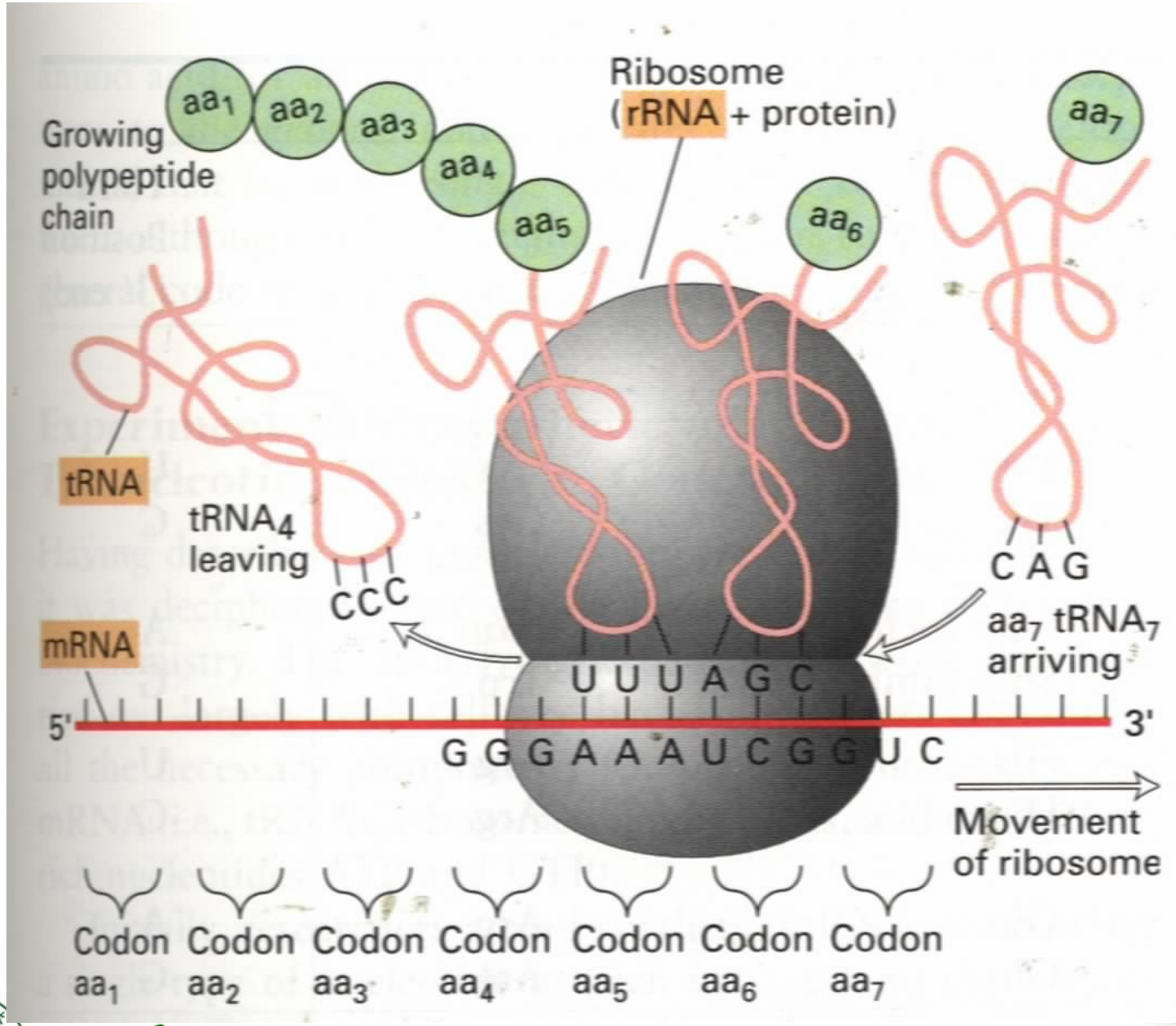
1. Three kinds of RNA molecules perform different but cooperative functions in protein synthesis

mRNAs carry the genetic information copied from DNA in the form of **genetic codons**.

tRNAs mediate the incorporation of specific **amino acids** according to genetic codons present on the mRNA molecules via their specific **anticodon triplets**.

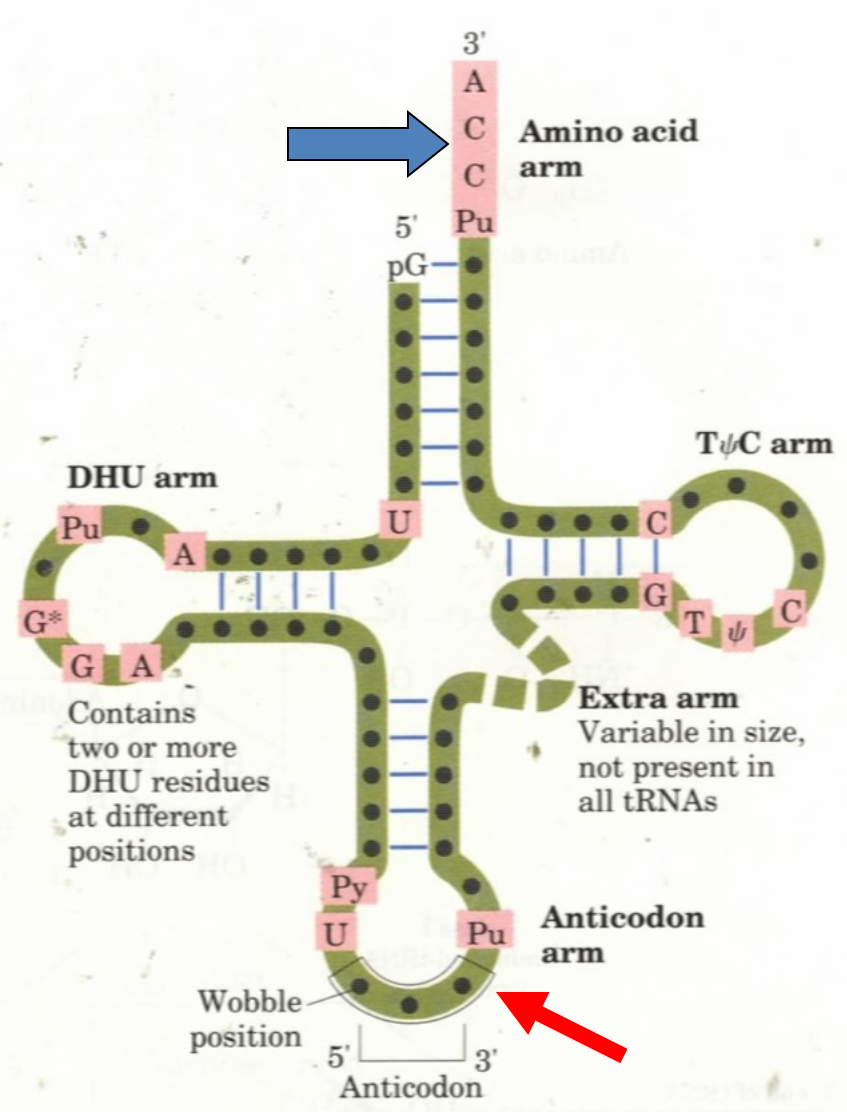
rRNAs associate with a set of proteins to form the protein-synthesizing machines (**ribosomes**) and probably **catalyze peptide bond formation** during protein synthesis.

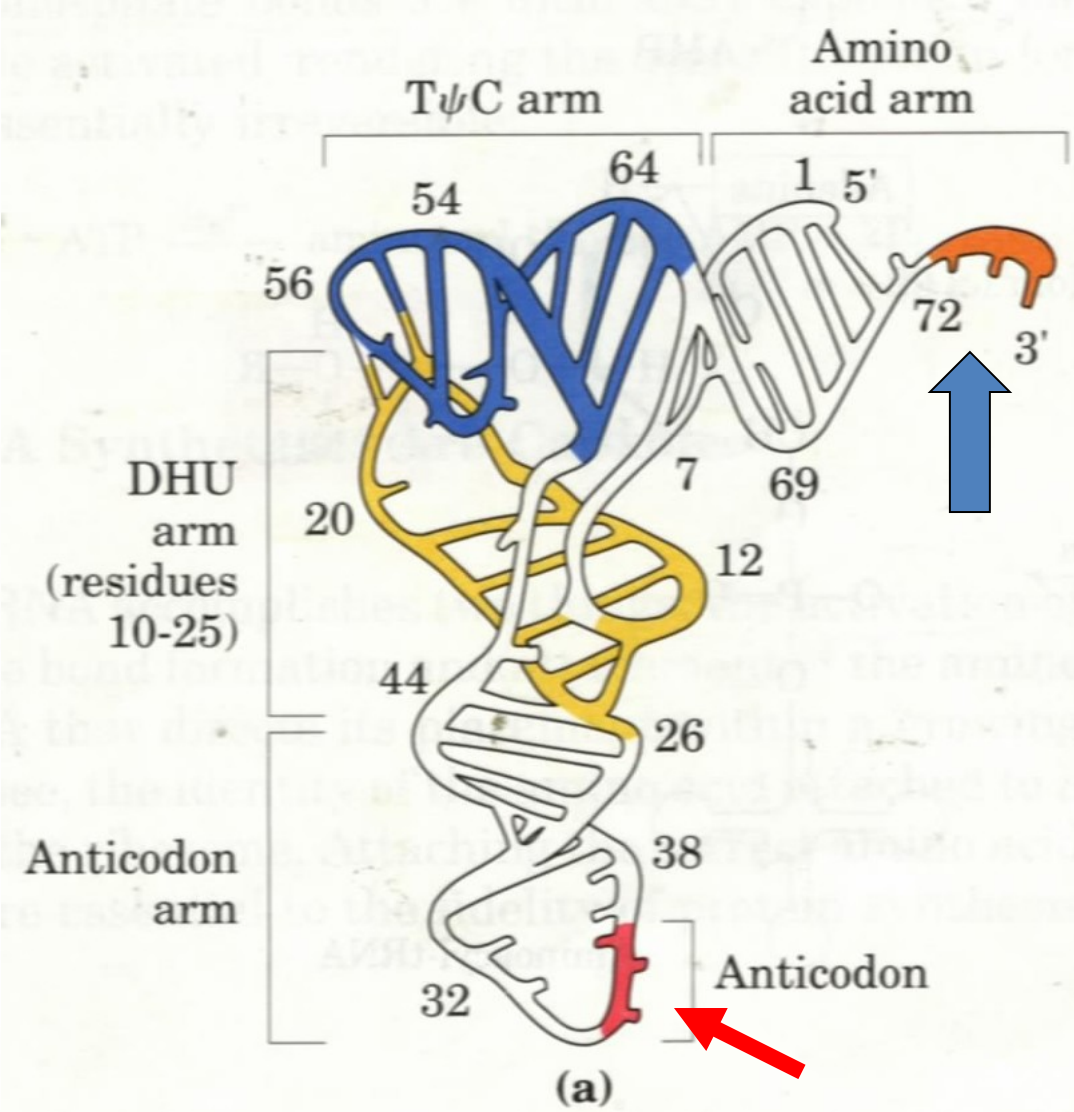



Ribosome
(rRNA + protein)
Growing polypeptide chain
aa1
aa2
aa3
aa4
aa5
aa6
aa7
tRNA
tRNA4 leaving
C C C
C A G
aa7 tRNA7 arriving
mRNA
5'
3'
U U U A G C
G G G A A A U C G G U C
Movement of ribosome
Codon aa1
Codon aa2
Codon aa3
Codon aa4
Codon aa5
Codon aa6
Codon aa7

2. All tRNAs have common structural features: cloverleaf in secondary, “L” in 3-D structures.




3'
A
C
C
Amino acid arm
Pu
5'
pG
TψC arm
DHU arm
Pu
A
U
C
G*
G
A
G
T
ψ
C
Contains two or more DHU residues at different positions
Extra arm
Variable in size, not present in all tRNAs
Py
U
Pu
Anticodon arm
Wobble position
5'
3'
Anticodon


TψC arm
Amino acid arm
64
54
1
5'
56
72
3'
DHU arm (residues 10-25)
7
69
20
12
44
26
Anticodon arm
38
Anticodon
32
(a)

3. Some tRNA molecules can recognize more than one codons via wobble pairing

The adapter tRNAs recognize the codons on a mRNA via a triplet called **anticodons.**

It was first proposed that a specific tRNA anticodon would exist for every of the **61 (or 64) codons**, **but less tRNAs were revealed**.

It was revealed that highly purified tRNA molecules (e.g., alanyl-$tRNA^{Ala}$) of known sequence could recognize several different codons.

Inosine （I, 次黄嘌呤）, which may form base pair with A, U, and C, was found to be present at the **first position** of the anticodons in some tRNAs.



Crick proposed the "**wobble hypothesis**" in 1966 to explain the pairing features between anticodons and codons:

The first two bases of a codon in mRNA confer most of the coding specificity, the third base can be **loosely** paired with the anticodons;

The **first** base of some **anticodons** can **wobble** and determines the number of codons a given tRNA can read (*A and C for one, U and G for two, I for three*);

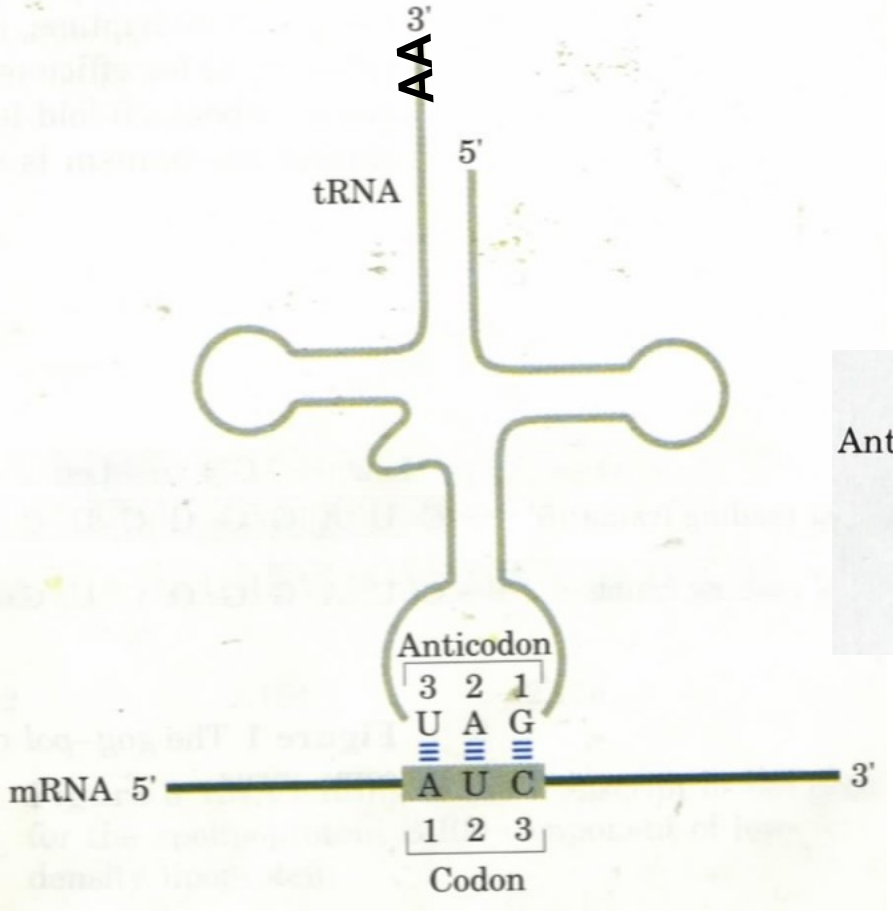
Codons that specify the same amino acid but differ in either of the first two bases need different tRNAs, i.e., a mininum of **31** tRNA are

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This hypothesis has been widely supported by all the evidence gathered (thus the "**wobble rule**"). This moderate pairing strength may serve to optimize both the accuracy and speed of polypeptide synthesis.





CGU
CGC
CGA
CGG

Arg

		3	2	1		3	2	1		3	2	1	
Anticodon	(3′)	G	C	I		G	C	I		G	C	I	(5′)
Codon	(5′)	C	G	A		C	G	U		C	G	C	(3′)
		1	2	3		1	2	3		1	2	3	

(b)

The codon-anticodon pairing between the a mRNA and a tRNA: the presence of an inosinate residue at position one in the anticodon allows the tRNA to recognize a few codons.

Table 26–5 The wobble base of the anticodon determines how many codons of a given amino acid a tRNA can recognize

In the following, X and Y denote complementary bases capable of strong Watson–Crick base pairing with each other. The bases in the wobble or 3′ position of the codons and 5′ position of the anticodons are shaded in red.

1. One codon recognized:

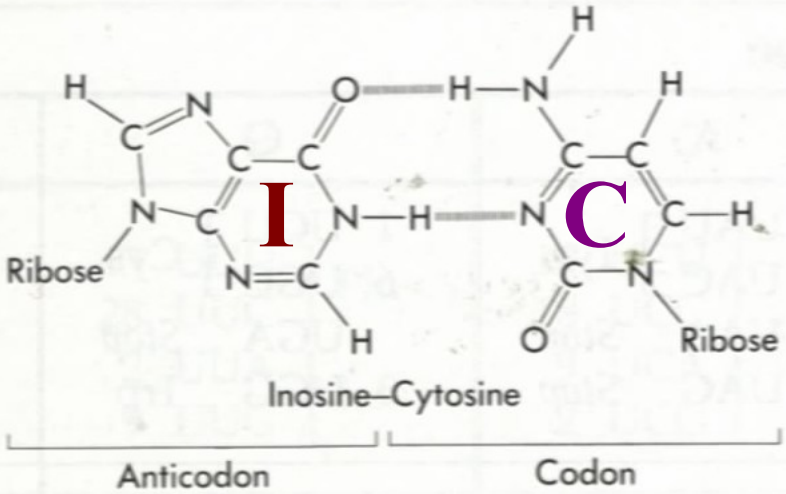
Anticodon	(3′) X–Y–**C** (5′)	(3′) X–Y–**A** (5′)
Codon	(5′) Y–X–**G** (3′)	(5′) Y–X–**U** (3′)

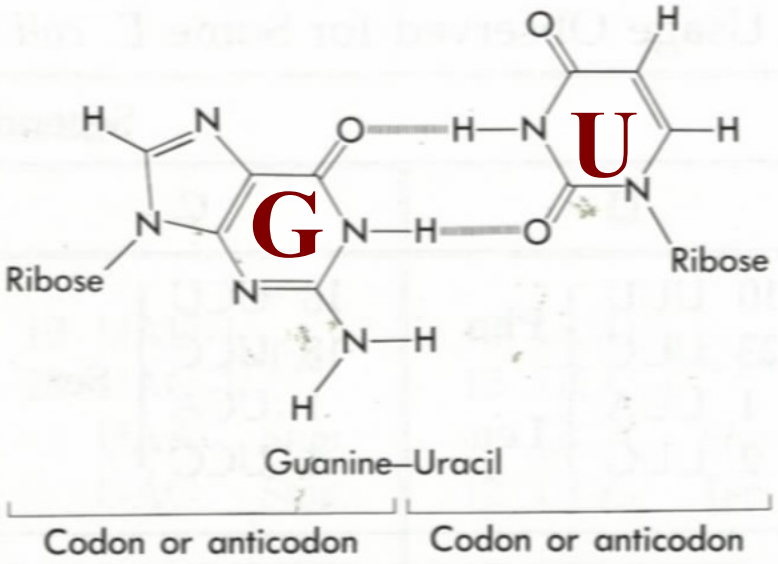
2. Two codons recognized:

Anticodon	(3′) X–Y–**U** (5′)	(3′) X–Y–**G** (5′)
Codon	(5′) Y–X–**A** / **G** (3′)	(5′) Y–X–**C** / **U** (3′)

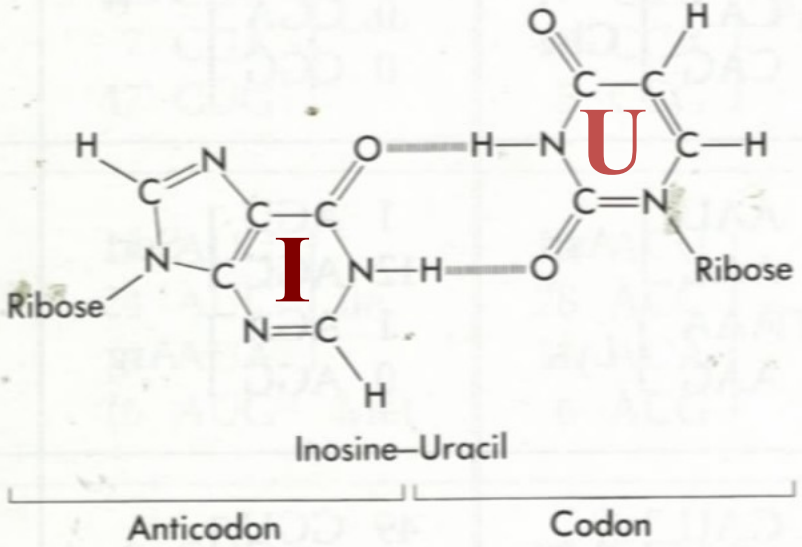
3. Three codons recognized:

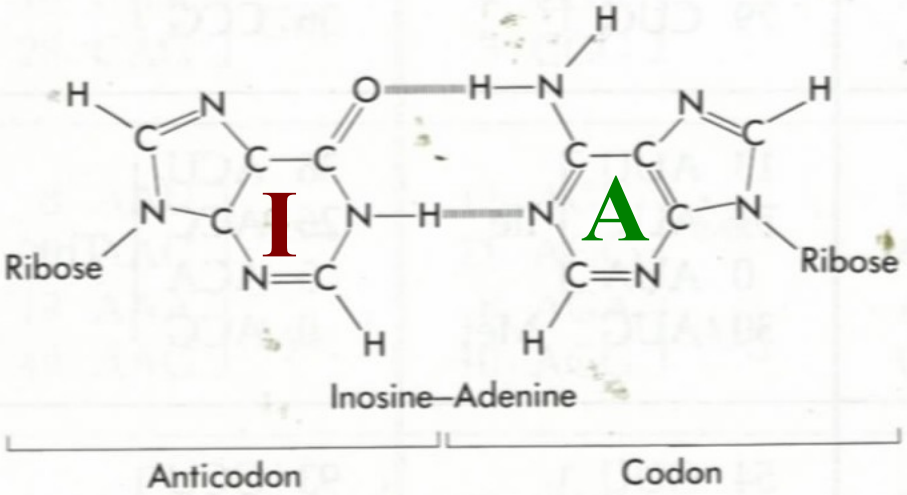
Anticodon	(3′) X–Y–**I** (5′)
Codon	(5′) Y–X–**A** / **U** / **C** (3′)





Possible wobble pairing between anticodon and codon.





它们之间的对应关系？

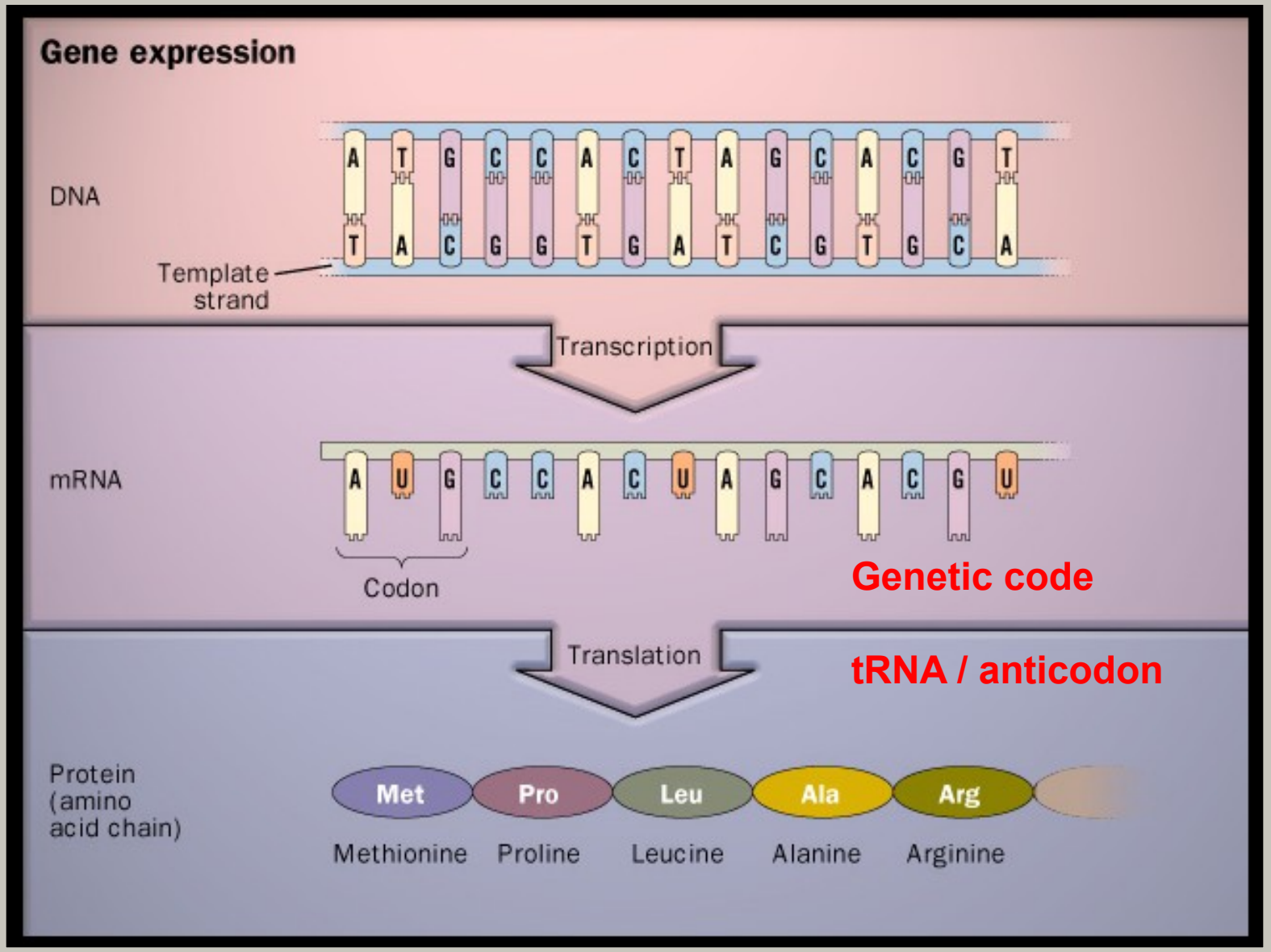
氨酰 tRNA 合成酶

codon　　tRNA　　AA



Table of Wobble Pairing

First (5') base of anticodon	Third (3') base of codon
C	G
A	U
U	A or G
G	U or C
I (Inosine)	U, C, or A


Gene expression
DNA
Template strand
A T G C C A C T A G C A C G T
T A C G G T G A T C G T G C A
Transcription
mRNA
A U G C C A C U A G C A C G U
Codon
Genetic code
Translation
tRNA / anticodon
Protein (amino acid chain)
Met
Pro
Leu
Ala
Arg
Methionine
Proline
Leucine
Alanine
Arginine

小结：

1. 遗传密码的基本性质

（the features of genetic codon）；

2. Anticodons wobble hypothesis



4. The synthesis of a protein can be divided into five stages

Each amino acid is first covalently attached to a specific tRNA molecule in a reaction catalyzed by a specific aminoacyl-tRNA synthetase (Stage 1 氨基酸的活化 （ Activation of Amino Acids ）).

The mRNA then binds to the smaller subunit of the ribosome, after which the initiating aminoacyl-tRNA and the large subunits of the ribosome will bind in turn to form the initiating complex (Stage 2 合成的起始（ initiation ）)



The first peptide bond is then formed after the second aminoacyl-tRNA is recruited with help of the elongation factors, and the chain is then further elongated (stage 3 合成的延伸（ elongation ）).

When a stop codon (UAA, UAG, and UGA) is met, the extension of the polypeptide chain will come to a stop and is released from the ribosome with help from release factors (Stage 4 合成的终止（ termination and release ）).

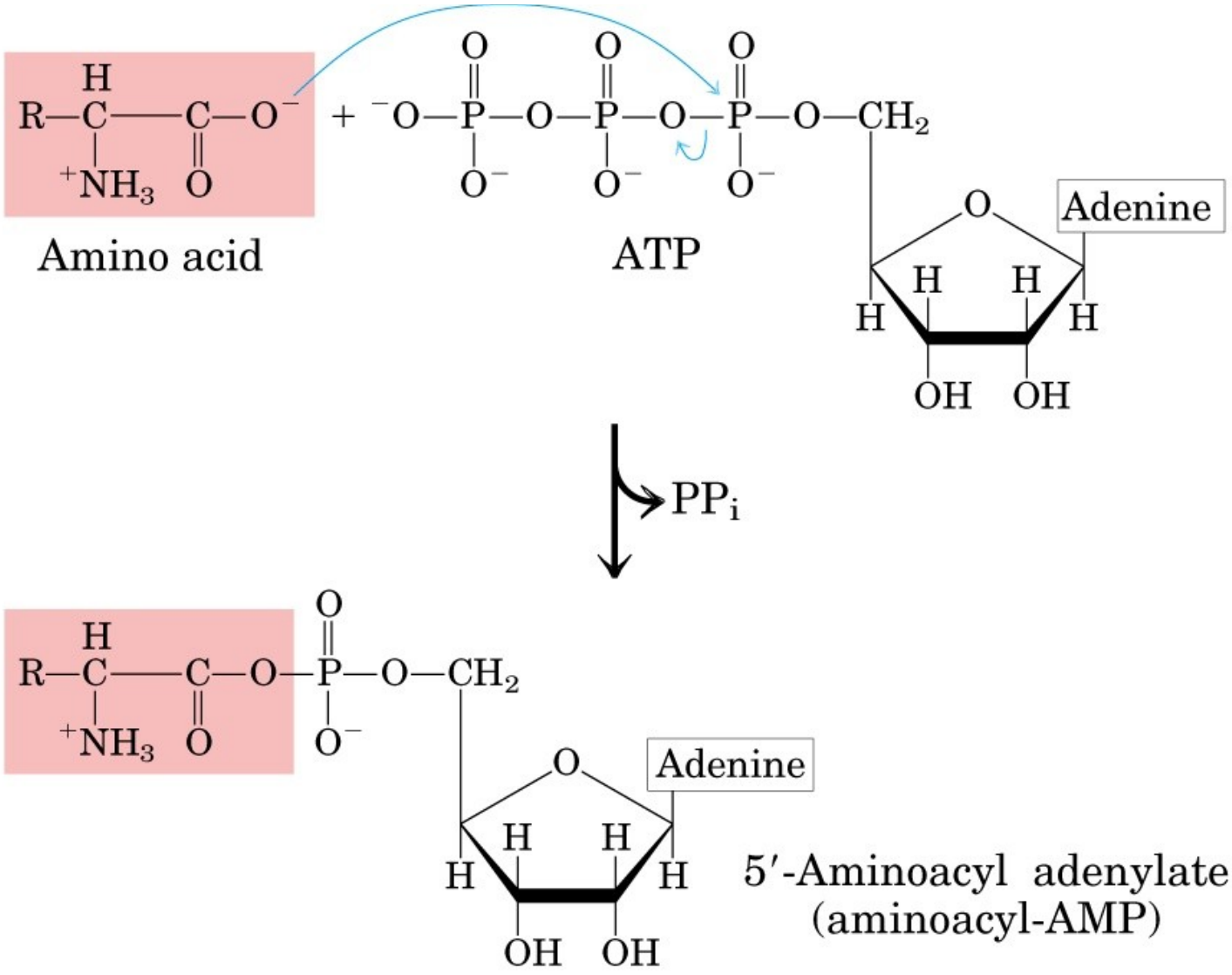
The newly synthesized polypeptide chain has to be folded and modified (in many cases) before becoming a functional protein (Stage 5 翻译后修饰（ Folding and Posttranslational Processing ）).

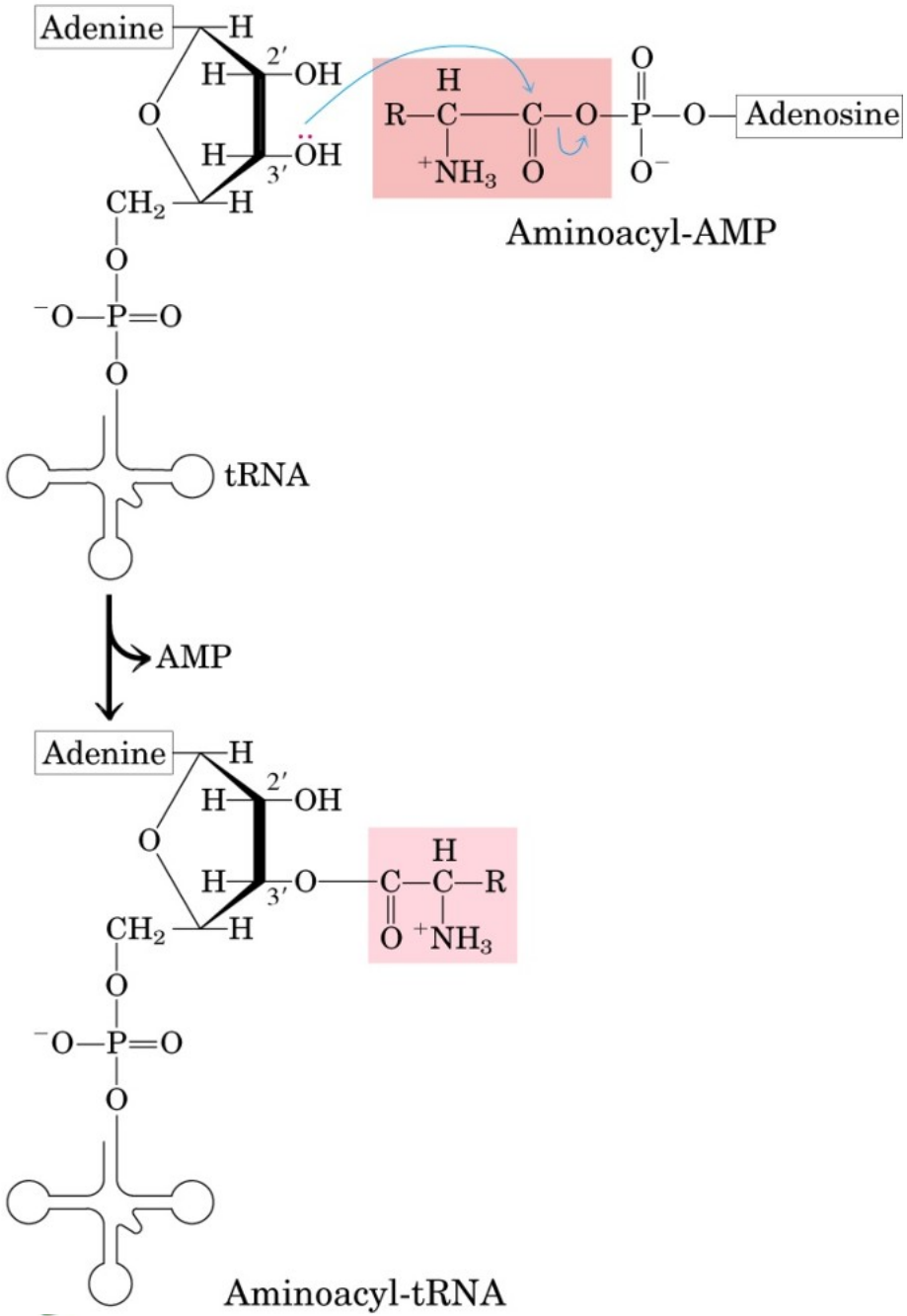


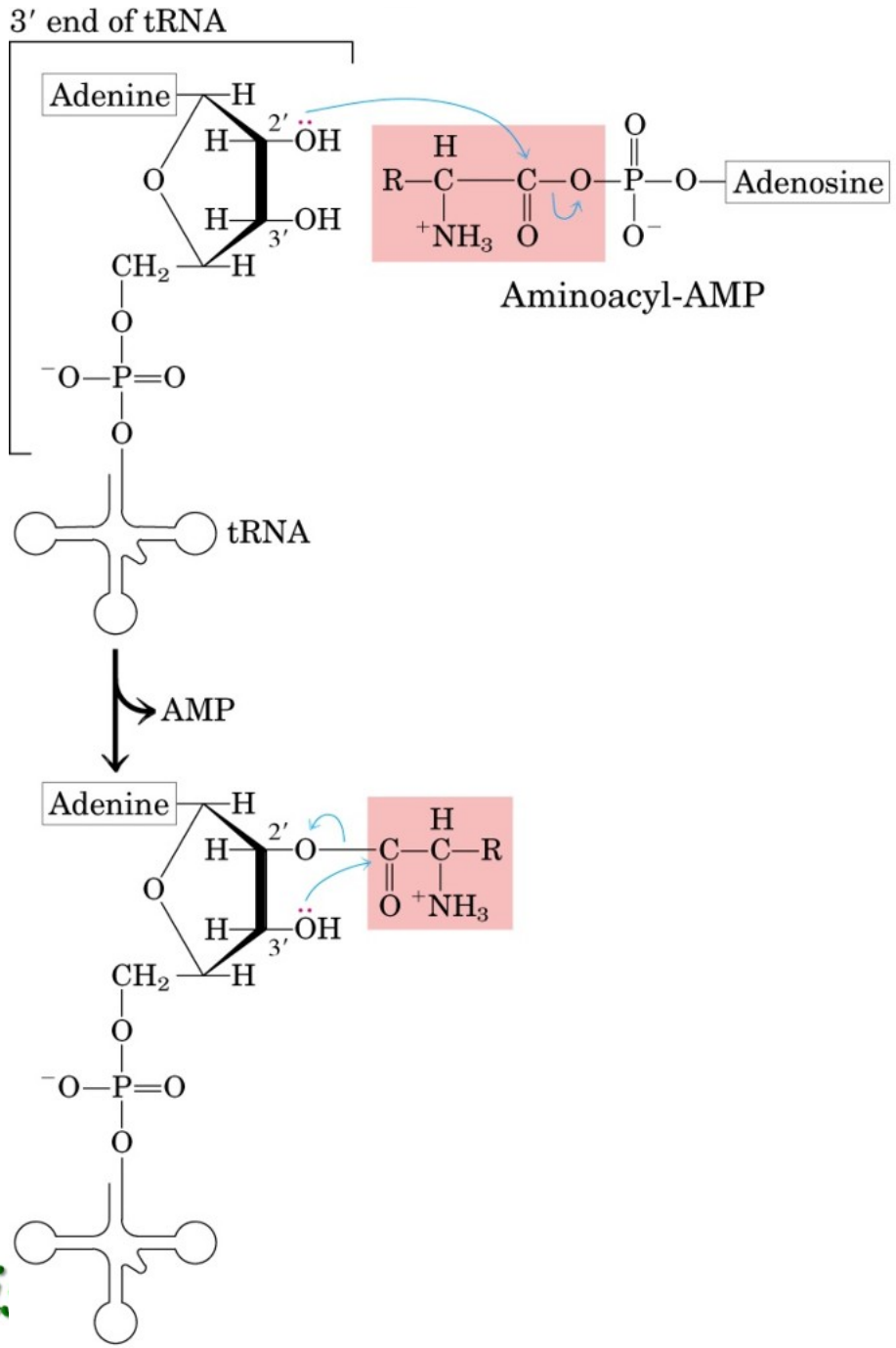
5. The 20 aminoacyl-tRNA synthetases attach the 20 amino acids to one or more specific tRNAs (stage 1)

An amino acid is first activated to form an **aminoacyl-AMP** intermediate, and is then charged to one or more specific tRNAs all catalyzed by one such specific aminoacyl-tRNA synthetase.

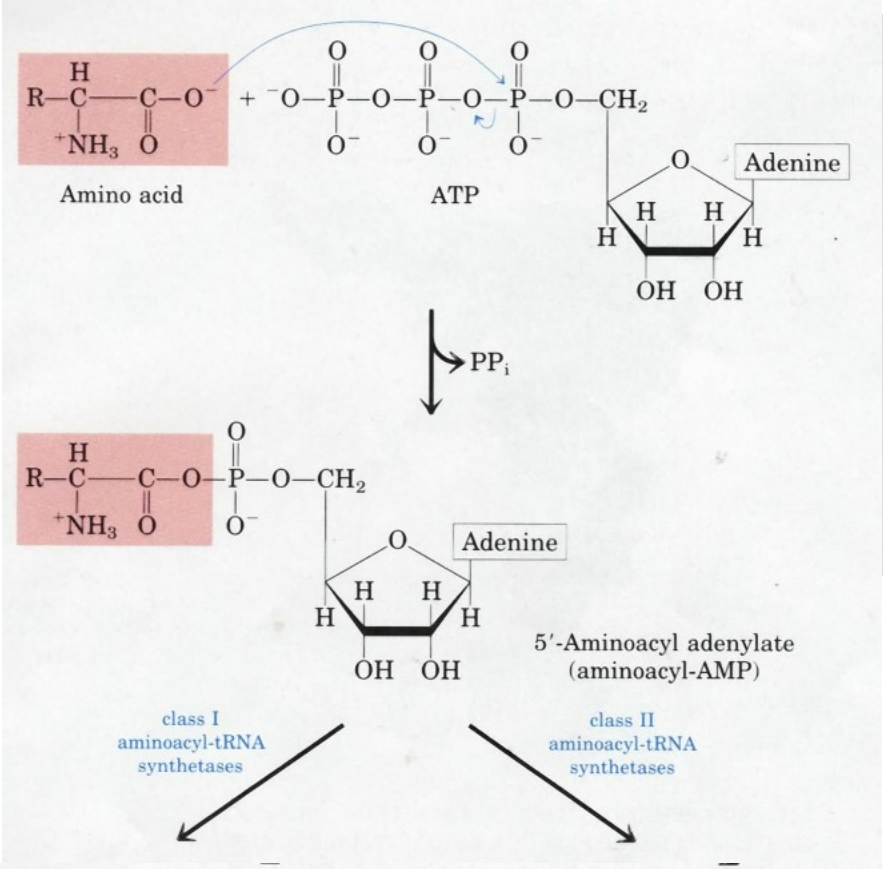


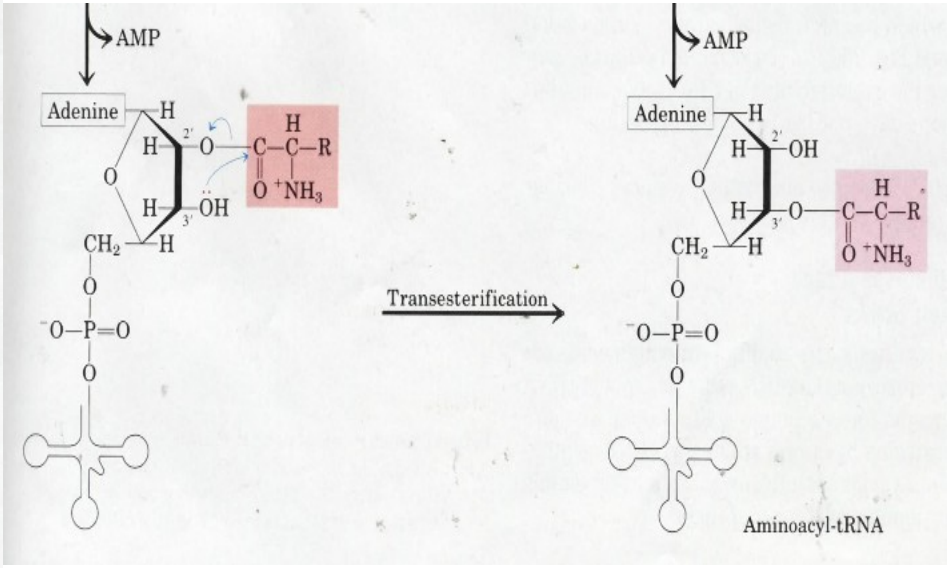

Amino acid
ATP
Adenine
PPi
Adenine
5′-Aminoacyl adenylate
(aminoacyl-AMP)


Adenine
Adenosine
Aminoacyl-AMP
tRNA
AMP
Adenine
Aminoacyl-tRNA

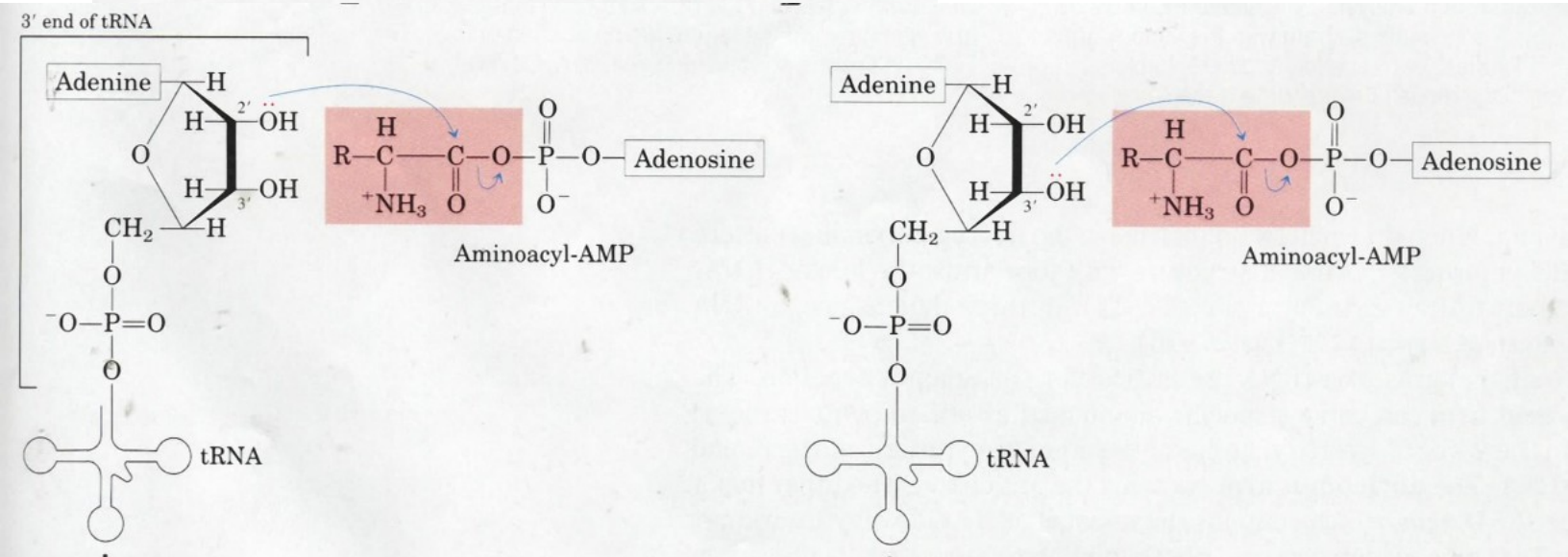

3′ end of tRNA
Adenine
H
2′
OH
H
O
H
OH
3′
H
CH2
O
⁻O—P=O
O
tRNA
H
R—C—C—O—P—O—Adenosine
+NH3
O
O
O⁻
Aminoacyl-AMP
AMP
Adenine
H
2′
O
H
O
H
OH
3′
H
CH2
O
⁻O—P=O
O
H
C—C—R
O
+NH3

Biochemi





Aminoacyl-tRNA synthetases can be divided into two classes based on differences in structure and reaction mechanisms.





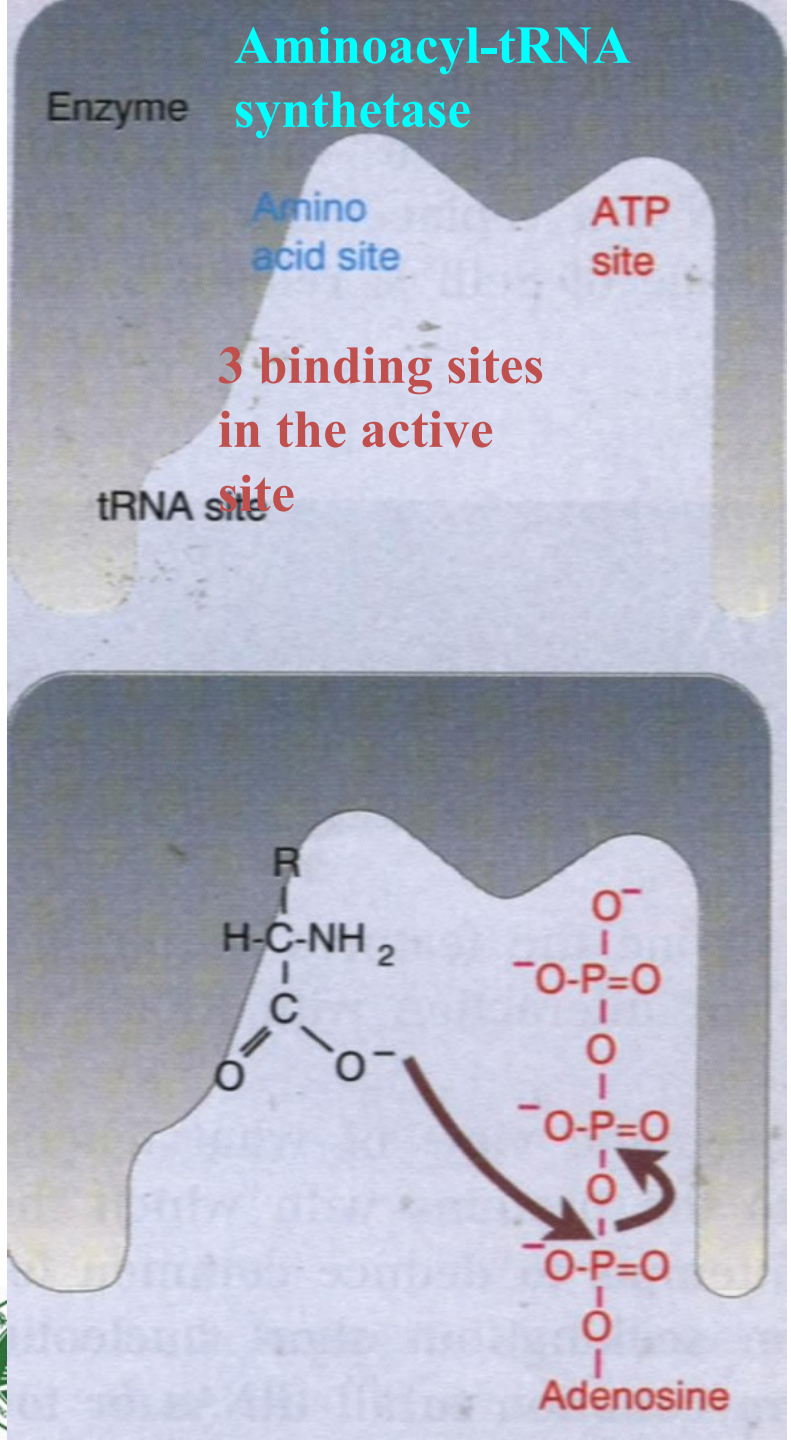
(1)Amino acid + ATP → aminoacyl-AMP +PPi

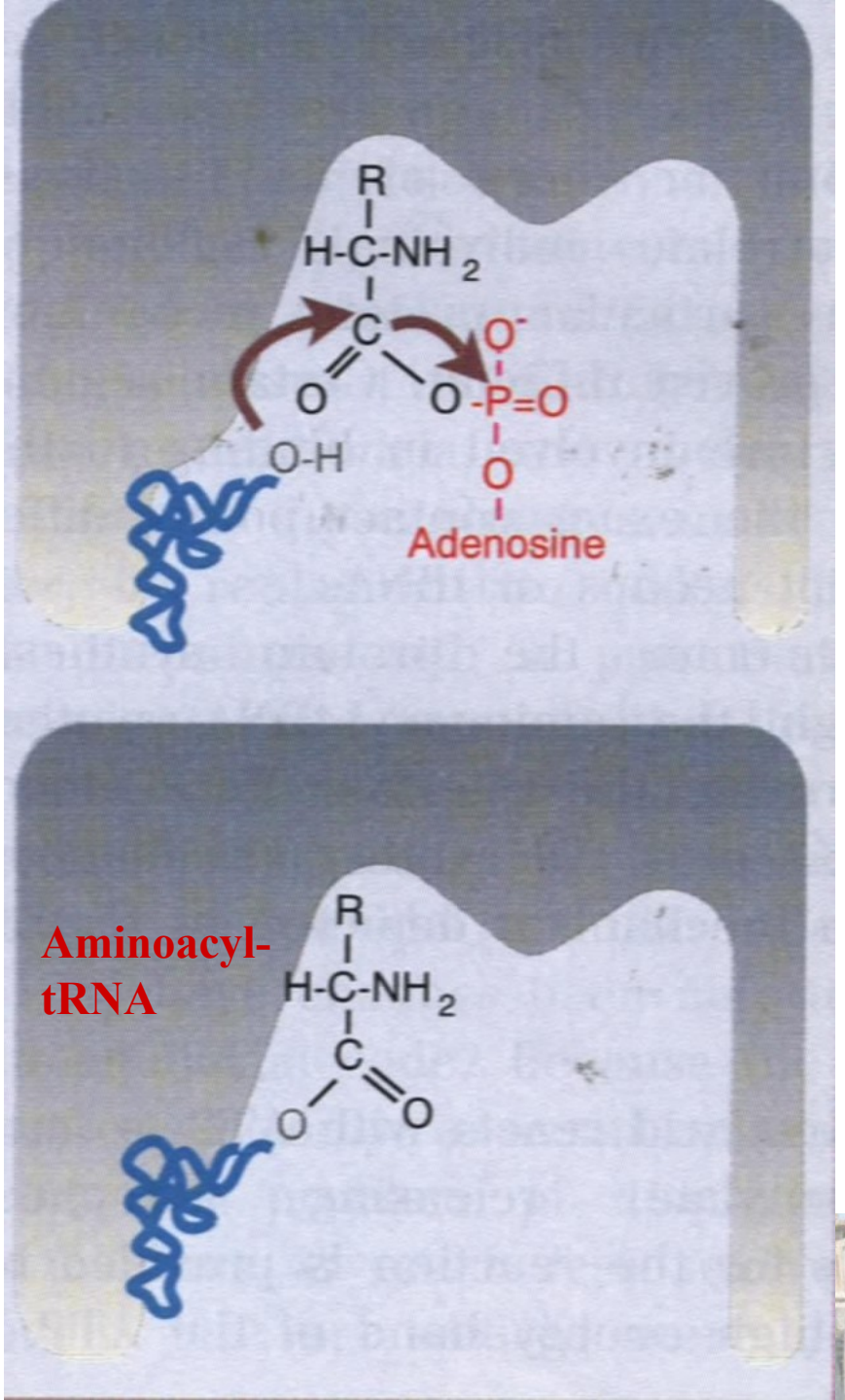
(2)Aminoacyl-AMP + tRNA → aminoacyl-tRNA + AMP

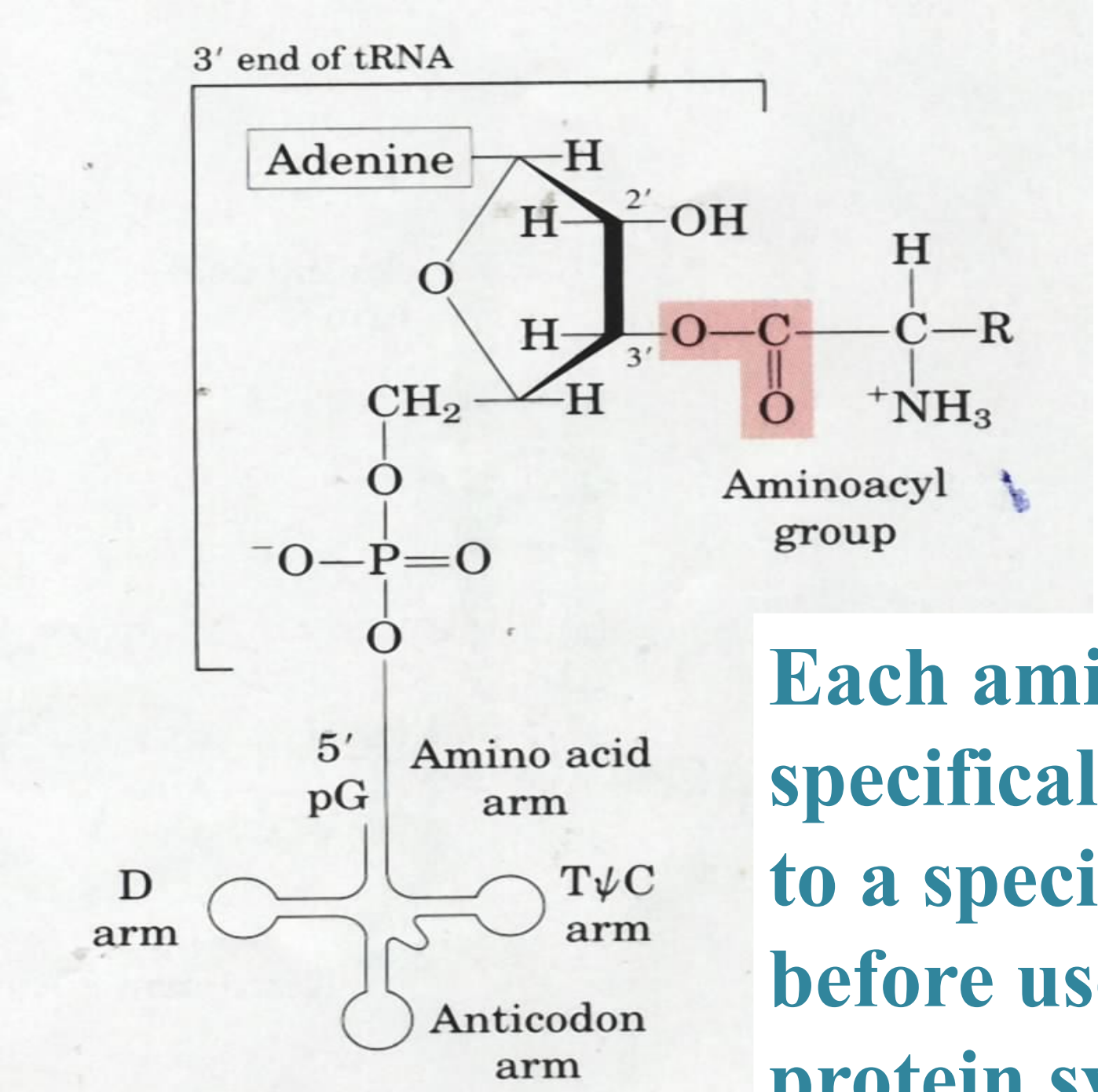
The overall reaction is

Amino acid + ATP + tRNA → aminoacyl-tRNA + AMP +PPi




Aminoacyl-tRNA synthetase
Enzyme
Amino acid site
ATP site
3 binding sites in the active site
tRNA site
R
H-C-NH 2
C
O
O
O-P=O
O
O-P=O
O
O-P=O
O
Adenosine


R
H-C-NH 2
C
O
O
O-P=O
O-H
O
Adenosine
Aminoacyl-tRNA
R
H-C-NH 2
C
O
O



Stage 1

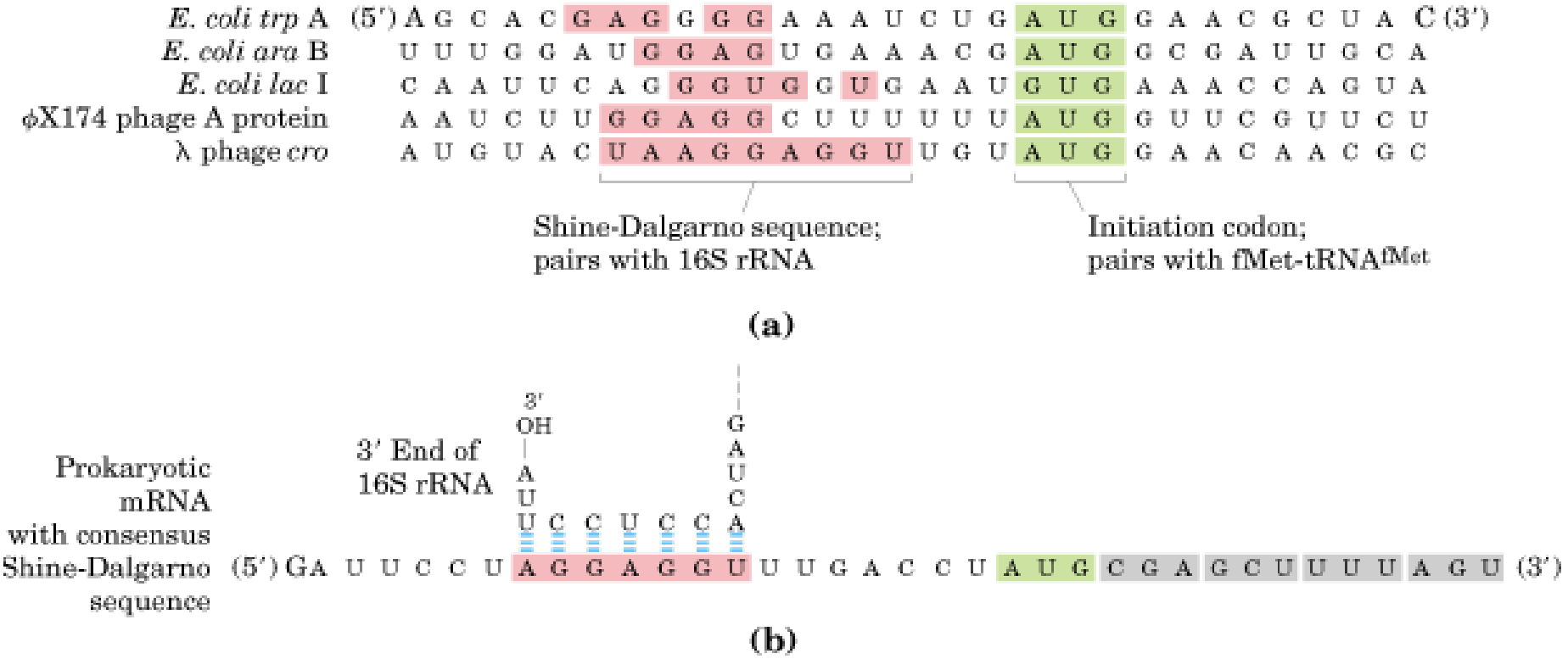
Each amino acid is specifically attached to a specific tRNA before used for protein synthesis

6. Translation in prokaryotes begins by the formation of a complex.

(Stage 2 Initiation of protein synthesis)

The Shine-Dalgarno sequence and a nearby AUG marks the start site of translation on a bacterial mRNA.



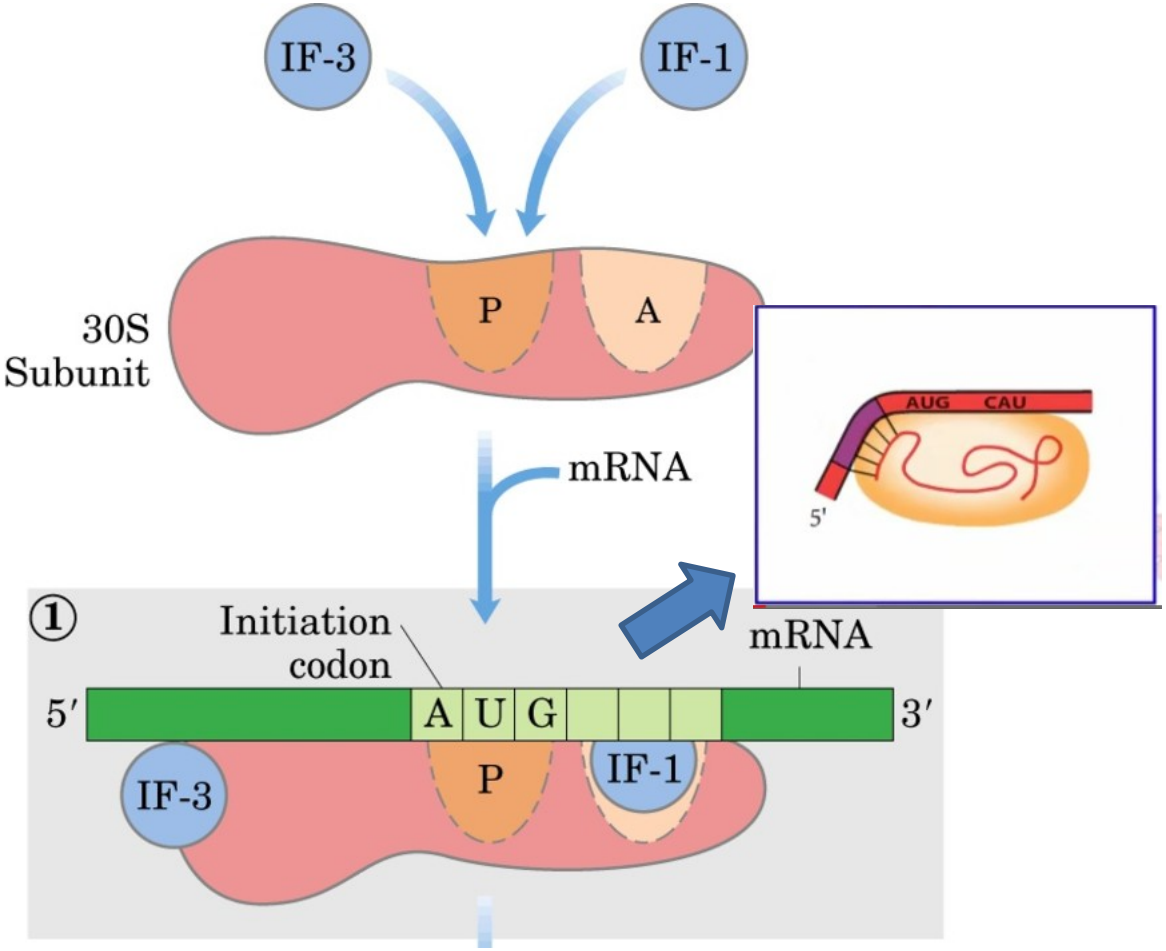


A purine-rich Shine-Dalgarno sequence and a AUG codon marks the start site of polypeptide synthesis on bacterial mRNA molecules.



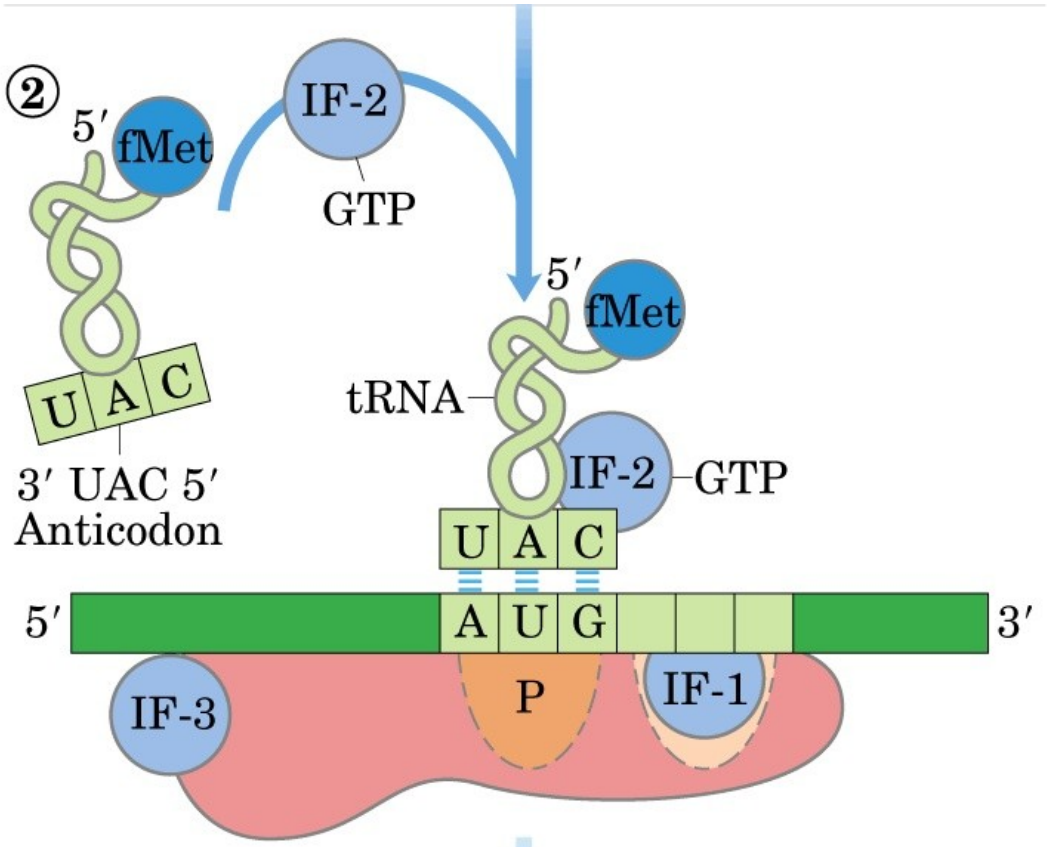
- This purine-rich sequence was found to be complementary to a pyrimidine-rich sequence at the 3`end of the 16S rRNA present in the 30S subunit of the ribosomes.
- The Shine-Dalgarno sequence may thus serve as the **ribosome binding site** on bacterial mRNAs.

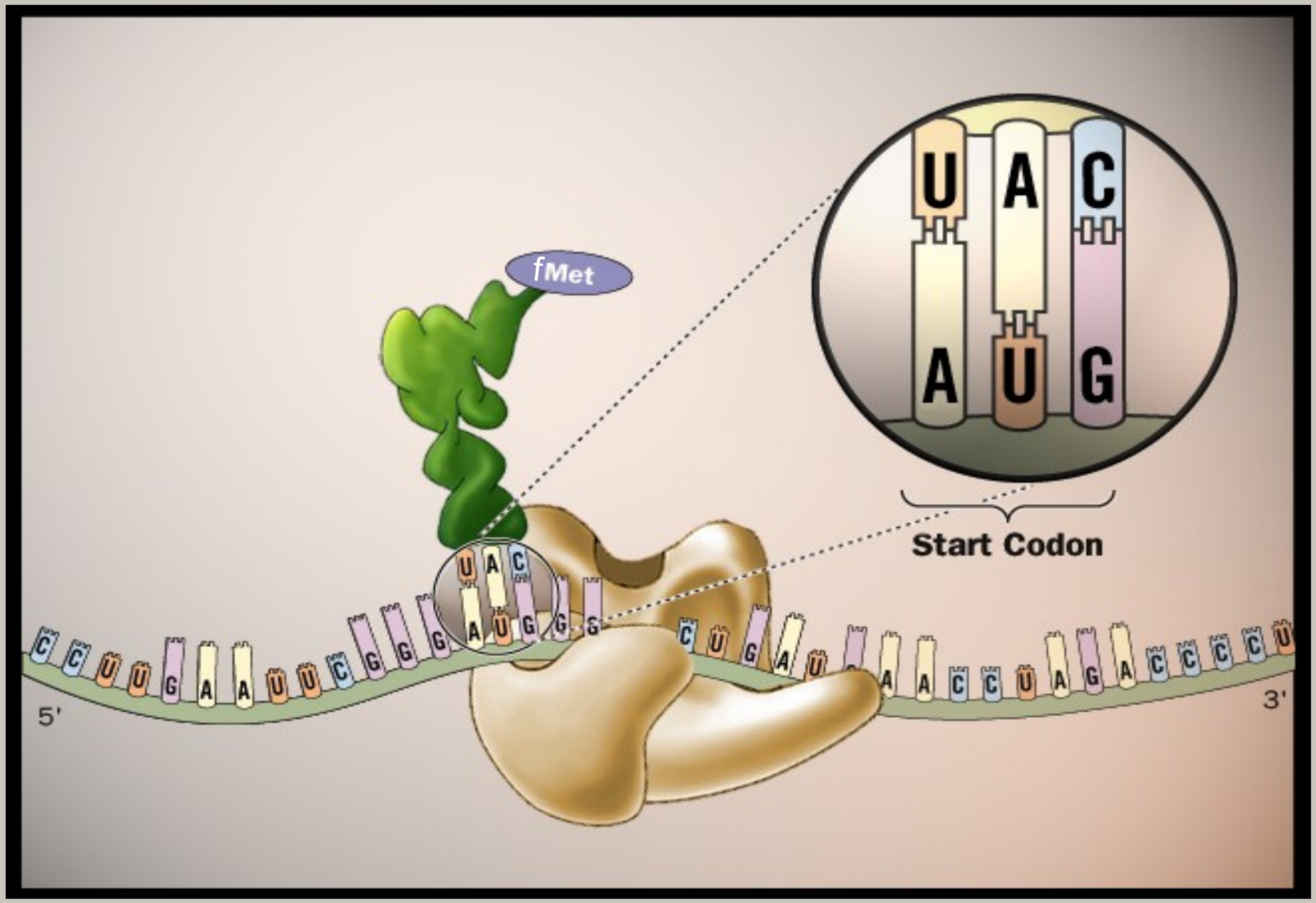


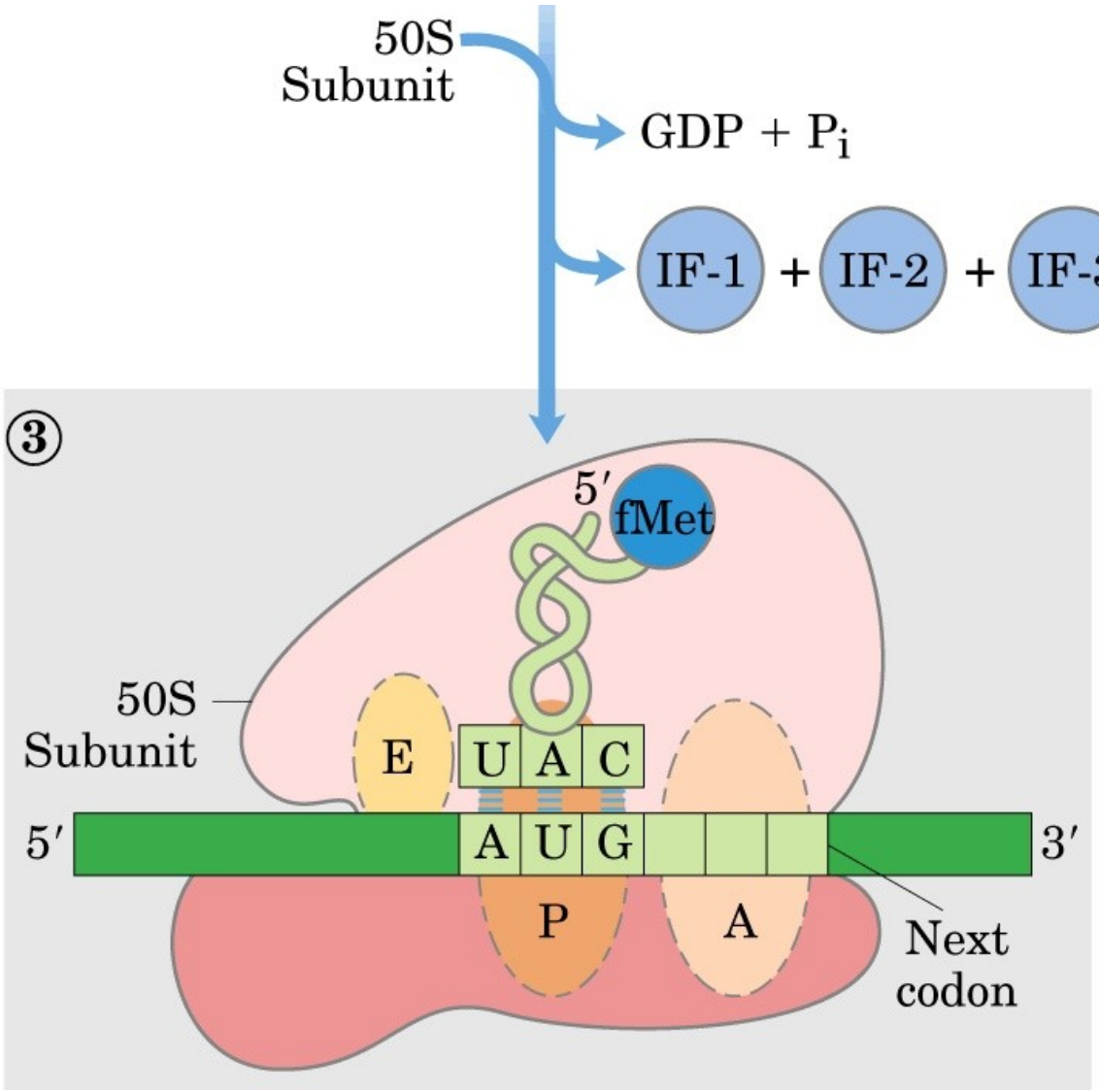


Factor IF-3 prevents the 30S and 50S subunits from combining prematurely;

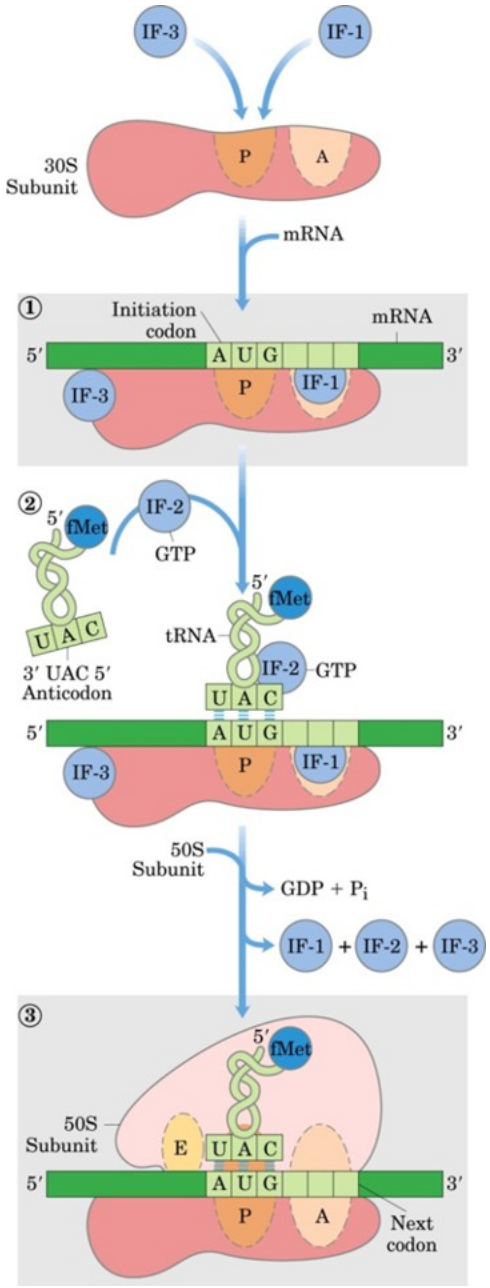
Factor IF-1 binds at the A site and prevents tRNA binding at this site during initiation.


②
5′
fMet
IF-2
GTP
U A C
3′ UAC 5′
Anticodon
5′
fMet
tRNA
IF-2
GTP
U A C
5′
A U G
3′
IF-3
P
IF-1


fMet
U A C
A U G
Start Codon
5'
3'



GTP bound to IF-2 is hydrolyzed to GDP and Pi, which are released from the complex. All three initiation factors depart from the ribosome at this point.



30S complex

initiation （70S） complex

Each prokaryotic ribosome has three binding sites for tRNA.

aminoacyl-tRNA binding site	**A site**
peptidyl-tRNA binding site	**P site**
exit site	**E site**

initiation factors	**IFs**
N-formylmethionine	**fMet**



起始氨酰tRNA：

$fMet\text{-}tRNA_f^{fMet}$

$Met\text{-}tRNA_i^{Met}$

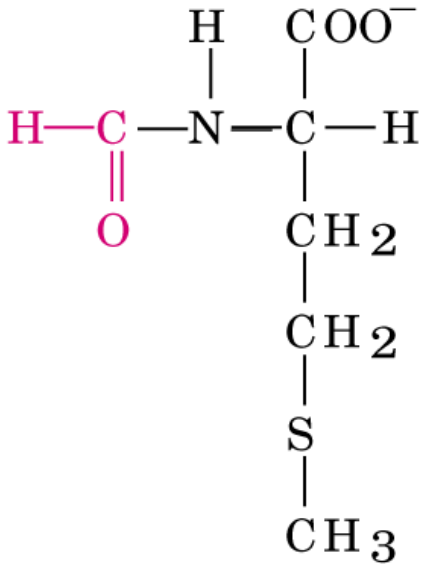
（真核生物）

蛋酰tRNA：

$Met\text{-}tRNA^{Met}$

起始氨基酸

甲酰蛋氨酸



N-Formylmethionine

$$N^{10}\text{-Formyltetrahydrofolate} + Met\text{-}tRNA^{fMet} \xrightarrow{\text{transformylase}} \text{tetrahydrofolate} + fMet\text{-}tRNA^{fMet}$$

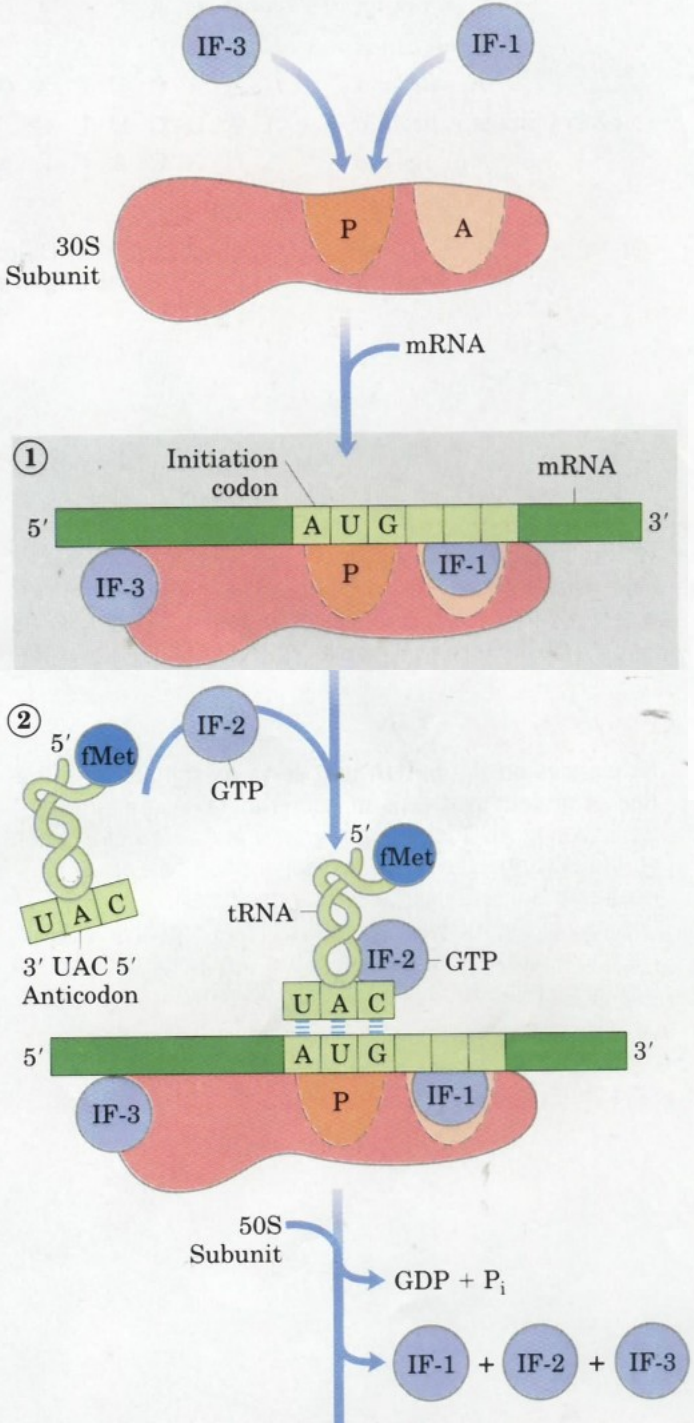
Biochemistry

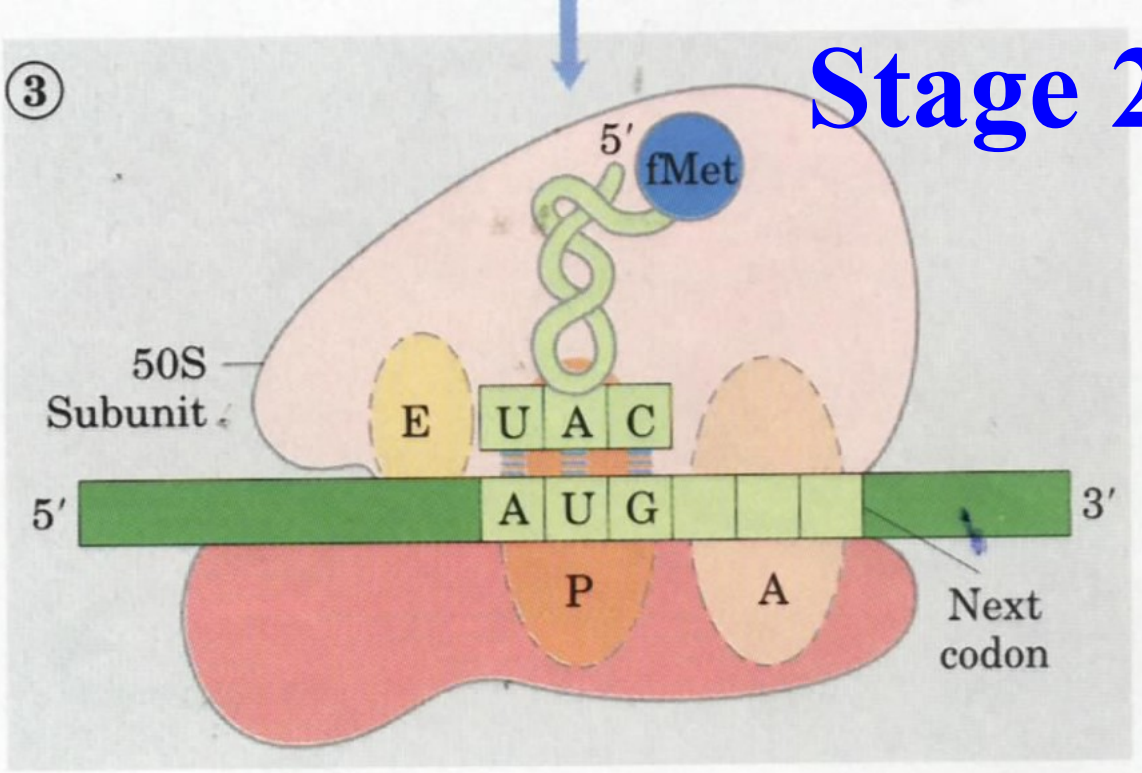


The Met charged on $tRNA^{Met}$ (by Met-tRNA synthetase) is specifically formylated by **a transformylase** to form fMet-$tRNA^{fMet}$ in bacteria (**Met-$tRNA^{Met}$ can not be formylated**). only fMet-$tRNA^{fMet}$ can enter the initiation site (**site P**) on the ribosome in *E.coli*.



Stage 2





The initiating complex is assembled from the small subunit of the ribosome, the mRNA, the initiating aminoacyl-tRNA (being fMet-tRNA^{fMet} in bacteria), and the large subunit of the ribosome.

7. The polypeptide chain is elongated via a repeating three-step reactions (stage3)

The elongation cycle consists of three steps

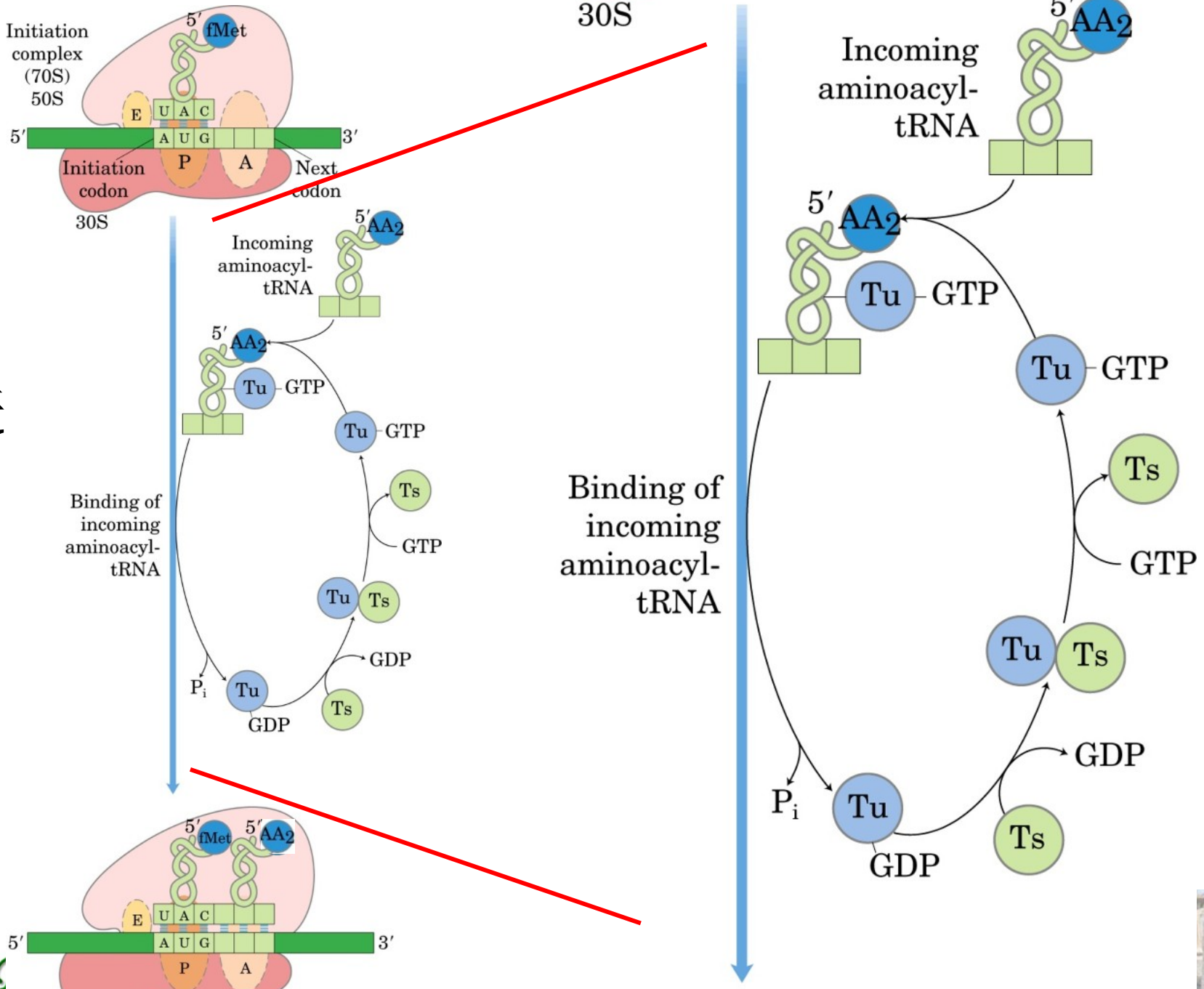
aminoacyl-tRNA binding （进位）

peptide bond formation （转肽）

Translocation （移位）



I 进位

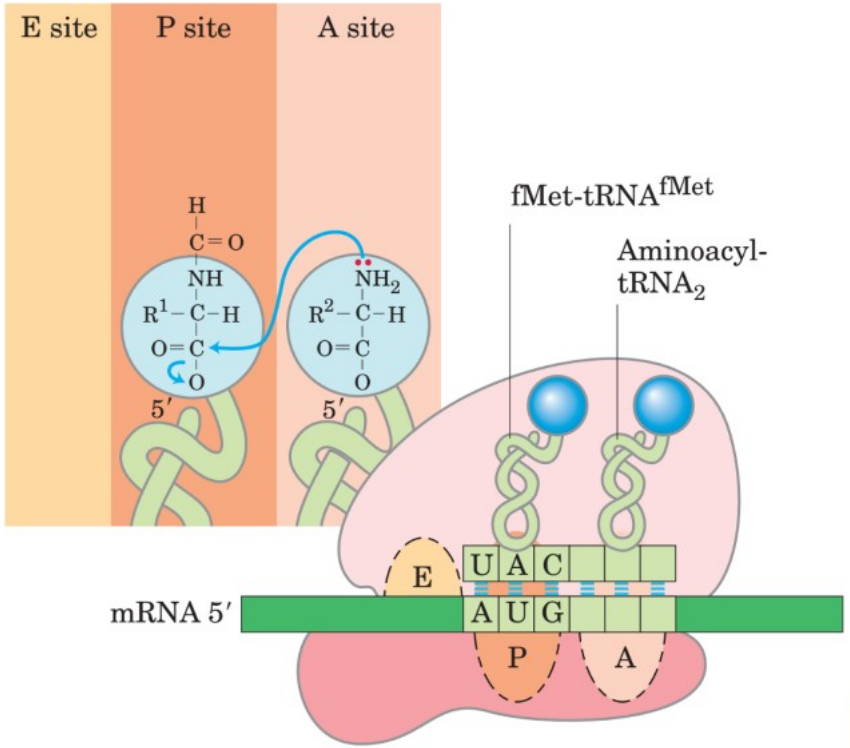

Initiation complex (70S)
50S
30S
5′ fMet
E
U A C
5′
A U G
3′
Initiation codon
P
A
Next codon
5′ AA2
Incoming aminoacyl-tRNA
Tu GTP
Binding of incoming aminoacyl-tRNA
Ts
GTP
Tu Ts
GDP
Pi
Tu
Ts
GDP
5′ fMet
5′ AA2

II 转肽

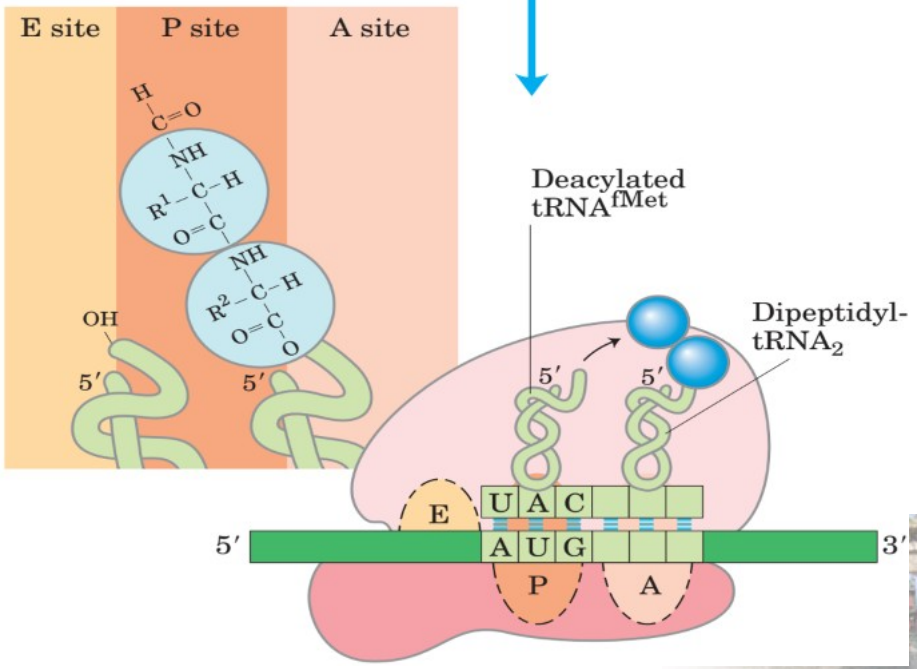
50S 大亚基中的 23S rRNA （核酶）

Peptidyl transferase

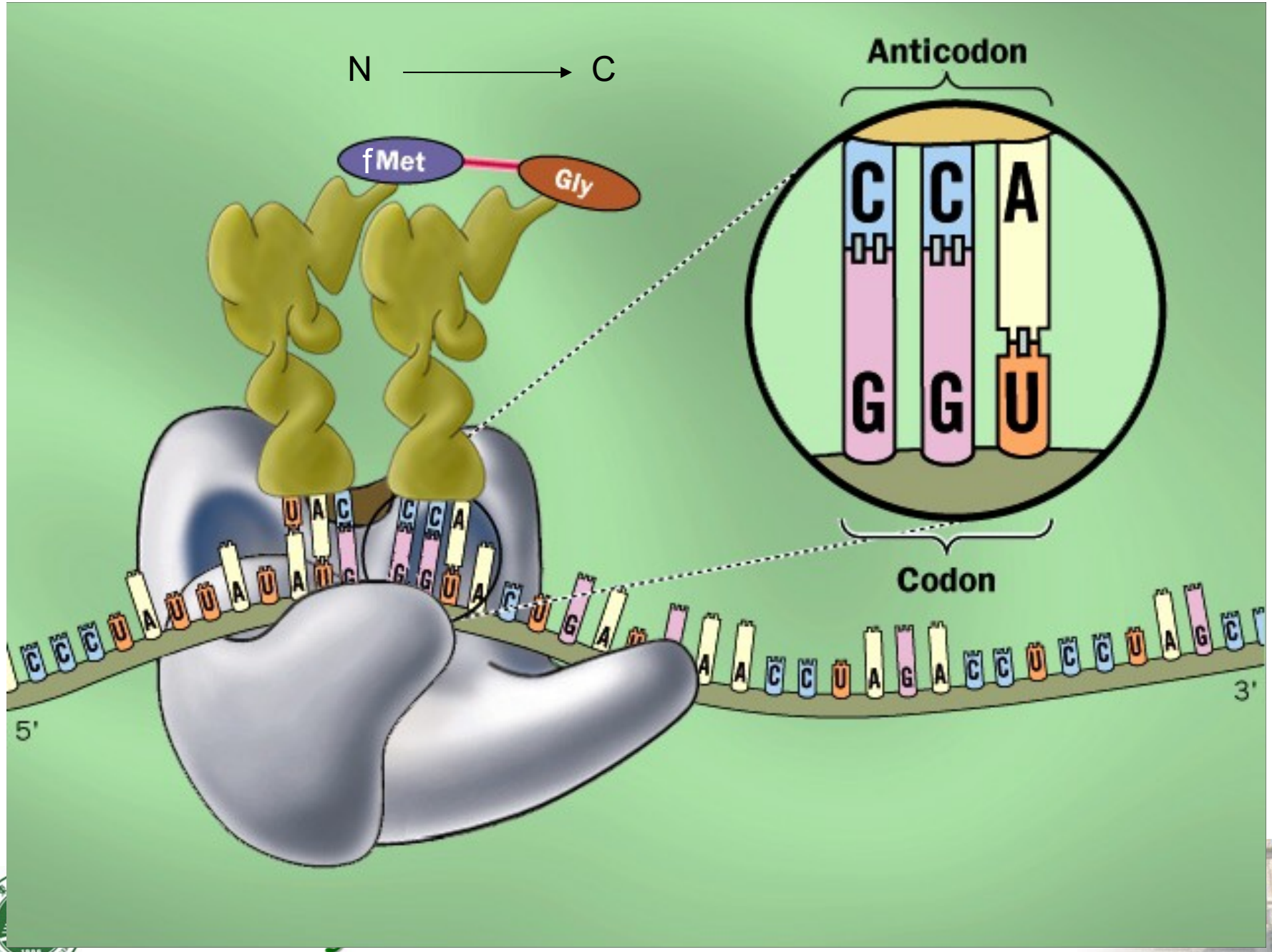
肽酰基转移酶



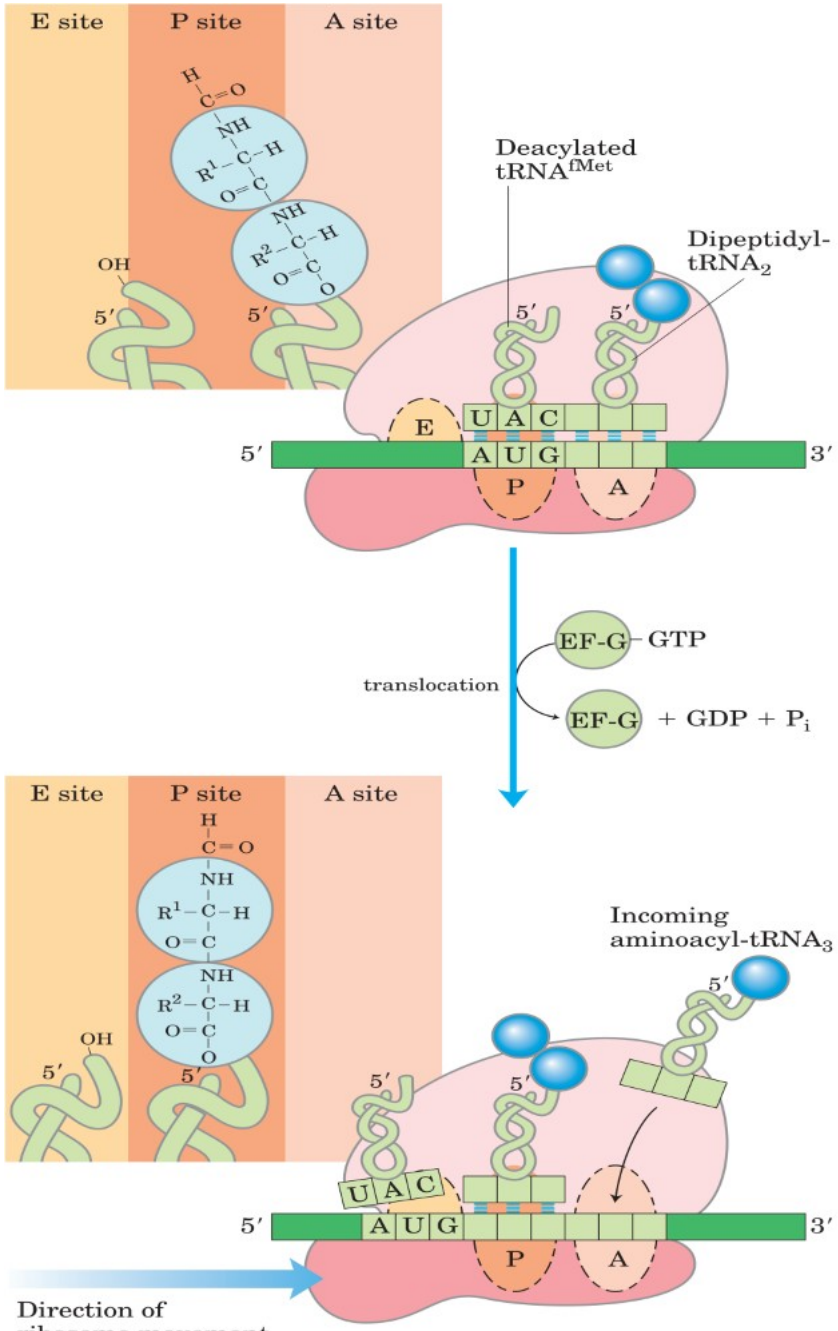
peptide bond formation



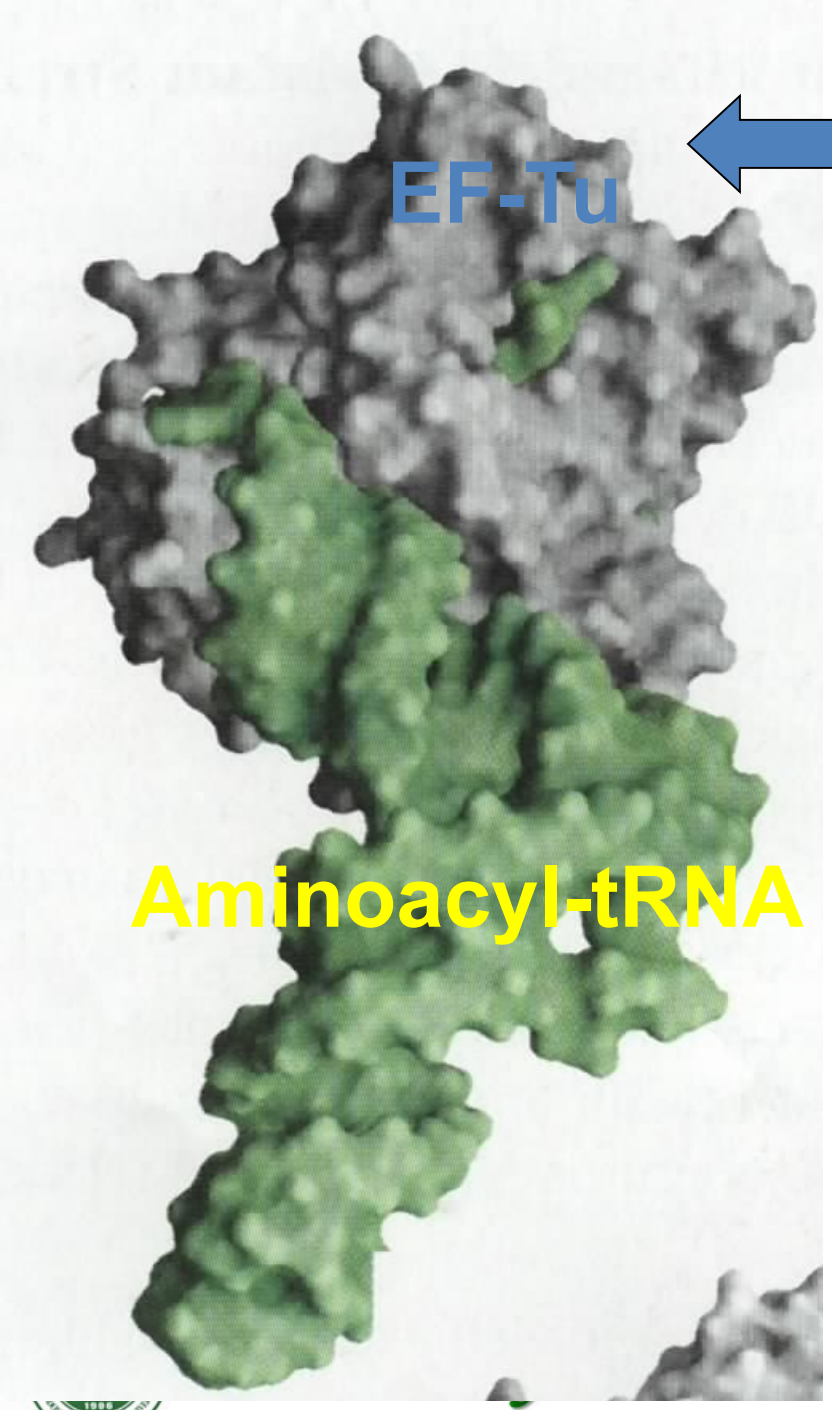
Biochemistry


N
C
fMet
Gly
Anticodon
C
C
A
G
G
U
Codon
5'
3'

III 移位

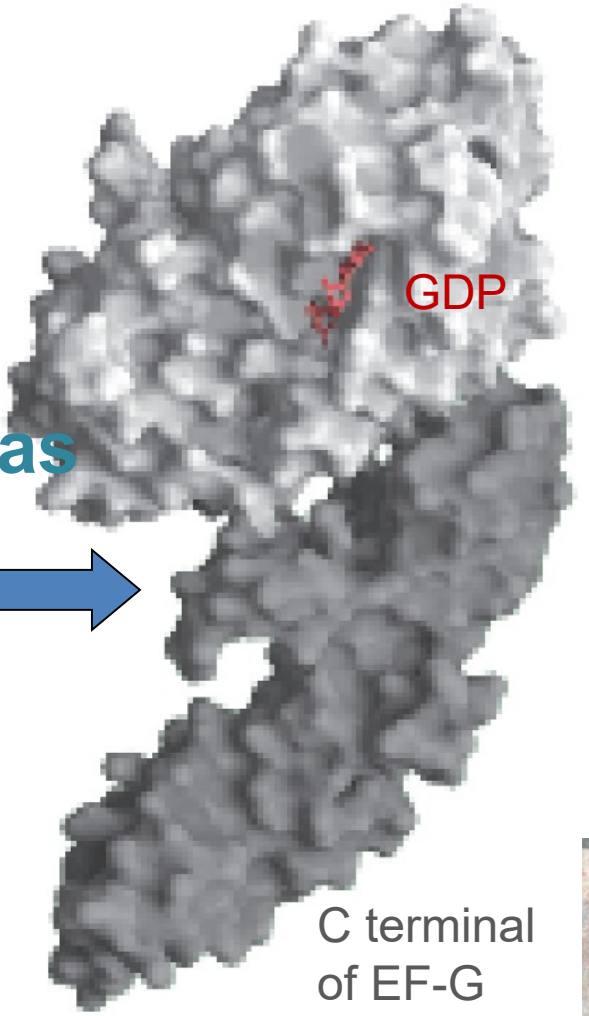


EF-G translocase



The EF-Tu-GTP-aminoacyl-tRNA complex

EF-G-GDP has a structure similar to EF-Tu-tRNA

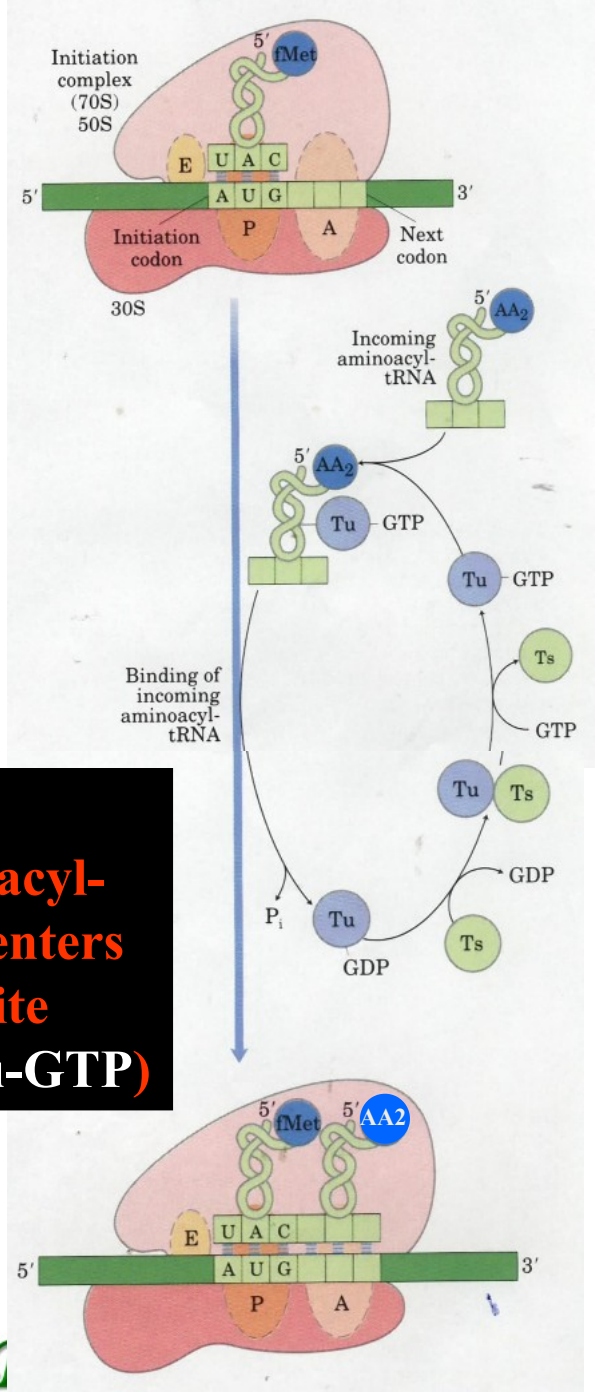


- The next **aminoacyl-tRNA** is delivered to the **A (aminoacyl) site** of the ribosome by **EF-Tu-GTP** (step 1).
- The fMet group is transferred to the second aminoacyl-tRNA in the **A** site to form a peptide bond, generating a dipeptidyl-tRNA (step 2).The **23S rRNA** of the large subunit of the ribosome seems to have the **peptidyl transferase** activity, thus being a ribozyme (Noller, 1992).

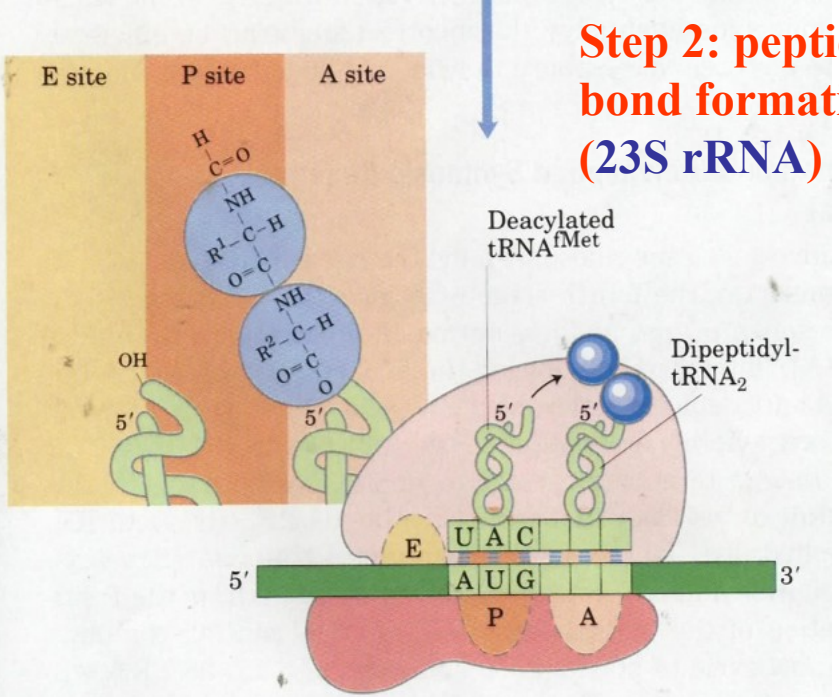


EF-G -GTP(the **translocase**) then promotes the translocation of the ribosome along the mRNA by the distance of one codon: the **deacylated tRNA^{fMet}** is released from the **E** (exit) site of the ribosome, the **dipeptidyl-tRNA** is relocated to the **P** site, and the **A** site is open for the **incoming** (the third) aminoacyl-tRNA (step 3).



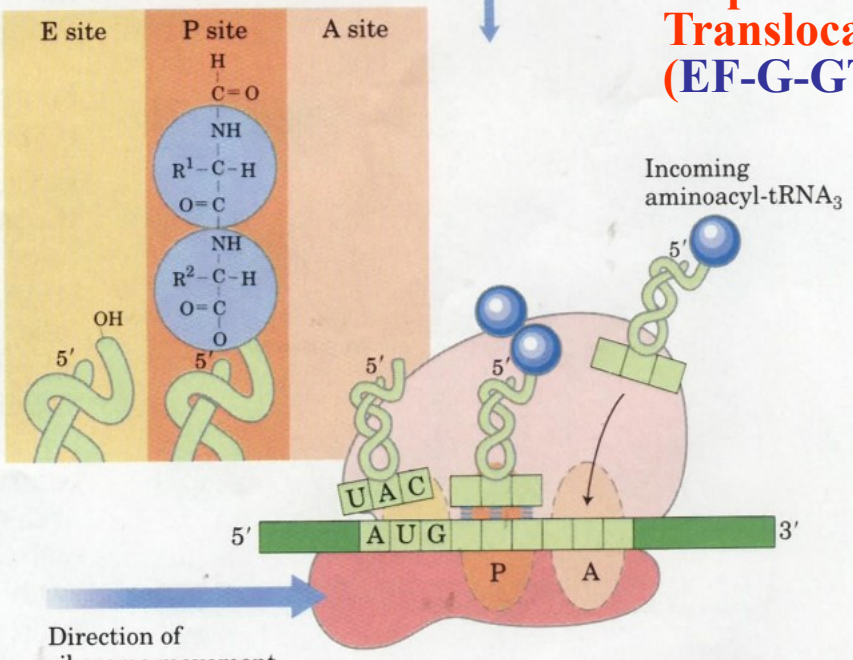


Step 1: Aminoacyl-tRNA enters the A site (EF-Tu-GTP)



Step 2: peptide bond formation (23S rRNA)

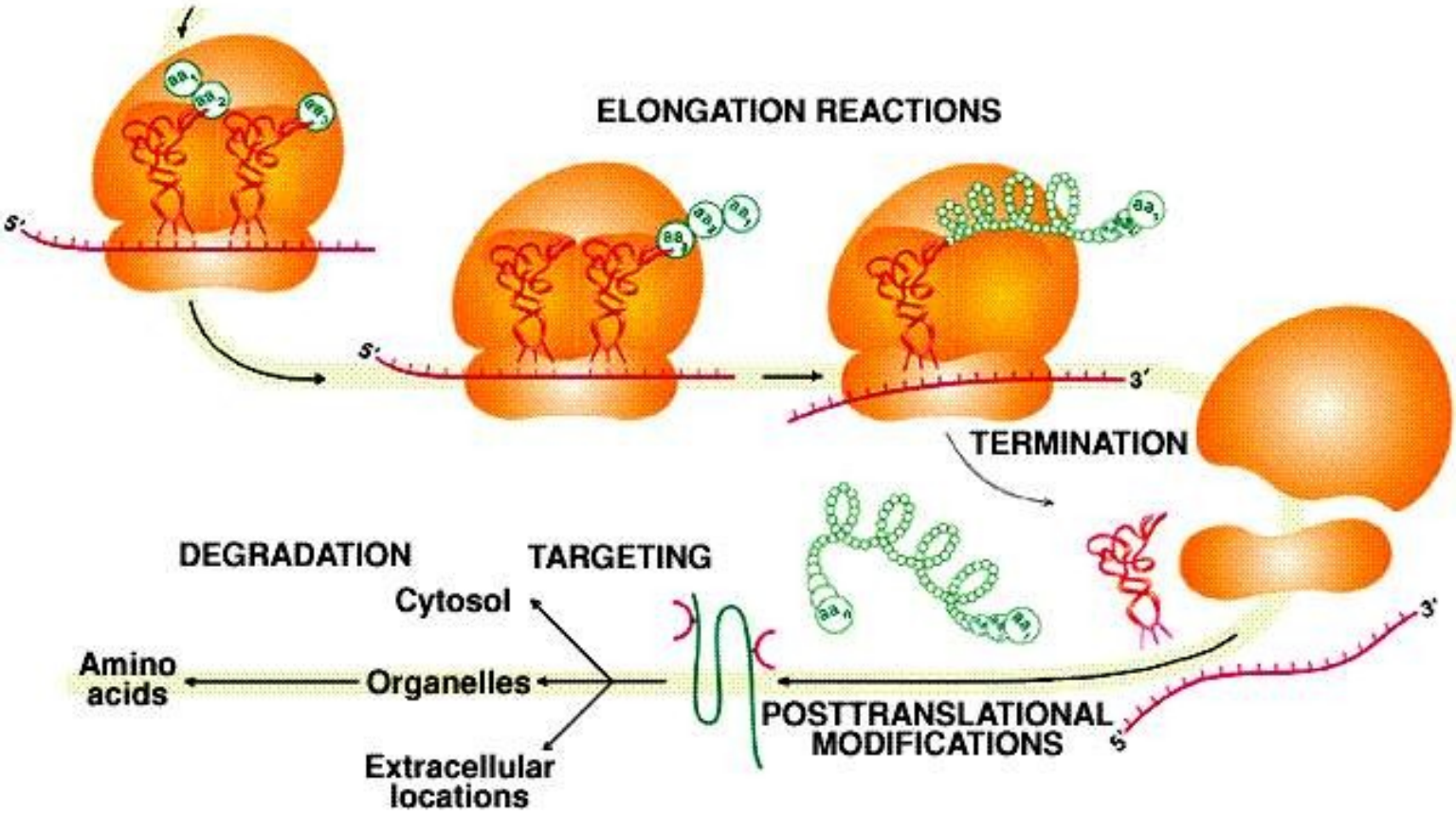
Stage 3



Step 3: Translocation (EF-G-GTP)



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8. Polypeptide synthesis is terminated by release factors that read the stop codons

Two bacterial protein factors called **RF1** and **RF2** recognize the stop codons (**RF1 for UAA and UAG; RF2 for UAA and UGA**).

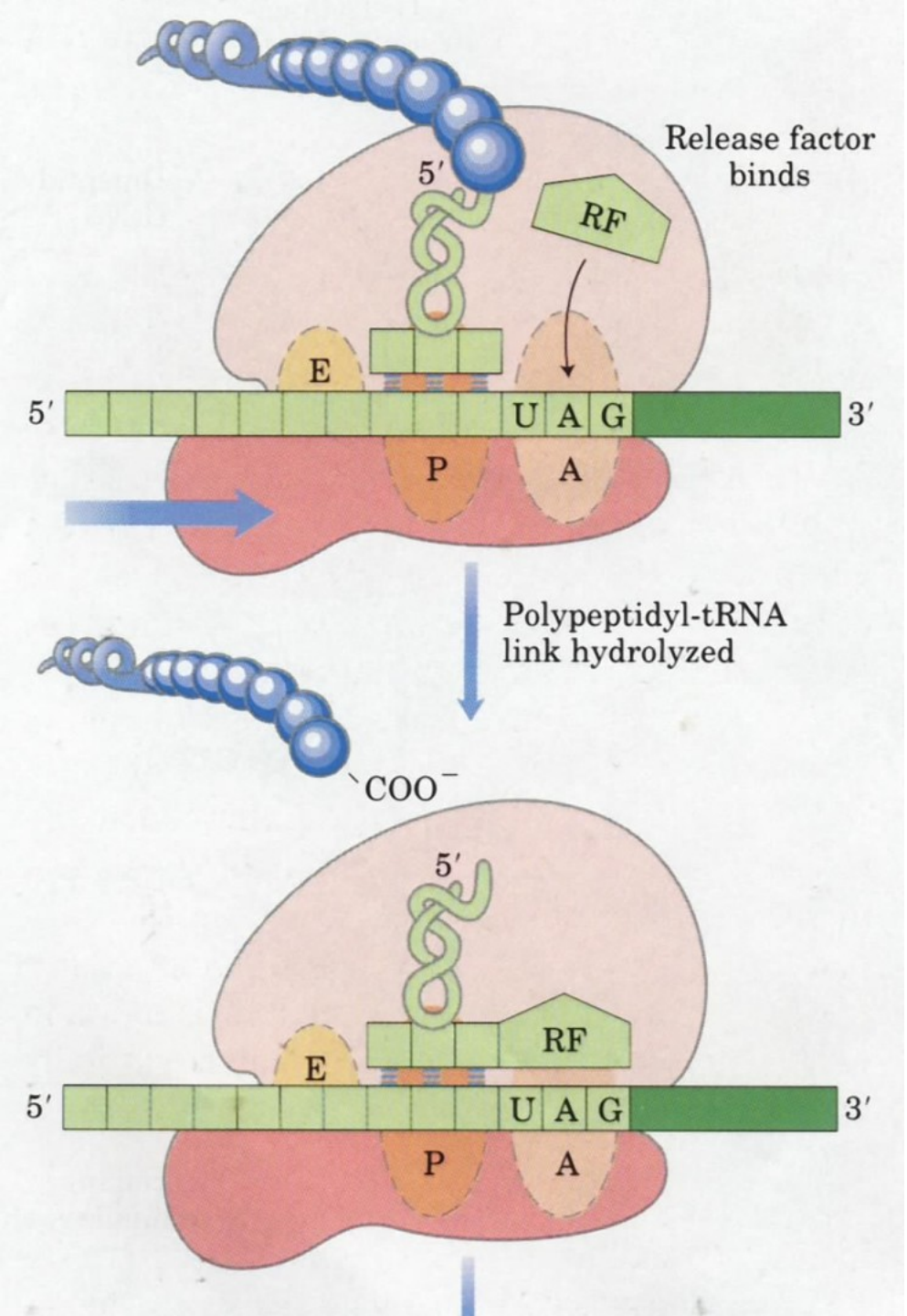
A third releasing factor **RF3** is a GTP-binding protein and seems to act in concert to promote the cleavage of the peptide from peptidyl-tRNA.

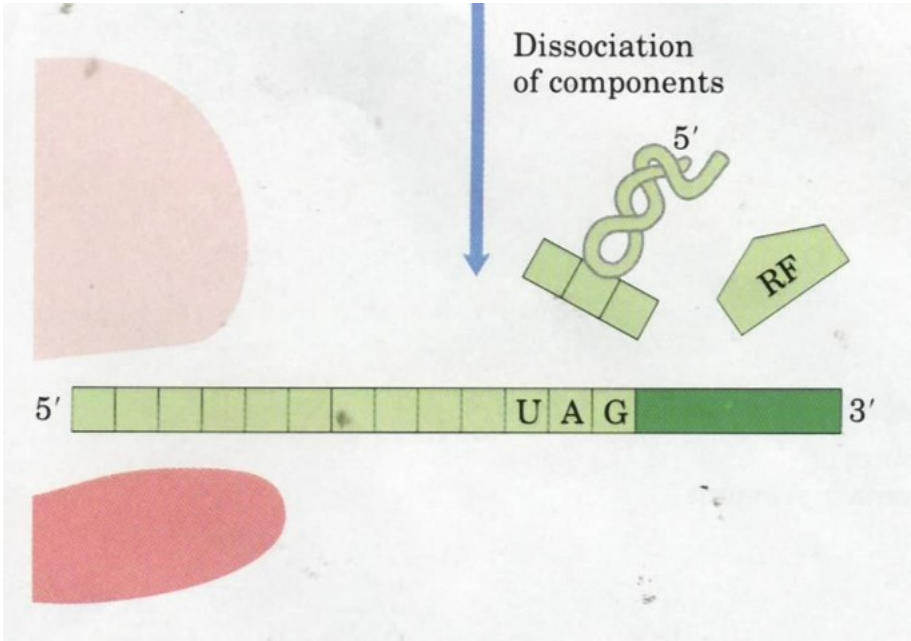
The specificity of the peptidyl transferase (23S rRNA) is altered by the release factor: water become the acceptor of the activated peptidyl moiety.



The polypeptide chain is released from the ribosome when meeting a stop codon (UAA, UGA, or UGA)

Stage 4





9. At least four high-energy bonds are ultimately consumed for the formation of each specific peptide bond

Two for activating each amino acid (ATP → AMP + $2P_i$).

One for EF-Tu to deliver the aminoacyl-tRNA to the **A** site of the ribosome (GTP → GDP + P_i).

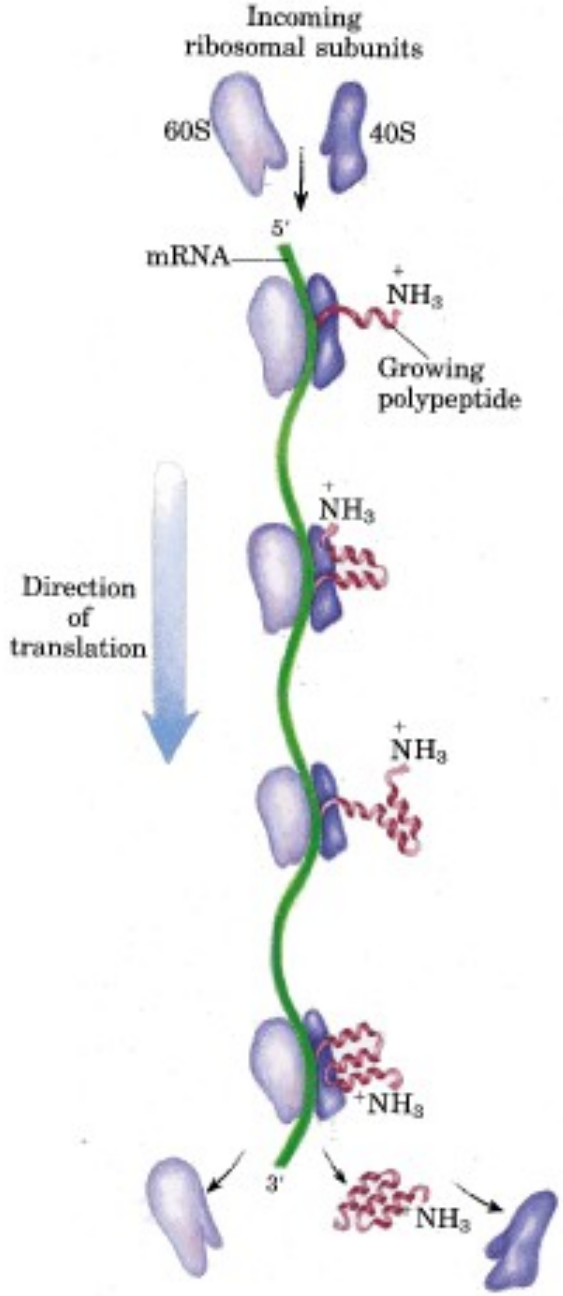
One for EF-G to translocate the ribosome after each peptide bond is formed (GTP → GDP + P_i).



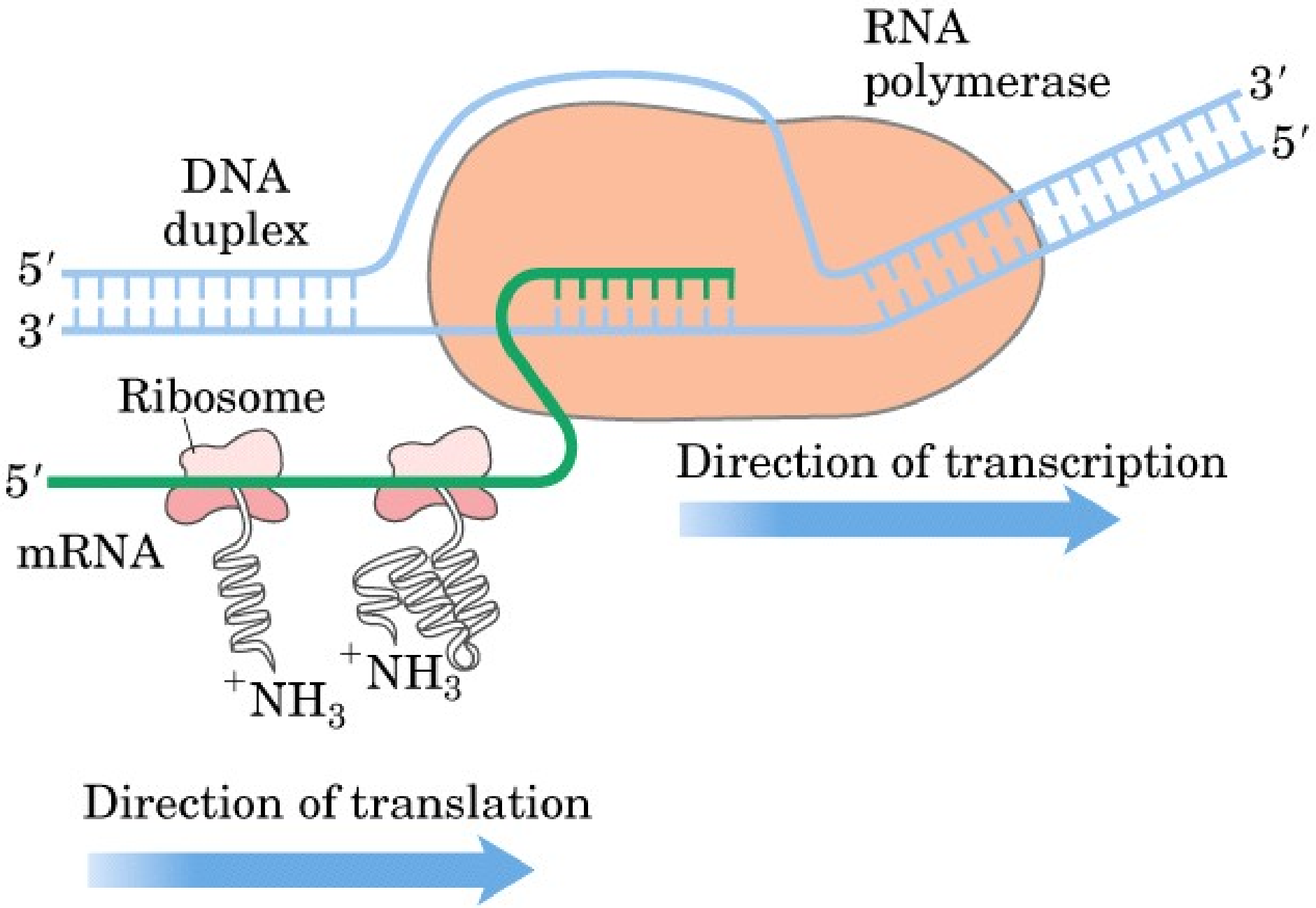
10. Each mRNA is usually translated simultaneously by many ribosomes (as polysomes)

This is observed in **both prokaryotic and eukaryotic** cells.
The guiding effectiveness of the short-lived mRNA molecules is thus dramatically increased.
In bacteria, the translation and transcription processes are tightly coupled!

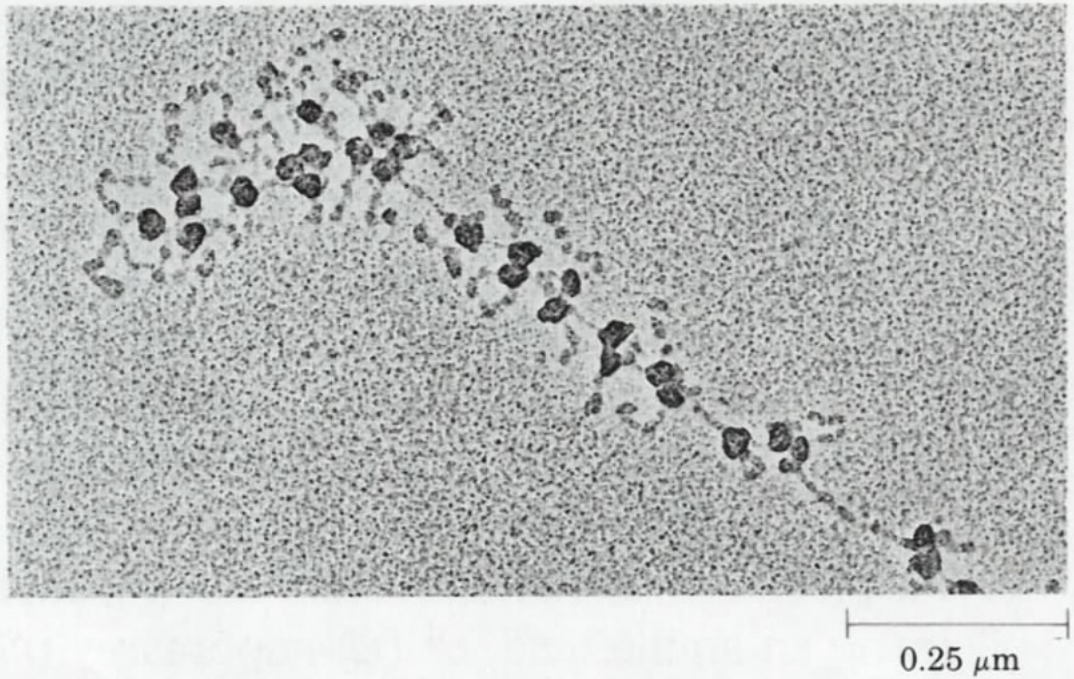


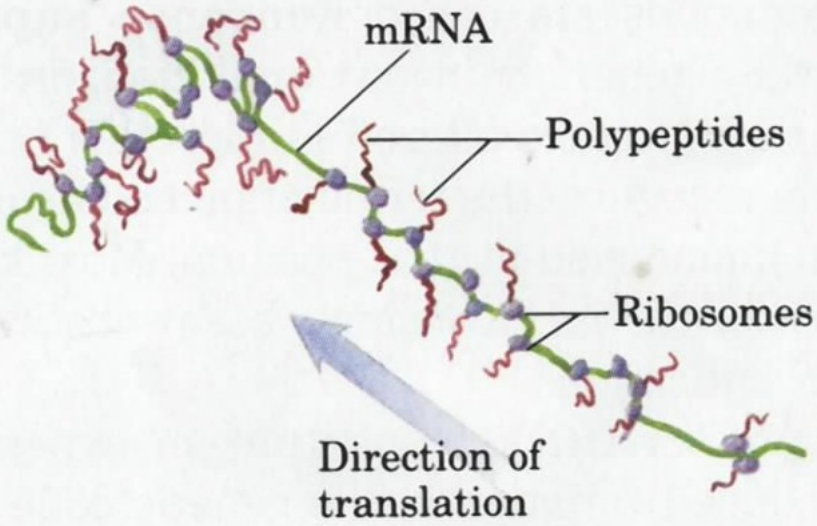

Incoming
ribosomal subunits
60S
40S
5'
mRNA
$^{+}NH_3$
Growing
polypeptide
$^{+}NH_3$
Direction
of
translation
$^{+}NH_3$
$^{+}NH_3$
3'
$^{+}NH_3$

(a)


RNA
polymerase
3′
5′
DNA
duplex
5′
3′
Ribosome
5′
mRNA
Direction of transcription
+NH3
+NH3
Direction of translation

Polysomes are regularly seen in cells





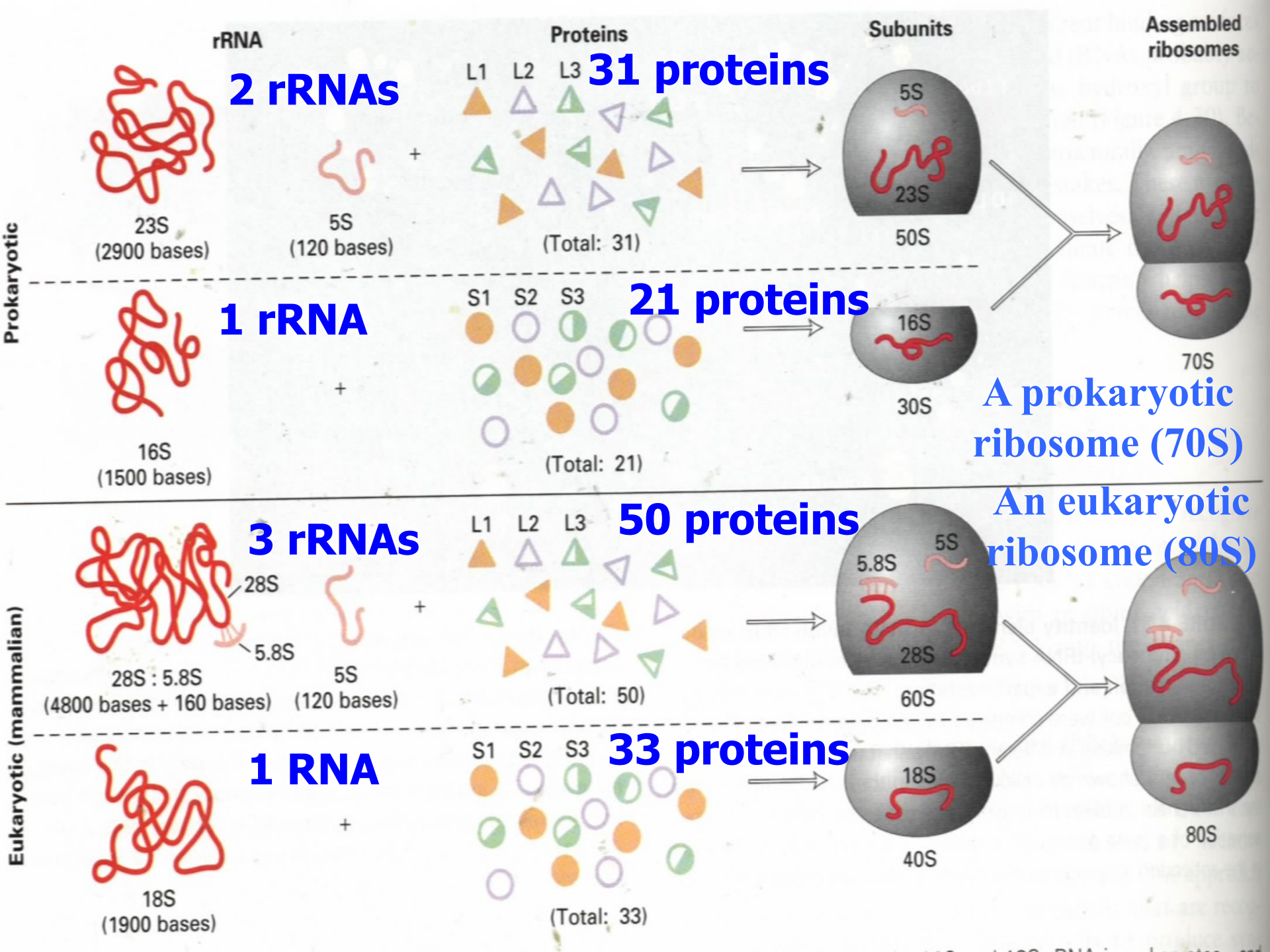
(b)

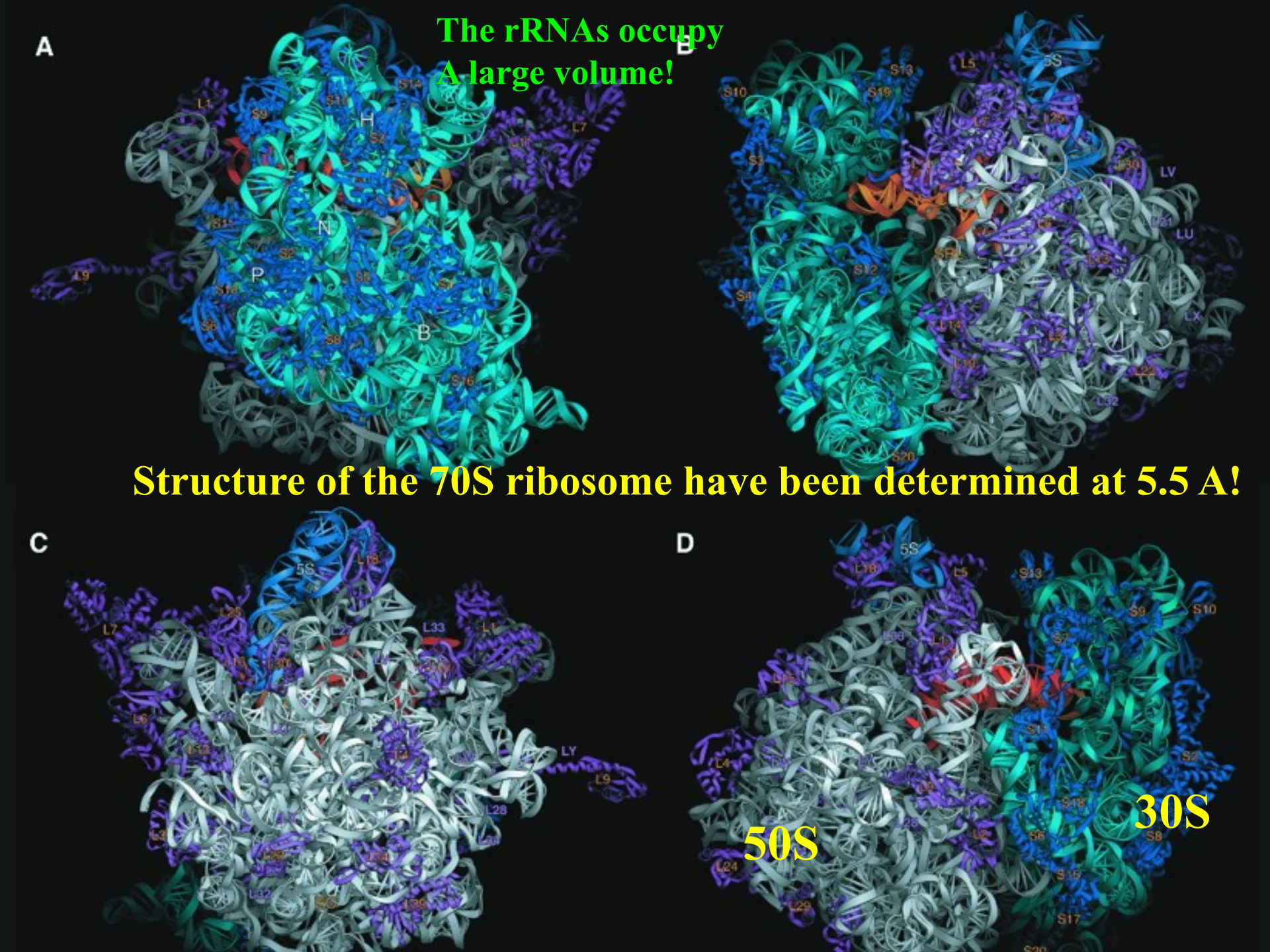


11. Translation in eukaryotes (H3)

- Eukaryotic ribosomes are larger than prokaryotic ribosomes.
- At least 9 initiation factors are involved.
- The initiating amino acid is Met, not fMet.
- Eukaryotic mRNA do not contain SD sequences.
- Eukaryotic mRNA is monocistronic (not polycistronic).
- Elongation factors
- Termination factors




rRNA
Proteins
Subunits
Assembled ribosomes
Prokaryotic
2 rRNAs
31 proteins
L1 L2 L3
23S
(2900 bases)
5S
(120 bases)
(Total: 31)
5S
23S
50S
1 rRNA
21 proteins
S1 S2 S3
16S
(1500 bases)
(Total: 21)
16S
30S
70S
A prokaryotic ribosome (70S)
An eukaryotic ribosome (80S)
Eukaryotic (mammalian)
3 rRNAs
50 proteins
L1 L2 L3
28S
5.8S
28S : 5.8S
(4800 bases + 160 bases)
5S
(120 bases)
(Total: 50)
5.8S
5S
28S
60S
1 RNA
33 proteins
S1 S2 S3
18S
(1900 bases)
(Total: 33)
18S
40S
80S


The rRNAs occupy
A large volume!
A
B
Structure of the 70S ribosome have been determined at 5.5 A!
C
D
50S
30S

12. The newly synthesized polypeptide chains need to be folded, assembled, and processed to make a functional protein (stage 5)

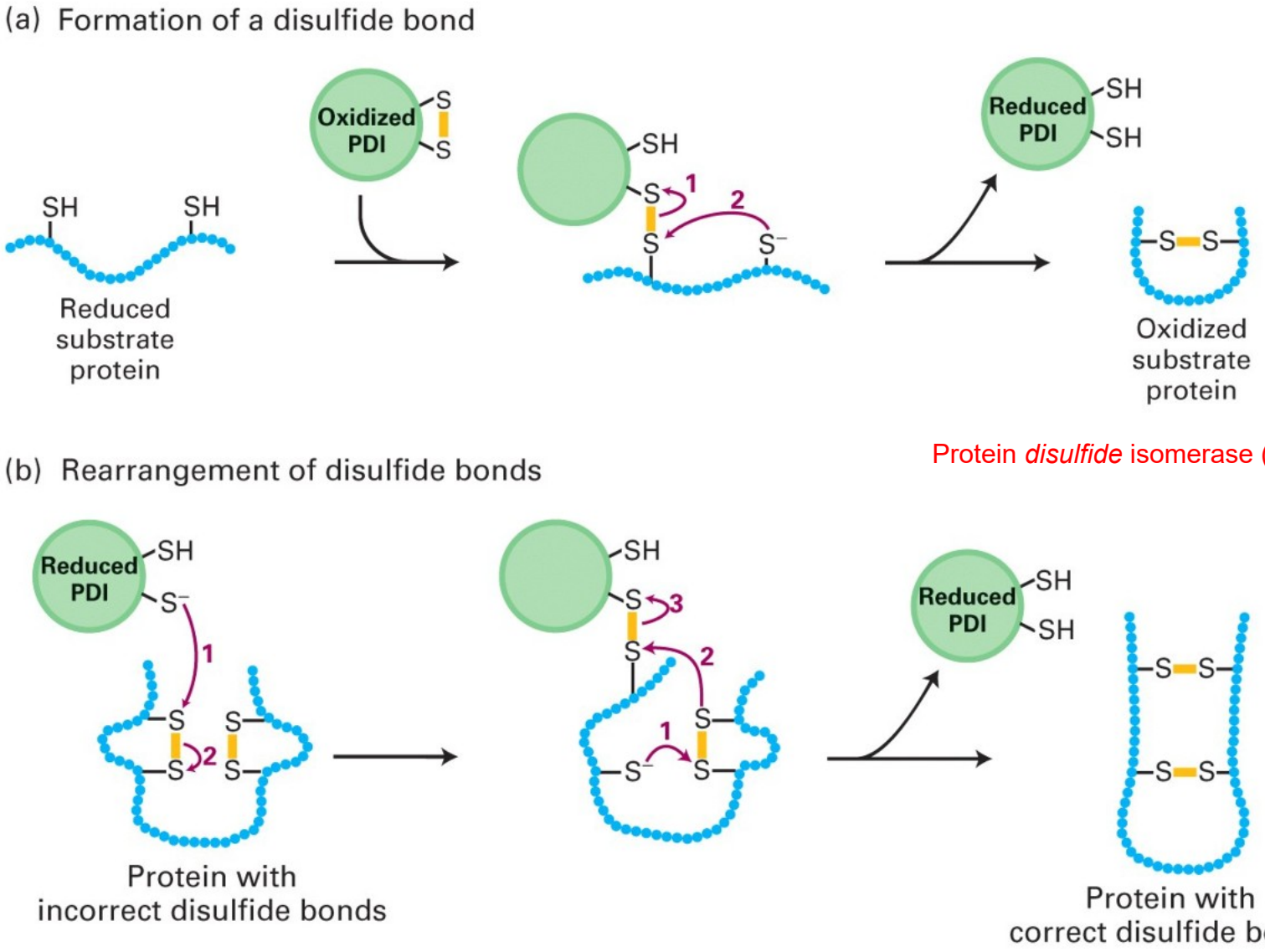
Newly synthesized polypeptide chains will assume their native conformation spontaneously or with help from other proteins (called **molecular chaperones**), with or without further **posttranslational modifications**.

Formation of **disulfide bonds**.

Addition of **cofactors** via covalent or noncovalent bonds.



Disulfide bonds are formed and rearranged in the ER lumen



Protein *disulfide* isomerase (*PDI*)

The N-terminal formyl, fMet or Met may be removed enzymatically.

Signal sequences (usually 15 to 30 residues in length) at the N-terminal of some proteins will be removed by specific peptidases after the protein reached their cellular destinations (protein targeting).

Proteolytic cleavage of larger precursor proteins.



The N-terminal amino and C-terminal carboxyl groups may be modified (e.g., acetylated). Specific amino acids in a protein may be enzymatically modified posttranslationally by groups like **phosphate**, **carboxylate**， or **methylate**.

Oligosaccharide groups are covalently attached to specific Asn, Thr, or Ser residues in some proteins (glycoproteins, often function extracellularly).



Epigenetics
-histone methylation

Histone 3 methylation	function
H3K4me3	Gene activation （protein coding gene）
H3K9me3	Gene repression （transposon）
H3K27me3	Gene repression （protein coding gene）



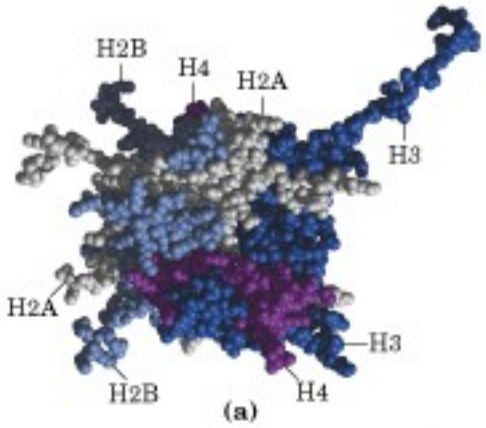
table 24–3

Types and Properties of Histones

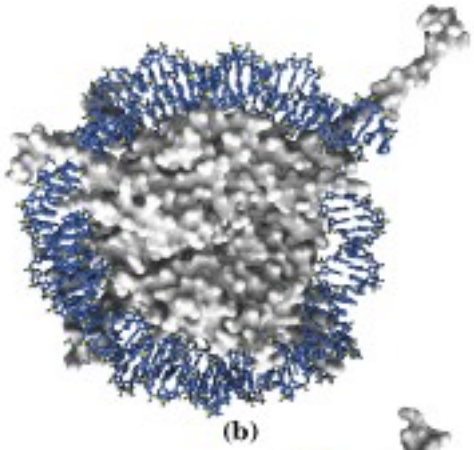
Histone	Molecular weight	Number of amino acid residues	Content of basic amino acids (% of total)	
			Lys	Arg
H1*	21,130	223	29.5	1.3
H2A*	13,960	129	10.9	9.3
H2B*	13,774	125	16.0	6.4
H3	15,273	135	9.6	13.3
H4	11,236	102	10.8	13.7

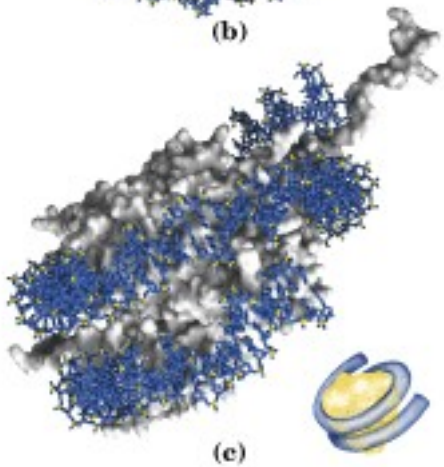
*The sizes of these histones vary somewhat from species to species. The numbers given here are for bovine histones.

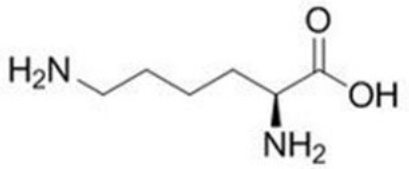




(a)


(b)


(c)



lysine

Epigenetics

-histone methylation

Histone 3 methylation	function	
H3K4me3	Gene activation （protein coding gene）	Euchromatin
H3K9me3	Gene repression （transposon）	Heterochromatin
H3K27me3	Gene repression （protein coding gene）	Euchromatin



Summary

Genetic and biochemical studies showed that the genetic codes are continuous triplets of nucleotides.

The genetic codes was deciphered within five years using artificial mRNA templates of various base composition, *in vitro* protein synthesis and filter binding assays.

A tRNA molecule can recognize one to three codons depending what the first (wobble) nucleotide of the anticodon is (C and A for one, U and G for two, I for three).

In certain viral DNAs, overlapping genes are found.



Protein synthesis occurs on ribosomes: having a large and small subunits, both composing one or two rRNA and many protein molecules.

Protein synthesis can be divided into five stages:

① **activation of amino acids** (ATP dependent, aminoacyl-tRNA synthetase catalyzed);

② **formation of the initiation complex** at the ribosome binding site and initiation codon (helped by specific initiation protein factors, IF-1, 2, and 3);

③ **elongation** of the peptide chain (helped by the elongation factors, EF-Tu, EF-Ts, EF-G and catalyzed by the 23S rRNA);

④ **termination** (helped by the releasing factors, RF_1, RF_2, and RF_3);

⑤ **folding and posttranslational modifications**.



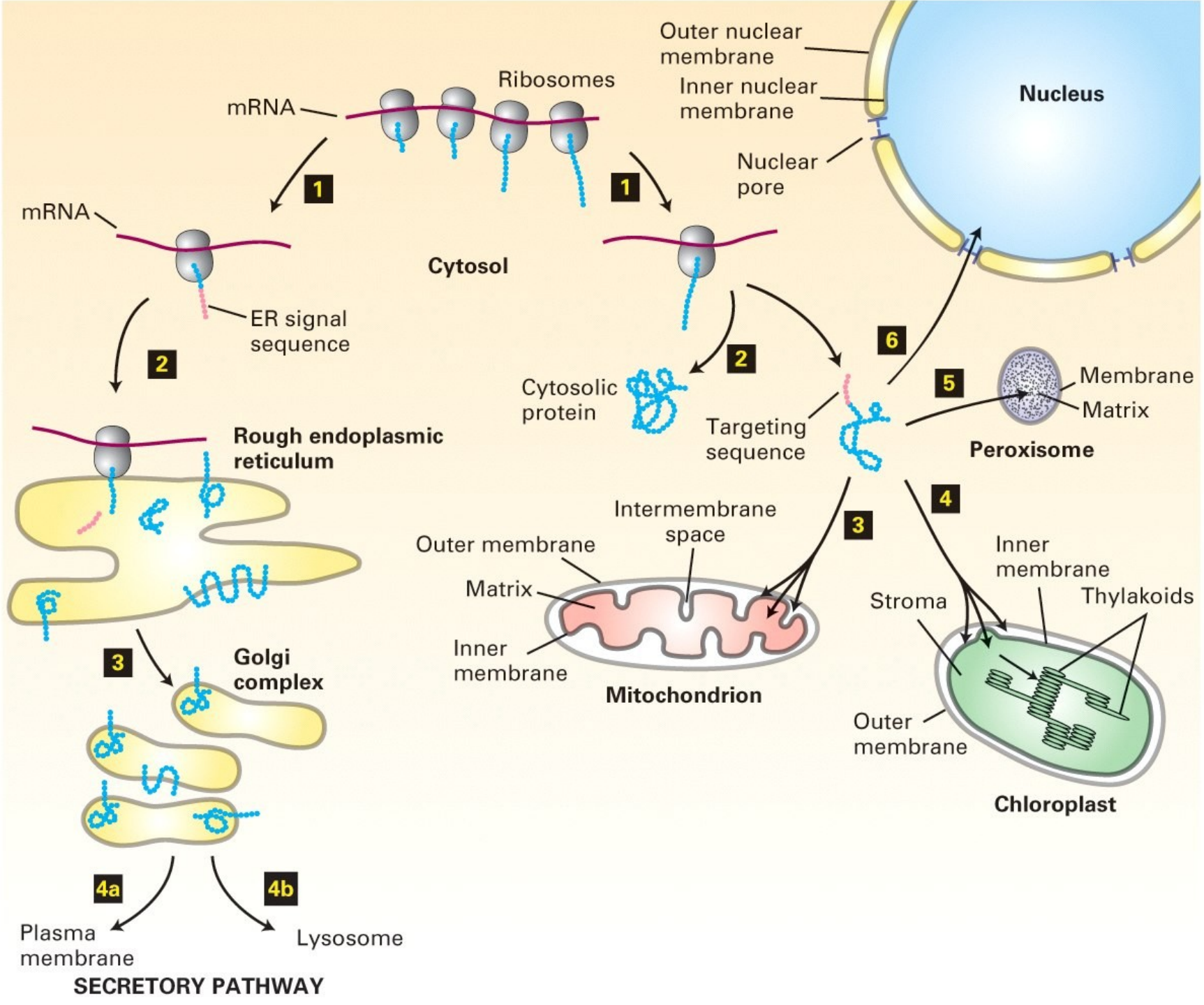
H4 Protein Targeting

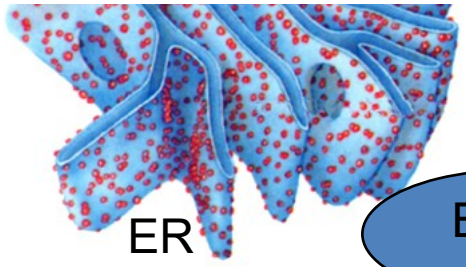


How does a protein get to the correct cellular location?

Proteins are targeted to different compartments in different ways.




Outer nuclear membrane
Inner nuclear membrane
Nucleus
Nuclear pore
Ribosomes
mRNA
1
mRNA
Cytosol
ER signal sequence
2
Cytosolic protein
2
6
5
Membrane
Matrix
Peroxisome
Targeting sequence
Rough endoplasmic reticulum
3
Golgi complex
Intermembrane space
Outer membrane
Matrix
Inner membrane
Mitochondrion
4
Inner membrane
Stroma
Thylakoids
Outer membrane
Chloroplast
4a
4b
Plasma membrane
Lysosome
SECRETORY PATHWAY



Ribosome

Bound

Free

Lysosomal
Secretory
Plasma Memb.

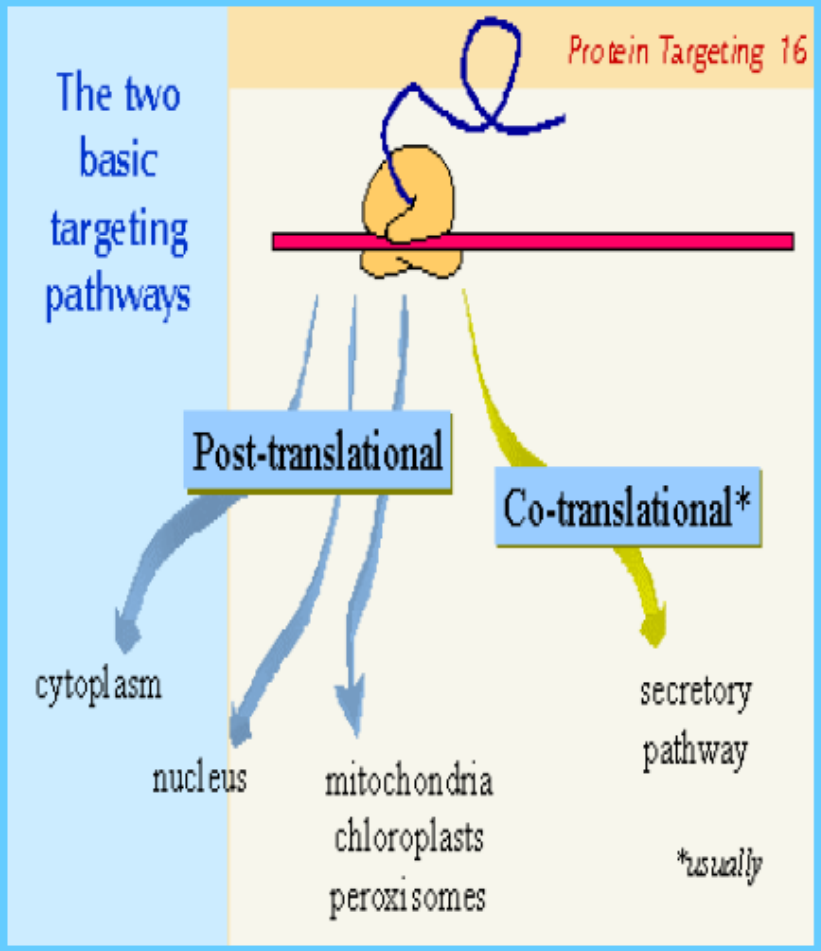
Soluble Proteins
Nucleus
Mitochondria
Chloroplasts
peroxisomes

Proteins that are targeted to the cell surface, Golgi and Lysosomes are synthesized on ER membrane bound ribosomes and move through the secretory pathway.

Proteins that are targeted to the nucleus, mitochondria, chloroplasts and peroxisomes are synthesized on free ribosomes as soluble polypeptides.



Two basic forms of targeting pathways



post-translational targeting:

- nucleus
- mitochondria
- chloroplasts
- Peroxisomes

co-translational targeting (***secretory pathway***):

- ER
- Golgi
- lysosomes
- plasma membrane
- secreted proteins

Protein Targeting:
How do secreted proteins get to the ER membrane?

Gunther Blobel
Nobel Prize 1999

"for the discovery that proteins have intrinsic signals that govern their transport and localization in the cell"



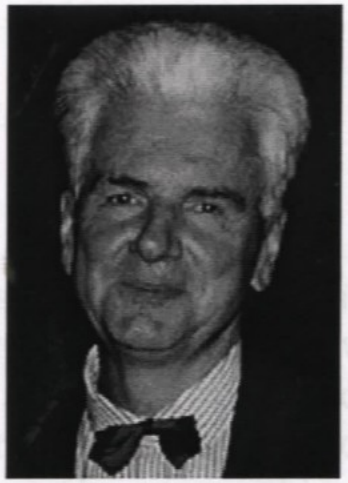


The Nobel Prize in Physiology or Medicine 1999

"for the discovery that proteins have intrinsic signals that govern their transport and localization in the cell"

Press release
Illustrated Presentation

G nter Blobel



USA

The Rockefeller University,
Howard Hughes Medical Institute,
New York, NY, USA

1936 -

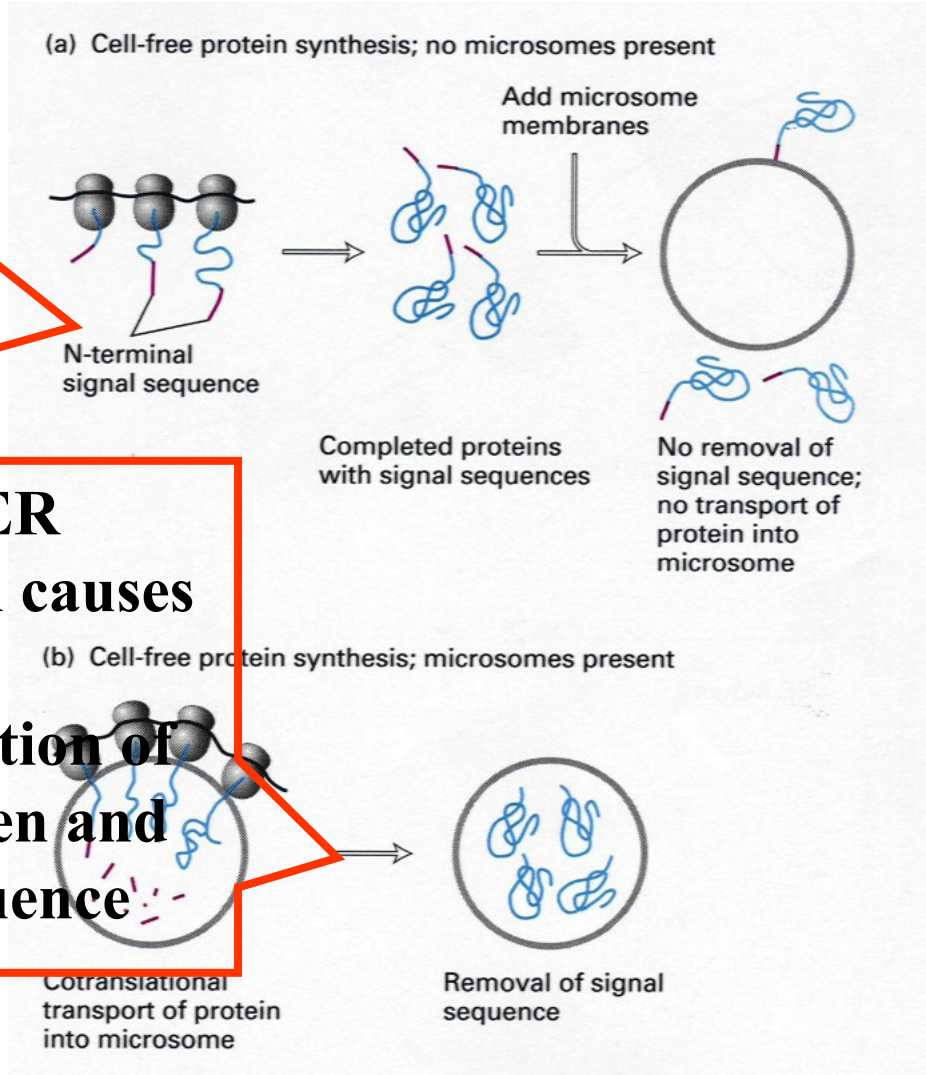
Curriculum Vitae
Nobel Lecture (video)
Interview (video)
Nobel Diploma
Prize Award Ceremony
Other Resources

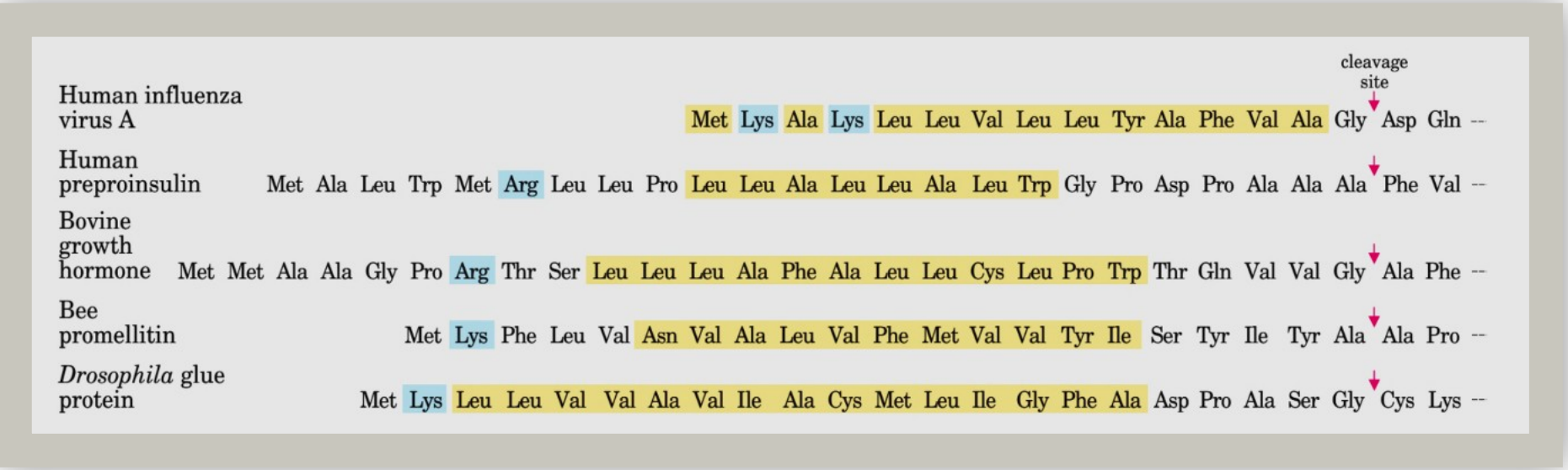


Early experiments: An N-terminal signal sequence on nascent secretory proteins targets synthesis to the ER and is then cleaved.

Translation of secretory mRNA's in cell free protein synthesis system produces full length proteins with intact signal sequence

Adding microsomes (ER membranes) to system causes ribosomes to bind to membranes, translocation of the protein to the lumen and cleavage of signal sequence





Σιγναλ σεθυενχεσ ϖαρψ ιν λενγτη φρομ 13 το 36 αμινο αχιδ ρεσιδυεσ, βυτ αλλ ηαϖε τηε φολλοωινγ φεατυρεσ:

(1) 10 το 15 ηψδροπηοβιχ αμινο αχιδ ρεσιδυεσ;

(2) ονε ορ μορε ποσιτιϖελψ χηαργεδ ρεσιδυεσ （ βασιχ ρεσιδυεσ ） ,υσυαλλψ νεαρ τηε αμινο τερμινυσ;

(3) Αν αμινο αχιδ ρεσιδυεσ ωιτη σηορτ σιδε χηαινσ (εσπεχιαλλψ Αλα) ατ τηε ποσιτιονσ χλοσεστ το τηε χλεαϖαγε σιτε.



What gets the ribosomes with secretory protein mRNA's to bind to the ER membranes?

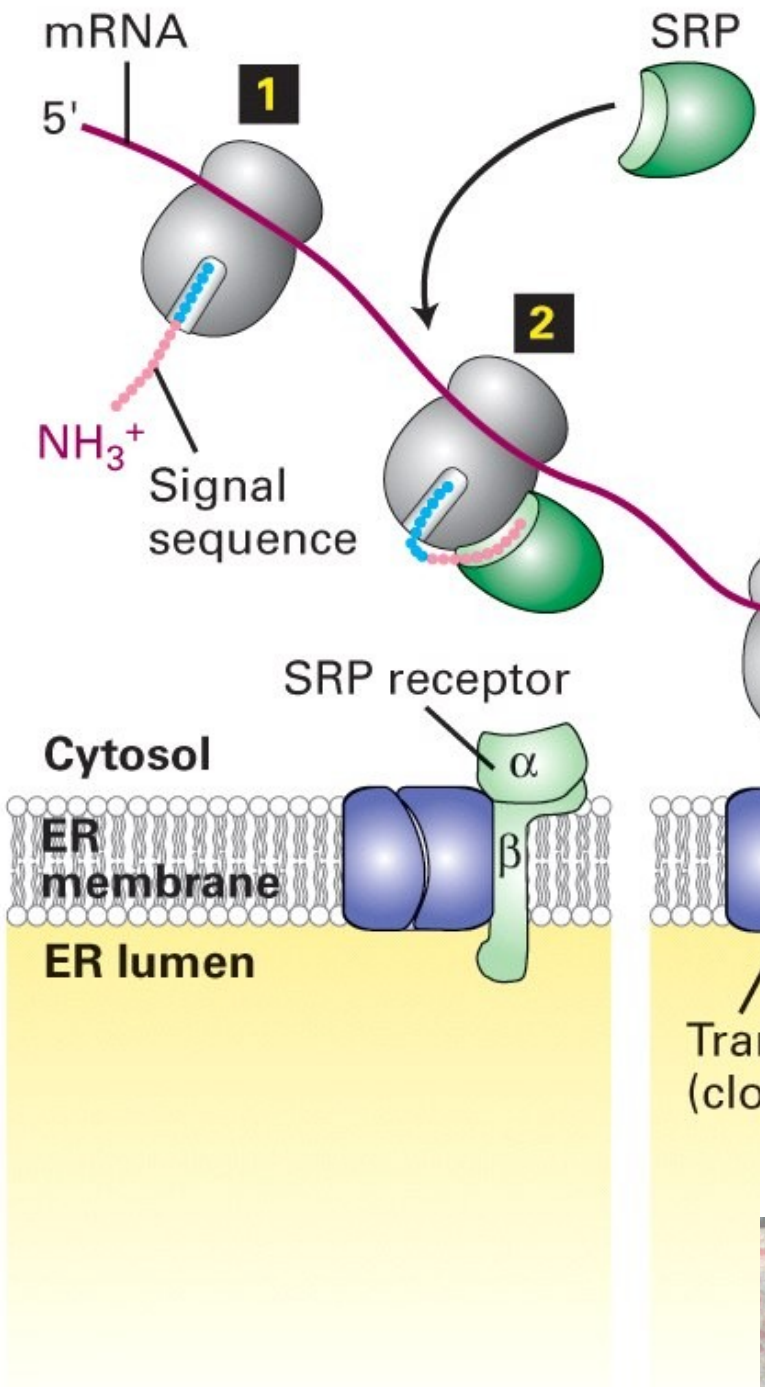
The ER targeting mechanism requires two special receptor proteins:

1. Signal recognition particle (SRP)
2. SRP receptor

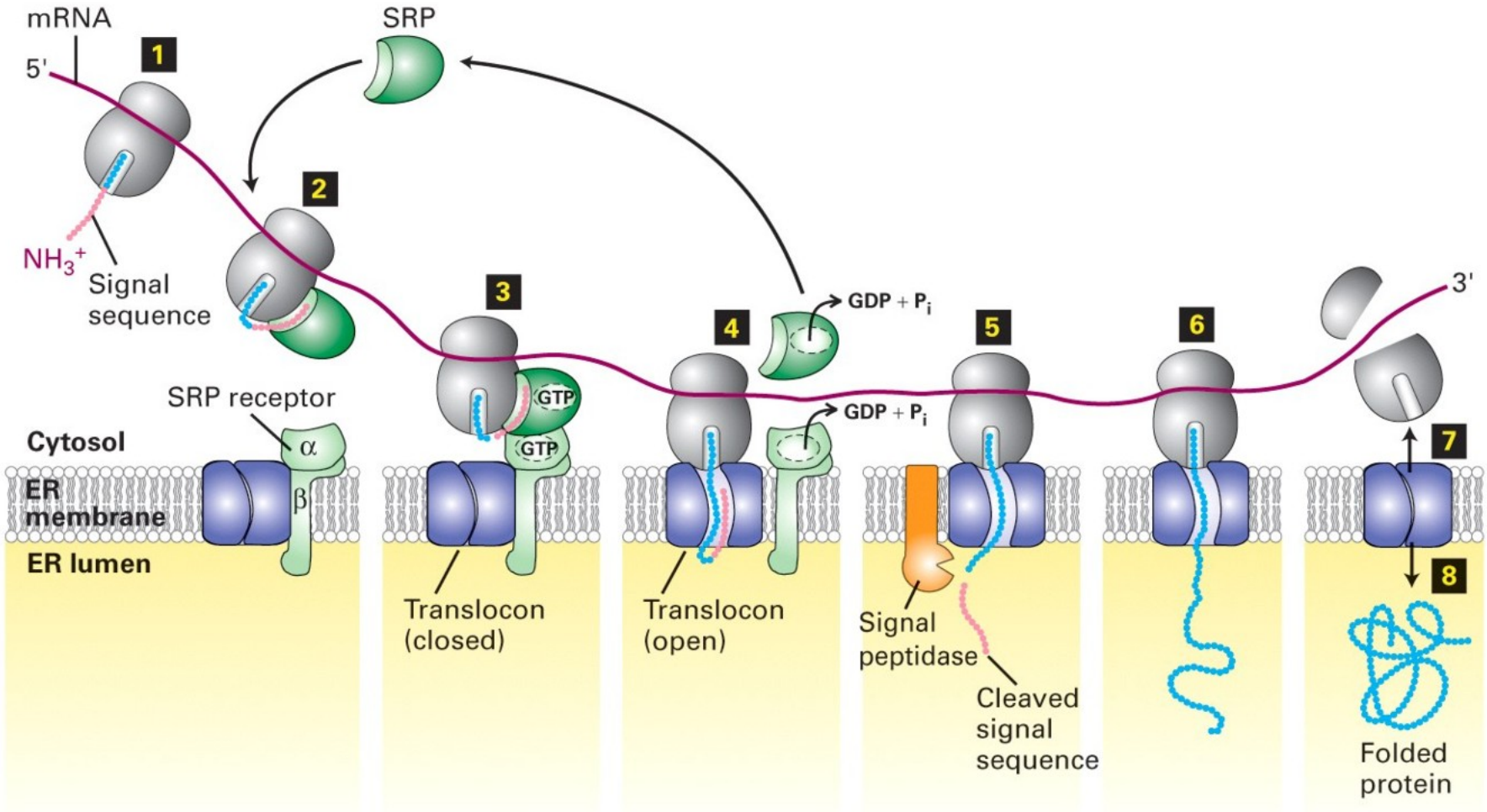


Translation of secretory mRNA begins on free ribosomes

- N-terminal signal sequence emerges from ribosome tunnel
- Signal recognition particle (SRP) binds to the emerging signal sequence from the ribosome

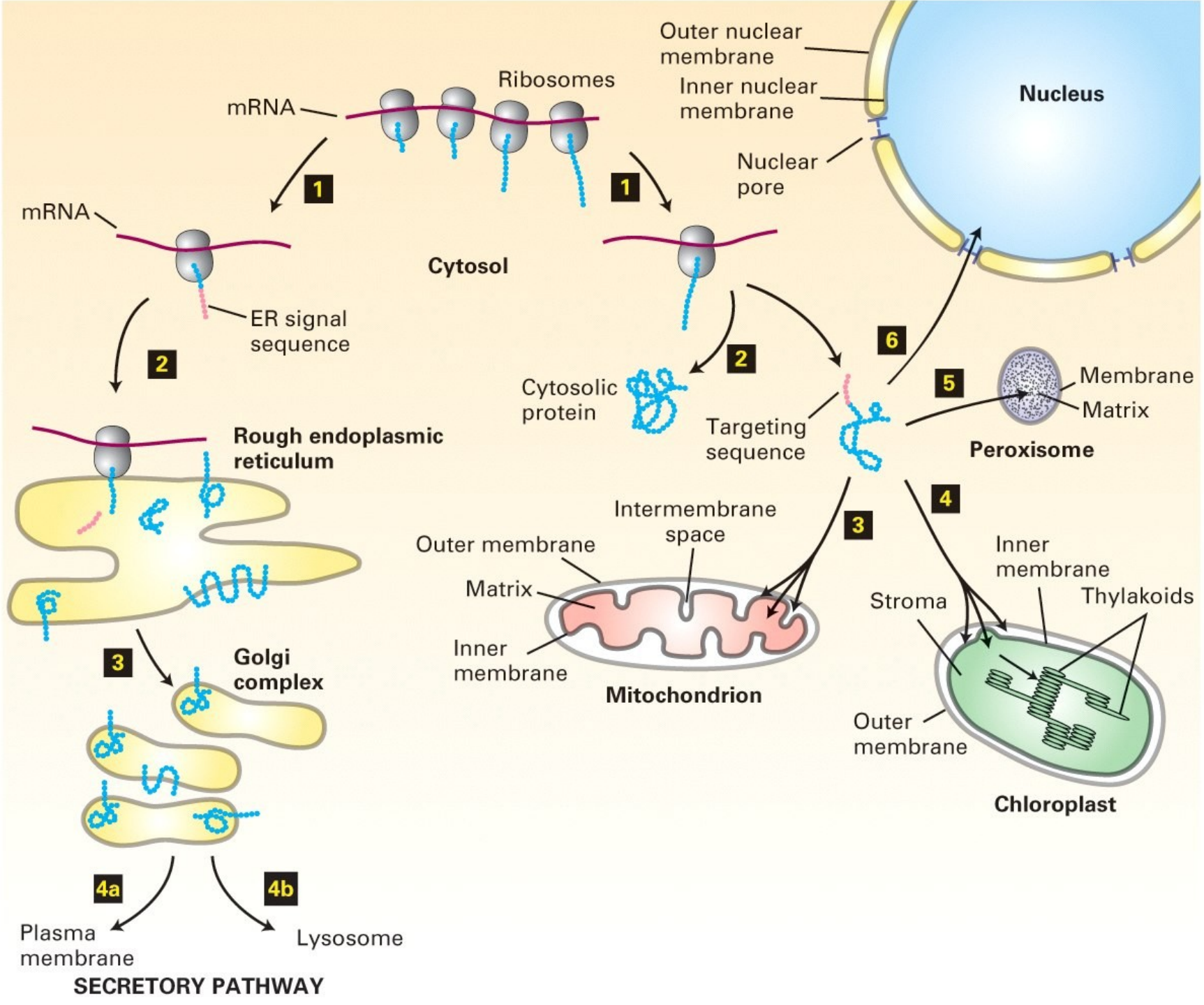


SRP receptor initiates the interaction of signal sequences with the ER membrane



Secretory proteins move from the Rough ER lumen through Golgi complex and then to cell surface by vesicle mediated transport




Outer nuclear membrane
Inner nuclear membrane
Nucleus
Nuclear pore
Ribosomes
mRNA
1
mRNA
Cytosol
ER signal sequence
2
Cytosolic protein
2
6
5
Membrane
Matrix
Peroxisome
Targeting sequence
Rough endoplasmic reticulum
3
Golgi complex
Intermembrane space
Outer membrane
Matrix
Inner membrane
Mitochondrion
4
Inner membrane
Stroma
Thylakoids
Outer membrane
Chloroplast
4a
4b
Plasma membrane
Lysosome
SECRETORY PATHWAY

Chloroplast targeting signals



1st Structure :
- contain hydroxylated aa.
- small hydrophobic aa.
- positively charged aa.
- few acidic aa.

2nd Structure:
- random coiled in solution
- β-sheet

The ability to perform conformational changes might be an important feature for cytosolic targeting

All proteins encoded by nuclear DNA are first translated on free cytoplasmic ribosomes

Soluble proteins and proteins targeted to the mitochondria, chloroplasts and peroxisomes are completely synthesized on free ribosomes

Translation of Integral membrane proteins, secreted proteins, and proteins in the ER, Golgi, and lysosomes are synthesized on ribosomes bound to the ER membrane

The subunits on free and ER bound ribosomes are identical



H5 Protein Glycosylation

Most proteins synthesized in the Rough ER are glycosylated by a core oligosaccharide.



Proteins are Glycosylated in ER and Golgi

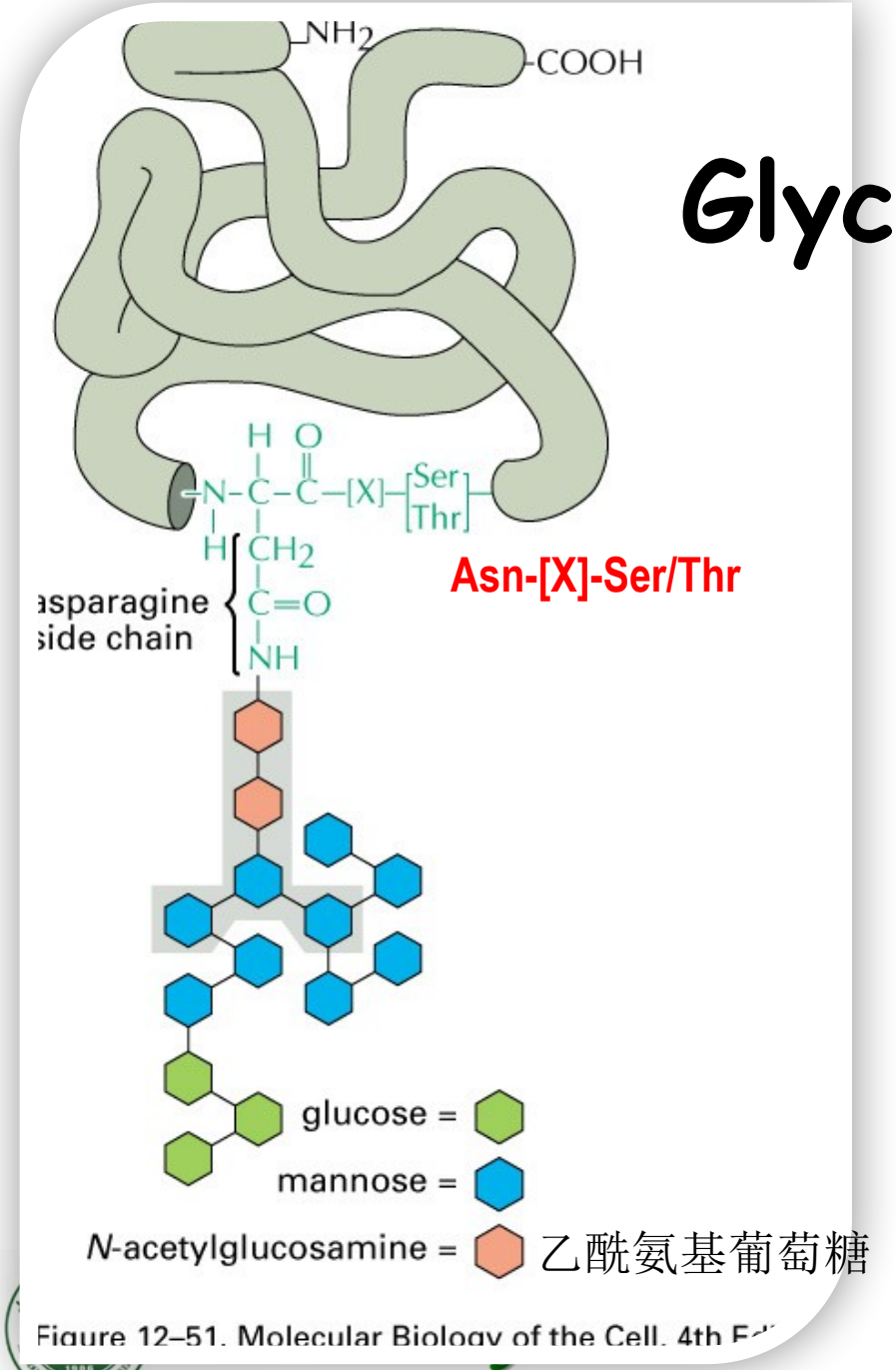


Figure 12–51. Molecular Biology of the Cell, 4th Ed.

-important for folding

-important for function

-important for targeting



N-linked on Asn and O-linked on Ser/Thr

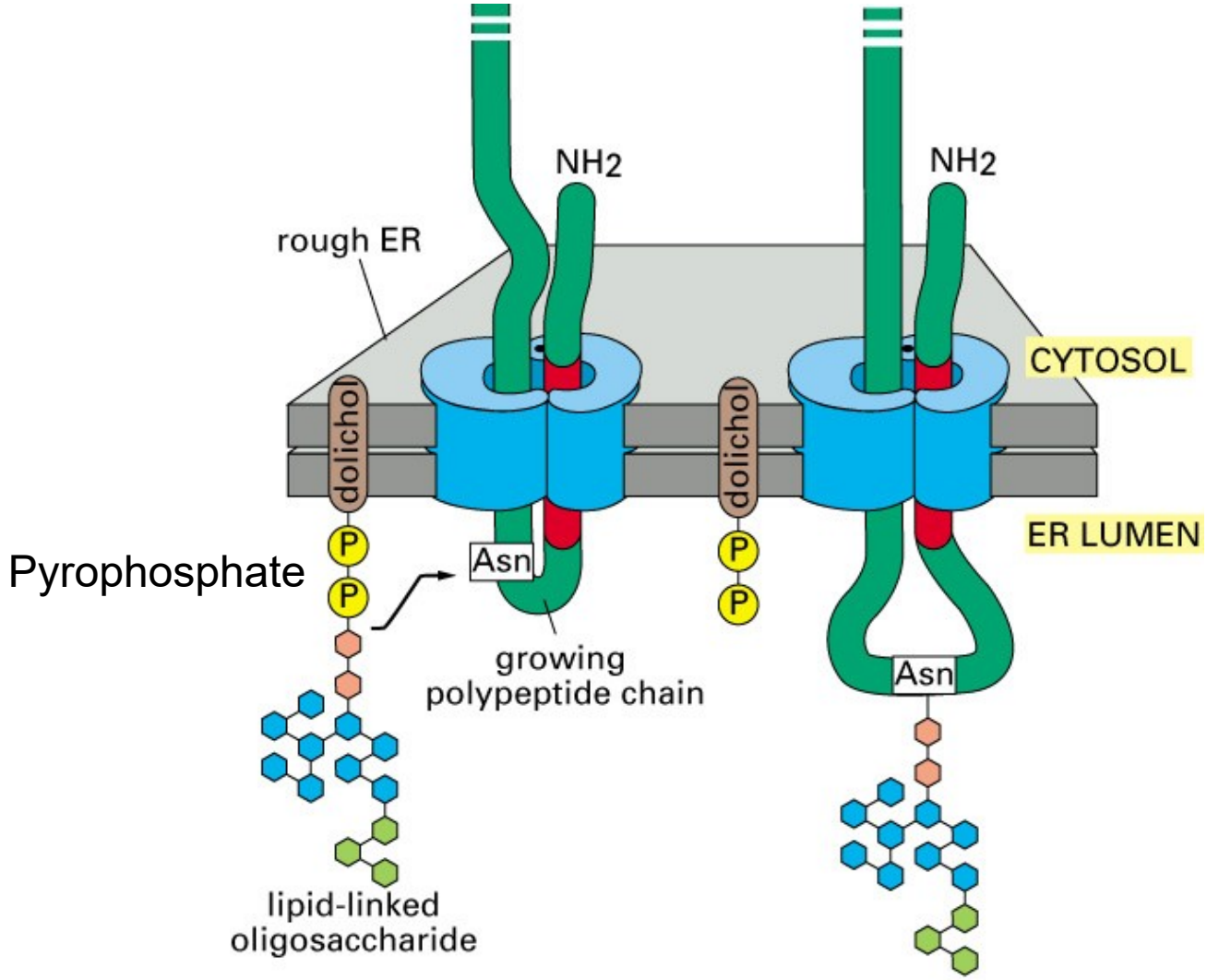
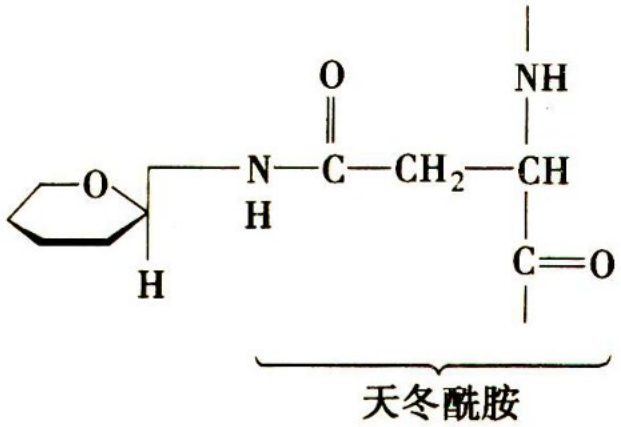


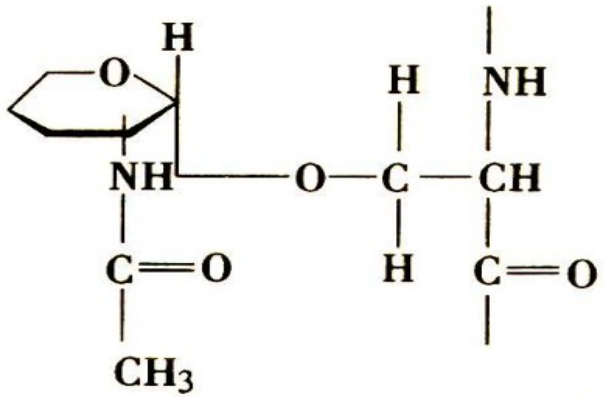
Figure 12–52. Molecular Biology of the Cell, 4th Edition.

- 连接方式
  - N- 连接：



天冬酰胺

  - O- 连接：

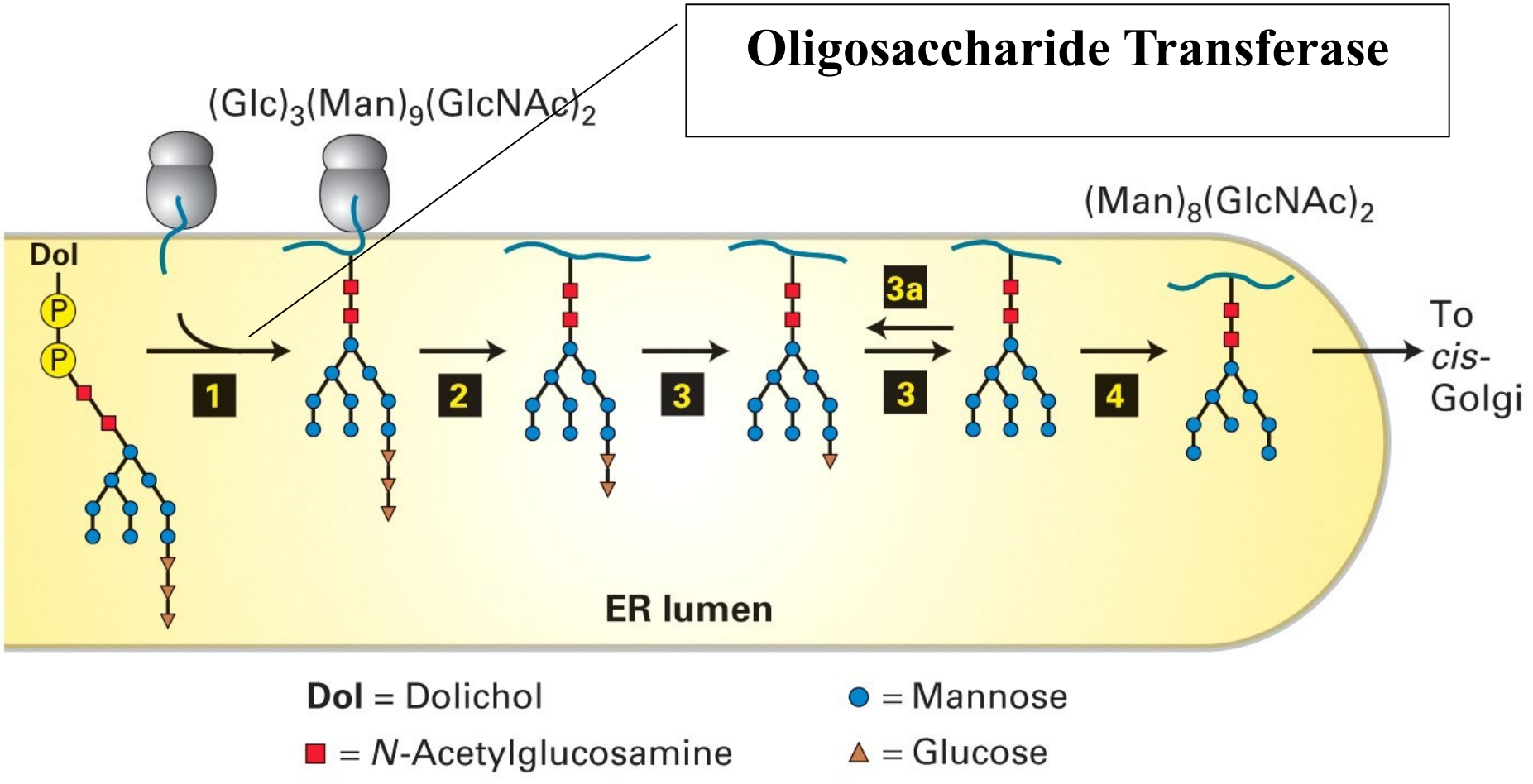


丝氨酸

The glycosylation signal is Asn-X- (Ser/Thr)



Core Glycosylation and Trimming in the ER lumen



The core oligosaccharide used for N-linked glycosylation is assembled onto the polyisoprenoid lipid, dolichol pyrophosphate.

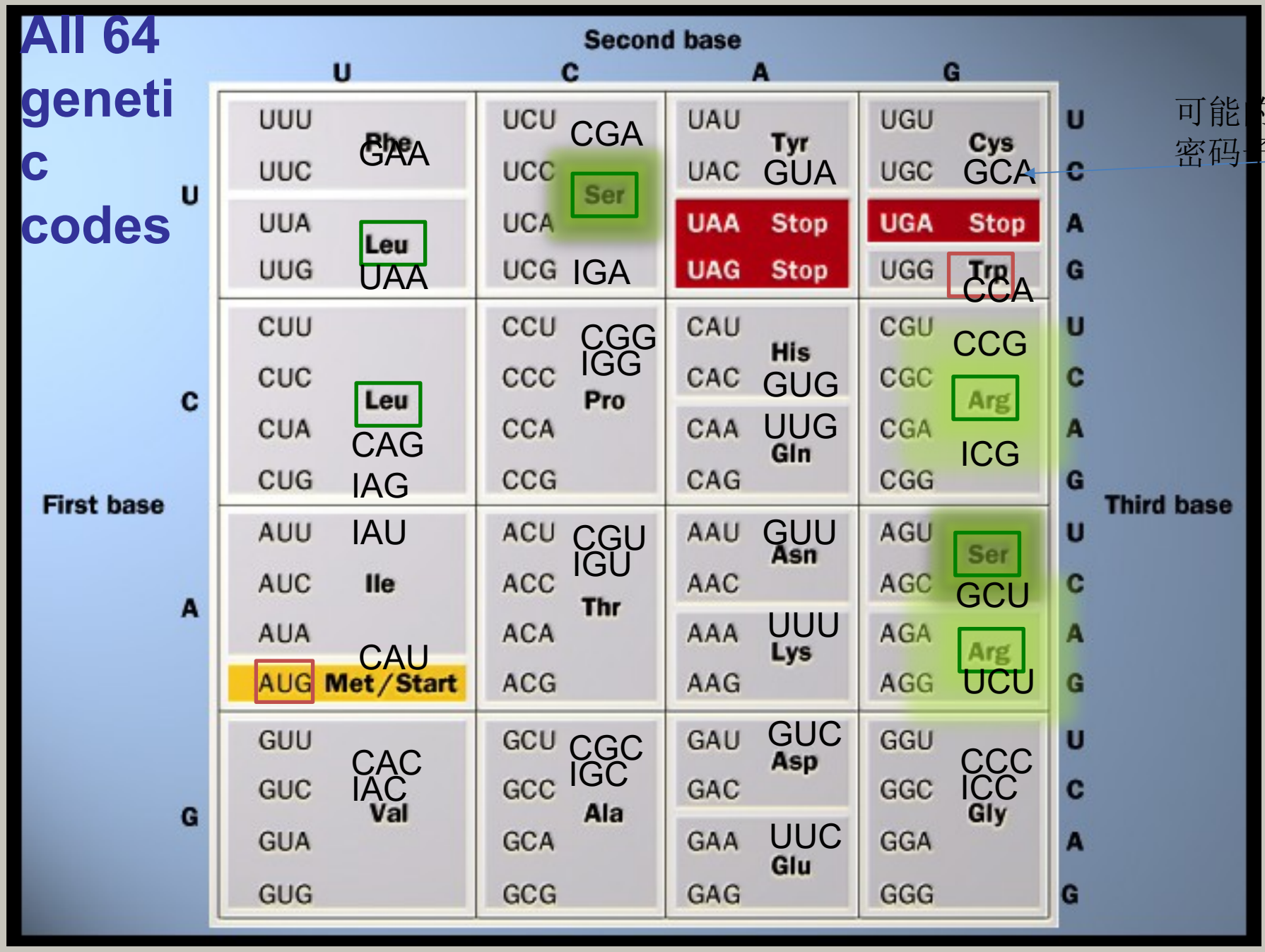


END

Thank you



All 64 genetic codes



可能的反密码子

All 64 genetic codes

